

Analysis of the impact of imported used vehicles on road safety and the effectiveness of policies

Part of the WP.29 “How it works – How to join it” series



UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE

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The publication **“Analysis of the impact of imported used vehicles on road safety and the effectiveness of policies”**, aims to analyse the policy impacts on trade of used vehicles and provide an analysis of road safety data linked to used vehicles globally. Includes a review of the impact of safer and cleaner vehicle policies that have been implemented in LMICs; and a review of road safety accidents and mortality data globally linked to used vehicles analysing the policy impacts on trade of used vehicles and analysis of road safety data linked to used vehicles globally. The publication was produced, initiated, and prepared by the UNECE Transport Division in cooperation with contracting parties and several stakeholders. The Transport Division wishes to express its sincere thanks to all those who contributed to this publication.

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List of abbreviations

AASA	Automobile Association of South Africa
ABS	Anti-lock Braking System
ADAS	Advanced Driver Assistance Systems
ARIMA	Auto-Regressive Integrated Moving Average
BEV	Battery Electric Vehicles
CCAM	Connected and Automated Mobility
CIF	Cost, Insurance and Freight
CITA	International Motor Vehicle Inspection Committee
CMF	Crash Modification Factor
CNG	Compressed Natural Gas
ECCAS	Economic Community of Central African States
EECCA	Eastern Europe, Caucasus and Central Asia
ECOWAS	Economic Community of West African States
ESC	Electronic Stability Control
ETSC	European Transport Safety Council
EUCARIS	EUropean CAR and driving licence Information System
Euro 1-7	Range of vehicle emissions standards set by the European Union
EV	Electric Vehicle
FIA	Fédération Internationale de l'Automobile
GLM	Generalised Linear Model
GRSF	Global Road Safety Facility
HEV	Hybrid Electric Vehicles
HDV	Heavy Duty Vehicle
HGV	Heavy Goods Vehicle
IDB	International Development Bank
ITC	Inland Transport Committee
ITF	International Transport Forum
LAC	Latin America and the Caribbean
LDVs	Light Duty Vehicles
LHD	Left Hand Drive
LMICs	Low- and Middle-Income Countries

LPG	Liquefied Petroleum Gas
MEM	Marginal Effects around the Mean
NAFTA	North American Free Trade Agreement
NB	Negative Binomial Model
NCAP	New Car Assessment Programme
NHTSA	National Highway Traffic Safety Administration
PHEV	Plug-In Hybrid Electric Vehicles
PTI	Periodic Technical Inspection
RHD	Right Hand Drive
SADC	Southern Africa Development Community
UNEP	United Nations Environmental Programme
VIN	Vehicle Identification Number
PTI	Vehicle Technical Inspection
VRUs	Vulnerable Road Users
WHO	World Health Organisation
WME	Weighted Mean Effect

Country classification by income

Economy	Region	Income group
American Samoa	East Asia & Pacific	High income
Andorra	Europe & Central Asia	High income
Antigua and Barbuda	Latin America & Caribbean	High income
Aruba	Latin America & Caribbean	High income
Australia	East Asia & Pacific	High income
Austria	Europe & Central Asia	High income
Bahamas, The	Latin America & Caribbean	High income
Bahrain	Middle East & North Africa	High income
Barbados	Latin America & Caribbean	High income
Belgium	Europe & Central Asia	High income
Bermuda	North America	High income
British Virgin Islands	Latin America & Caribbean	High income
Brunei Darussalam	East Asia & Pacific	High income
Bulgaria	Europe & Central Asia	High income
Canada	North America	High income
Cayman Islands	Latin America & Caribbean	High income
Channel Islands	Europe & Central Asia	High income
Chile	Latin America & Caribbean	High income
Croatia	Europe & Central Asia	High income
Curaçao	Latin America & Caribbean	High income
Cyprus	Europe & Central Asia	High income
Czechia	Europe & Central Asia	High income
Denmark	Europe & Central Asia	High income
Estonia	Europe & Central Asia	High income
Faroe Islands	Europe & Central Asia	High income
Finland	Europe & Central Asia	High income
France	Europe & Central Asia	High income
French Polynesia	East Asia & Pacific	High income
Germany	Europe & Central Asia	High income
Gibraltar	Europe & Central Asia	High income
Greece	Europe & Central Asia	High income
Greenland	Europe & Central Asia	High income
Guam	East Asia & Pacific	High income
Guyana	Latin America & Caribbean	High income
Hong Kong SAR, China	East Asia & Pacific	High income
Hungary	Europe & Central Asia	High income
Iceland	Europe & Central Asia	High income
Ireland	Europe & Central Asia	High income
Isle of Man	Europe & Central Asia	High income
Israel	Middle East & North Africa	High income
Italy	Europe & Central Asia	High income
Japan	East Asia & Pacific	High income

Economy	Region	Income group
Korea, Rep.	East Asia & Pacific	High income
Kuwait	Middle East & North Africa	High income
Latvia	Europe & Central Asia	High income
Liechtenstein	Europe & Central Asia	High income
Lithuania	Europe & Central Asia	High income
Luxembourg	Europe & Central Asia	High income
Macao SAR, China	East Asia & Pacific	High income
Malta	Middle East & North Africa	High income
Monaco	Europe & Central Asia	High income
Nauru	East Asia & Pacific	High income
Netherlands	Europe & Central Asia	High income
New Caledonia	East Asia & Pacific	High income
New Zealand	East Asia & Pacific	High income
Northern Mariana Islands	East Asia & Pacific	High income
Norway	Europe & Central Asia	High income
Oman	Middle East & North Africa	High income
Palau	East Asia & Pacific	High income
Panama	Latin America & Caribbean	High income
Poland	Europe & Central Asia	High income
Portugal	Europe & Central Asia	High income
Puerto Rico	Latin America & Caribbean	High income
Qatar	Middle East & North Africa	High income
Romania	Europe & Central Asia	High income
Russian Federation	Europe & Central Asia	High income
San Marino	Europe & Central Asia	High income
Saudi Arabia	Middle East & North Africa	High income
Seychelles	Sub-Saharan Africa	High income
Singapore	East Asia & Pacific	High income
Sint Maarten (Dutch part)	Latin America & Caribbean	High income
Slovak Republic	Europe & Central Asia	High income
Slovenia	Europe & Central Asia	High income
Spain	Europe & Central Asia	High income
St. Kitts and Nevis	Latin America & Caribbean	High income
St. Martin (French part)	Latin America & Caribbean	High income
Sweden	Europe & Central Asia	High income
Switzerland	Europe & Central Asia	High income
Taiwan, China	East Asia & Pacific	High income
Trinidad and Tobago	Latin America & Caribbean	High income
Turks and Caicos Islands	Latin America & Caribbean	High income
United Arab Emirates	Middle East & North Africa	High income
United Kingdom	Europe & Central Asia	High income
United States	North America	High income
Uruguay	Latin America & Caribbean	High income
Virgin Islands (U.S.)	Latin America & Caribbean	High income

Economy	Region	Income group
Albania	Europe & Central Asia	Upper middle income
Algeria	Middle East & North Africa	Upper middle income
Argentina	Latin America & Caribbean	Upper middle income
Armenia	Europe & Central Asia	Upper middle income
Azerbaijan	Europe & Central Asia	Upper middle income
Belarus	Europe & Central Asia	Upper middle income
Belize	Latin America & Caribbean	Upper middle income
Bosnia and Herzegovina	Europe & Central Asia	Upper middle income
Botswana	Sub-Saharan Africa	Upper middle income
Brazil	Latin America & Caribbean	Upper middle income
China	East Asia & Pacific	Upper middle income
Colombia	Latin America & Caribbean	Upper middle income
Costa Rica	Latin America & Caribbean	Upper middle income
Cuba	Latin America & Caribbean	Upper middle income
Dominica	Latin America & Caribbean	Upper middle income
Dominican Republic	Latin America & Caribbean	Upper middle income
Ecuador	Latin America & Caribbean	Upper middle income
El Salvador	Latin America & Caribbean	Upper middle income
Equatorial Guinea	Sub-Saharan Africa	Upper middle income
Fiji	East Asia & Pacific	Upper middle income
Gabon	Sub-Saharan Africa	Upper middle income
Georgia	Europe & Central Asia	Upper middle income
Grenada	Latin America & Caribbean	Upper middle income
Guatemala	Latin America & Caribbean	Upper middle income
Indonesia	East Asia & Pacific	Upper middle income
Iran, Islamic Rep.	Middle East & North Africa	Upper middle income
Iraq	Middle East & North Africa	Upper middle income
Jamaica	Latin America & Caribbean	Upper middle income
Kazakhstan	Europe & Central Asia	Upper middle income
Kosovo	Europe & Central Asia	Upper middle income
Libya	Middle East & North Africa	Upper middle income
Malaysia	East Asia & Pacific	Upper middle income
Maldives	South Asia	Upper middle income
Marshall Islands	East Asia & Pacific	Upper middle income
Mauritius	Sub-Saharan Africa	Upper middle income
Mexico	Latin America & Caribbean	Upper middle income
Moldova	Europe & Central Asia	Upper middle income
Mongolia	East Asia & Pacific	Upper middle income
Montenegro	Europe & Central Asia	Upper middle income
Namibia	Sub-Saharan Africa	Upper middle income
North Macedonia	Europe & Central Asia	Upper middle income
Paraguay	Latin America & Caribbean	Upper middle income
Peru	Latin America & Caribbean	Upper middle income
Serbia	Europe & Central Asia	Upper middle income
South Africa	Sub-Saharan Africa	Upper middle income

Economy	Region	Income group
St. Lucia	Latin America & Caribbean	Upper middle income
St. Vincent and the Grenadines	Latin America & Caribbean	Upper middle income
Suriname	Latin America & Caribbean	Upper middle income
Thailand	East Asia & Pacific	Upper middle income
Tonga	East Asia & Pacific	Upper middle income
Türkiye	Europe & Central Asia	Upper middle income
Turkmenistan	Europe & Central Asia	Upper middle income
Tuvalu	East Asia & Pacific	Upper middle income
Ukraine	Europe & Central Asia	Upper middle income

Economy	Region	Income group
Angola	Sub-Saharan Africa	Lower middle income
Bangladesh	South Asia	Lower middle income
Benin	Sub-Saharan Africa	Lower middle income
Bhutan	South Asia	Lower middle income
Bolivia	Latin America & Caribbean	Lower middle income
Cabo Verde	Sub-Saharan Africa	Lower middle income
Cambodia	East Asia & Pacific	Lower middle income
Cameroon	Sub-Saharan Africa	Lower middle income
Comoros	Sub-Saharan Africa	Lower middle income
Congo, Rep.	Sub-Saharan Africa	Lower middle income
Côte d'Ivoire	Sub-Saharan Africa	Lower middle income
Djibouti	Middle East & North Africa	Lower middle income
Egypt, Arab Rep.	Middle East & North Africa	Lower middle income
Eswatini	Sub-Saharan Africa	Lower middle income
Ghana	Sub-Saharan Africa	Lower middle income
Guinea	Sub-Saharan Africa	Lower middle income
Haiti	Latin America & Caribbean	Lower middle income
Honduras	Latin America & Caribbean	Lower middle income
India	South Asia	Lower middle income
Jordan	Middle East & North Africa	Lower middle income
Kenya	Sub-Saharan Africa	Lower middle income
Kiribati	East Asia & Pacific	Lower middle income
Kyrgyz Republic	Europe & Central Asia	Lower middle income
Lao PDR	East Asia & Pacific	Lower middle income
Lebanon	Middle East & North Africa	Lower middle income
Lesotho	Sub-Saharan Africa	Lower middle income
Mauritania	Sub-Saharan Africa	Lower middle income
Micronesia, Fed. Sts.	East Asia & Pacific	Lower middle income
Morocco	Middle East & North Africa	Lower middle income
Myanmar	East Asia & Pacific	Lower middle income
Nepal	South Asia	Lower middle income
Nicaragua	Latin America & Caribbean	Lower middle income
Nigeria	Sub-Saharan Africa	Lower middle income
Pakistan	South Asia	Lower middle income

Economy	Region	Income group
Papua New Guinea	East Asia & Pacific	Lower middle income
Philippines	East Asia & Pacific	Lower middle income
Samoa	East Asia & Pacific	Lower middle income
São Tomé and Príncipe	Sub-Saharan Africa	Lower middle income
Senegal	Sub-Saharan Africa	Lower middle income
Solomon Islands	East Asia & Pacific	Lower middle income
Sri Lanka	South Asia	Lower middle income
Tajikistan	Europe & Central Asia	Lower middle income
Tanzania	Sub-Saharan Africa	Lower middle income
Timor-Leste	East Asia & Pacific	Lower middle income
Tunisia	Middle East & North Africa	Lower middle income
Uzbekistan	Europe & Central Asia	Lower middle income
Vanuatu	East Asia & Pacific	Lower middle income
Vietnam	East Asia & Pacific	Lower middle income
West Bank and Gaza	Middle East & North Africa	Lower middle income
Zambia	Sub-Saharan Africa	Lower middle income
Zimbabwe	Sub-Saharan Africa	Lower middle income

Economy	Region	Income group
Afghanistan	South Asia	Low income
Burkina Faso	Sub-Saharan Africa	Low income
Burundi	Sub-Saharan Africa	Low income
Central African Republic	Sub-Saharan Africa	Low income
Chad	Sub-Saharan Africa	Low income
Congo, Dem. Rep.	Sub-Saharan Africa	Low income
Eritrea	Sub-Saharan Africa	Low income
Ethiopia	Sub-Saharan Africa	Low income
Gambia, The	Sub-Saharan Africa	Low income
Guinea-Bissau	Sub-Saharan Africa	Low income
Korea, Dem. People's Rep.	East Asia & Pacific	Low income
Liberia	Sub-Saharan Africa	Low income
Madagascar	Sub-Saharan Africa	Low income
Malawi	Sub-Saharan Africa	Low income
Mali	Sub-Saharan Africa	Low income
Mozambique	Sub-Saharan Africa	Low income
Niger	Sub-Saharan Africa	Low income
Rwanda	Sub-Saharan Africa	Low income
Sierra Leone	Sub-Saharan Africa	Low income
Somalia	Sub-Saharan Africa	Low income
South Sudan	Sub-Saharan Africa	Low income
Sudan	Sub-Saharan Africa	Low income
Syrian Arab Republic	Middle East & North Africa	Low income
Togo	Sub-Saharan Africa	Low income
Uganda	Sub-Saharan Africa	Low income
Yemen, Rep.	Middle East & North Africa	Low income

Executive summary

Road traffic crashes are responsible for 1.19 million fatalities globally (WHO, 2023). **Over 90% of the global burden of traffic injuries occurs in Low- and Middle-Income Countries (LMICs)**, where the rate of fatalities per population is the highest among the other regions. Over the last decades, intensive efforts have been devoted to developing new vehicle designs, higher crashworthiness and in-vehicle safety features. However, these developments have not been deployed across the world region in a systematic and equitable way. In fact, the reality of existing vehicles and existing road networks that have been developed in challenging road environments demonstrates that the regions needing the most urgent interventions in this respect have not receive adequate attention.

The unsafe state of vehicles has been identified as a key crash risk factor in many countries (WHO, 2023b). There are several UN 'core' regulations on vehicle safety (WHO, 2023) that, if applied to countries' manufacturing, production and import standards, would potentially save many lives. In parallel, the global trade in used vehicles plays a significant role in the mobility landscape of many countries. **The influx of older, less safe, and higher-emission vehicles into these regions has long raised serious concerns regarding road safety, environmental impact, and public health.**

In this context, the **UNRSF Project "Safer and Cleaner Used Vehicles for Africa"** aimed to create a platform for the major exporters of used vehicles, including the European Union, United States, and Japan, along with African importing countries, and to transform the current approach to the export and import of used vehicles. More specifically, the UNECE, along with its project partners, established minimum criteria for safety and environmental standards in used vehicles to be harmonized across regions, particularly benefiting African countries (UNECE, 2022a). This project phase raised awareness of the poor regulatory standards surrounding used vehicles and led to the creation of regional standards within African economic communities such as ECOWAS and EAC. Building on the successes and lessons learned from the first phase of the UNRSF project "Safer and Cleaner Used Vehicles for Africa", the currently ongoing second phase of the project aims to address these challenges by expanding its focus to the Central and Southern African regions, as well as selected Asian countries.

The objective of this study is the **development of recommendations based on the analysis of the policy impacts on trade of used vehicles, and the analysis of road safety data linked to used vehicles globally**. The project includes a review of the impact of safer and cleaner vehicle policies that have been implemented in LMICs; and a review of road safety accidents and mortality data globally linked to used vehicles, with a dedicated analysis aiming to estimate the road safety impacts of relevant policies.

First, **a systematic review on the existing regulations and standards** in place for the imported used vehicles in LMICs globally was carried out. The international and regional regulations and frameworks were described, following by analyses on the regional level and subsequently on the country level. At the same time, the adherence of LMICs to the UN core requirements on vehicle safety was examined. In total, **information for 136 LMICs was compiled**, based on the scientific literature, other international publications and related projects, as well as of additional sources such as national websites, exporter's data portals, blogs, media reports etc. The results were summarised in a comparative analysis, in which **the main features of existing regulations and standards** are included, namely: **year of regulation, age limit adopted, technical standards included, environmental standards included, and adherence to the UN core vehicle safety regulations**.

At the same time, an international literature review on the effects of vehicle characteristics on road safety, confirmed that **vehicle age is the most popular policy used to date to discriminate the state of imported used vehicles in LMICs and its safety implications**. Older vehicles are associated with increased crash and fatality risk. Nevertheless, the impact of vehicle age on crash risk and crash severity is strongly associated with other factors, such as the presence of restraint systems, the crashworthiness of the vehicle, the Periodic Technical Inspection status, and the active safety / driver assistance systems in place. It is therefore concluded that countries' used vehicles policies need to be assessed not only in terms of age limits set, but also in terms **of additional indicators (variables) of interest such as the adherence to UN vehicle regulations and the use of restraint systems in the country**.

Data collection included the compiling of key indicators on road traffic mortality, vehicle fleet size, UN vehicle regulations adopted, imports / exports of used vehicles, and socioeconomic indicators of countries around the globe. Moreover, a stakeholders' survey was launched among the members of the UNECE WP.29 (World Forum for Harmonisation of Vehicle Regulations) to collect information on their knowledge, experience and recommendations with respect to the safety of imported used vehicles.

The analysis methods that were implemented for the purposes of the study were:

- **A macroscopic analysis**, aiming to associate the rate of imported used vehicles with the traffic mortality rates of countries around the world. A statistical model, i.e. a Generalised Linear Model (GLM) with a Negative Binomial (NB) distribution for mortality rates was developed for all countries, with a separate model for LMICs. This analysis provides the effect of used vehicle imports on mortality rates, and its Marginal Effect around the Mean (MEM).
- **A cross-sectional before-and-after analysis**, in which the road safety performance of a country in the years following a policy intervention is compared with a scenario that shows what would have happened in the country had the intervention not been implemented. For that purpose, a ‘comparison group’ of similar countries where no intervention took place is used. This analysis provides the safety effect (percentage reduction of fatalities) that can be credibly attributed to the policy intervention, without any of the bias that may be involved in a simple before/after comparison. Six case studies were conducted for that purpose, representing the different regions and the different types of vehicle regulations, also considering data availability and UN vehicle regulations adherence. **The selected countries for the case studies are: Malaysia, Pakistan, Egypt, Romania, Georgia and Nigeria.**

The stakeholders’ survey results revealed that the imports of used vehicles are considered a valuable resource to meet the mobility needs and contribute to the economic growth of LMICs. The survey participants highlighted the lack of evidence supporting the potentially detrimental effect of imported used vehicles on road safety and the need to adopt a holistic policy approach to the road safety challenges in these countries. Furthermore, they placed the policy focus on technical inspections of imported used vehicles and the adoption of UN core vehicle regulation, rather than on the age restrictions or complete bans. It was also highlighted that **the quality of imported used vehicles should be a shared responsibility of exporters and importers.**

The results of the macroscopic analysis showed that the size of the vehicle fleet (total registered vehicles per population) and the average number of used vehicle imports per population over the last five years are significantly associated with increased mortality rates. In particular, the marginal effects of the model estimates (all other things kept equal) are as follows:

- In all countries, each 1 unit increase of total registered vehicles per 108 of population leads to an increase of about 0.091 of the crash mortality rate.
- In LMICs, each 1 unit increase of total registered vehicles per 108 of population leads to an increase of 0.086 of the crash mortality rate.
- In all countries, a safety belt wearing compliance above 70% decreases mortality rates by more than 2.9 on average.
- In LMICs, **each 1 unit increase of average used vehicle imports (years 2017-2021) per population leads to an increase of 0.476 of the crash mortality rate.**

The results of the case studies showed that the impact of policies on the imports of used vehicles have been diverse, which is intuitive given that the type of regulation and the time of regulation varied considerably across the examined countries. The below Table summarises the results of the conducted case studies. The ‘before’ period was taken as 5 years before the year of regulation introduction, and the ‘after’ period was taken as 5 years after the year of regulation (the year of regulation itself was not included in these periods). The selected control groups were based on a comparison of basic geographical, socioeconomic and cultural indicators. In some cases, more than one control groups were tested.

Country*	Year of regulation	Type of regulation	Control groups	CMF**	Safety effect (% reduction)	Statistical significance
Malaysia	2017	Age limit (<5 years)	India	0.843	15.7	Significant
Pakistan	2006	Age limit (<3 years)	India	0.994	0.6	Non-significant
Egypt	2017	Ban (<1 year)	i. India, ii. Several African countries	0.674	32.6	Significant
Romania	2007	UN core vehicle regulations	Russia	0.783	21.7	Significant
Georgia	2017	Age and vehicle size taxation	i. Russia, ii. Ukraine	1.368	-36.8	Significant
Nigeria	2010	Age limit (<15 years)	Several African Countries	0.701	29.9	Significant

* All countries currently adhere to UN vehicle regulations

** CMF: Crash Modification Factor

The results showed that **in most of the examined countries, the introduction of used vehicles imports regulation resulted in a statistically significant reduction of traffic fatalities within the first 5 years after implementation.** It can be said that countries that introduced an age limit on vehicle imports had a positive effect on road fatalities, even if this age limit was quite lenient (i.e. >15 years in Nigeria). On the other hand, taxation based regulation (e.g. Georgia) does not seem to be effective in the short term. In Romania, the accession to the EU likely brought additional road safety regulations that contributed to the positive short-term benefit.

It is also worth noting that the estimated safety effect may also depend on the trend and magnitude of vehicle imports before the regulation. For instance, Pakistan introduced the regulation at a time when used vehicles imports started to increase sharply, and the influx of vehicles possibly limited the effectiveness of the policy. On the other hand, Nigeria introduced additional policies to reinforce national vehicle industry at approximately the same time as the used vehicles regulation, and this led to a sharp decrease in the imports, which probably facilitated the impact of the policy.

For three out of the six case studies, the policy was introduced in 2017, and may be too early to fully evaluate. In these cases, data after 2019 had to be collected from informal web resources (national or international).

From the above results, a Weighted Mean Effect was calculated. It was found that **the introduction of a policy on the imports of used vehicles may result in a 18.8% reduction of traffic fatalities within 5 years of implementation**, on average. This indicates a substantial benefit to be expected from introducing such regulations, while also adopting the UN 'core' vehicle regulations. Of course, individual case studies suggest that the effect can be higher or lower, depending on the timing of regulation, the magnitude of vehicle imports and the time of regulation, and the other accompanying policies that may affect road safety at the same time. Moreover, the policies can be ineffective or even have a negative effect on traffic fatalities; the conducted case studies suggest that, controlling the influx of used vehicles and the motorisation rate / traffic growth in the country are key to ensuring the effectiveness of such policies.

Based on the above, **the following recommendations are made:**

- **Recommendation 1: Countries should monitor the state of the vehicle fleet in the country**, by promoting newer, environmentally friendly and technically robust vehicles through the adoption of UN core vehicle regulations, high environmental standards, and the implementation and monitoring of Periodic Technical Inspections (PTI) for vehicles
- **Recommendation 2: Countries should control the influx of used vehicles imports**, through the adoption of a national strategy for addressing the transport demand, including the provision of safe and sustainable public and private transport options in the medium-term, to reduce the low quality private transport, and through the adoption of minimum technical and environmental standards for imported used vehicles.
- **Recommendation 3: Countries, especially LMICs, should introduce effective regulations of imported used vehicles**, through the adoption of at least all of the following: (i) the UN core regulations for all imported used vehicles, (ii) age limit of 8 years or lower for all imported used vehicles, (iii) EURO 4 or higher environmental standards for all imported used vehicles, and (iv) mandatory technical inspections of all used vehicles prior to their deployment on the country's road network.
- **Recommendation 4: International standards should be established for the imports of used vehicles in LMICs**, through relevant studies and stakeholder consultations to consolidate the national experience, identify best practice and facilitate the transfer of knowledge; international organisations should take leadership in developing appropriate standards, and promoting these standards among their member states. Countries should work closely with the UNECE, UNEP and other international stakeholders to benefit from international evidence and receive strategic and technical assistance in the development of their used vehicle policies
- **Recommendation 5: Countries should establish and monitor the presence and use of restraint systems and protective devices in all vehicles in the country**, through the establishment of a national law on safety belts, helmets and standardised child restraints use for all passengers (front and rear seats) with UN safety belt standards, and the enforcement of this law by means of systematic checks, random in time and space, by means of traffic police roadside checks, and effective enforcement of fines for offenders.
- **Recommendation 6: Countries should work on improving their data systems regarding road safety, vehicle fleet and used vehicle imports**, through the creation of databases on the basis of international definitions (e.g. the UN Glossary for Transport statistics), and the regular update of these databases, through the reporting of their data to international organisations' databases or relevant questionnaires and through national contact points with good knowledge of data quality and limitations.

1. INTRODUCTION

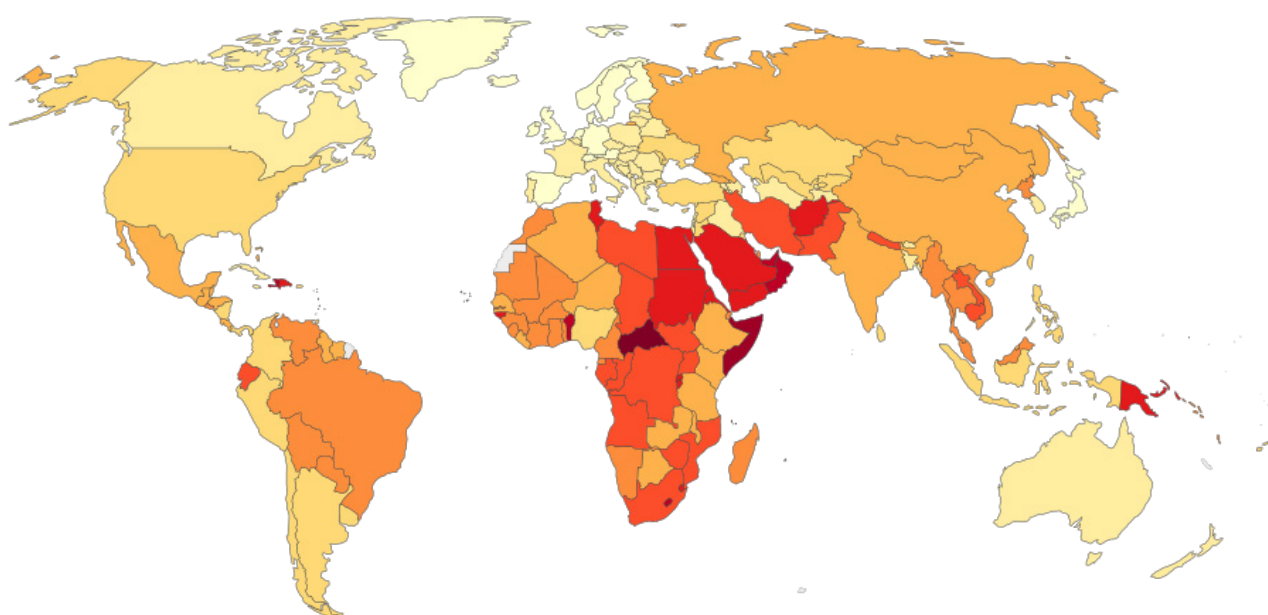
1.1. Background

Road traffic crashes are responsible for 1.19 million fatalities globally (WHO, 2023); this number is reduced compared to previous years (e.g. 1.35 million on 2018); however it remains the leading cause of death among people younger than 29 years old, and among the top 10 leading causes of death overall. The United Nations' Sustainable Development Goal target 3.6 aims to reduce traffic fatalities by 50% by 2030, a target which has been adopted by most countries globally. Moreover, the European Commission had adopted the Vision Zero approach (Johansson, 2009), as the new road safety paradigm that aims to eliminate road fatalities in the region by 2050 (EU, 2019).

However, over 90% of the global burden of injuries occurs in Low- and Middle-Income Countries (LMICs), where the rate of fatalities per population is the highest among the other regions, as shown in Figure 1.1 (WHO, 2023). More specifically, there is a large disparity in traffic mortality rates between regions, with the European average being equal to 8, the Southeast Asia average being equal to 16, and the African average being equal to 21 fatalities per 100,000 inhabitants. It has thus become imperative to continue searching for niche, targeted actions to improve both the global and LMIC-specific road safety 'footprint'.

Figure 1.1

Traffic Fatalities per 100,000 Population in Countries around the World, (2017)



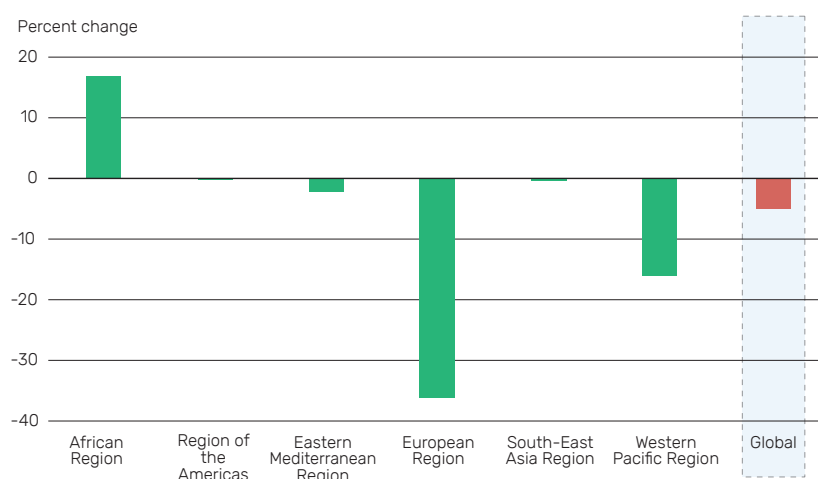
Source: IHME, Global Burden of Disease (GBD)

Note: To allow comparisons between countries and over time this metric is age-standardized

In addition, while in High- and High-to-Middle income countries there has been a constantly decreasing trend over the last decades, road safety outcomes appear to have plateaued in many countries of these regions (see Figure 1.2). In the decade of 2010-2021, there has been a reduction in road fatalities of ~35% in the European region, ~15% in the Western Pacific region, ~5% in the East Mediterranean region and ~2% in the Southeast Asia region – leading to a global reduction of 5%. However, at the same time, there were several countries in each of these regions where reductions were negligible, or even an increase in fatalities was observed. Notably, in the African region there was an increase of 18% in road fatalities over the last decade. This increase can be explained by the economic development and the resulting increase in the level of motorisation (e.g. number of vehicles per 1,000 inhabitants), together with an unequally developed road infrastructure, vehicle regulation framework and safety culture among road users as well as among national and local governments.

Looking into the regional data in more detail, there is a clear association between country income level, road safety outcomes and road safety progress (see Figure 1.3). In the African region, most Low income countries presented an increase in fatalities. In Southeast Asia and in the Americas, all low income countries presented an increase in road fatalities. Low middle income countries had mixed outcomes: in Southeast Asia one third of the countries presented an increase, while in the African region most of the countries presented an increase. On the other hand, it is clearly observed the upper middle- and high-income countries have been largely able to decrease their traffic fatalities, especially in the European region.

Figure 1.2
Percentage Change in Estimated Road Fatalities per Region, 2010 - 2021



Source: WHO, 2023

Figure 1.3
Number of Countries where a Change in Road Traffic Deaths has been Observed, by Region and Country Income Level, 2010-2021



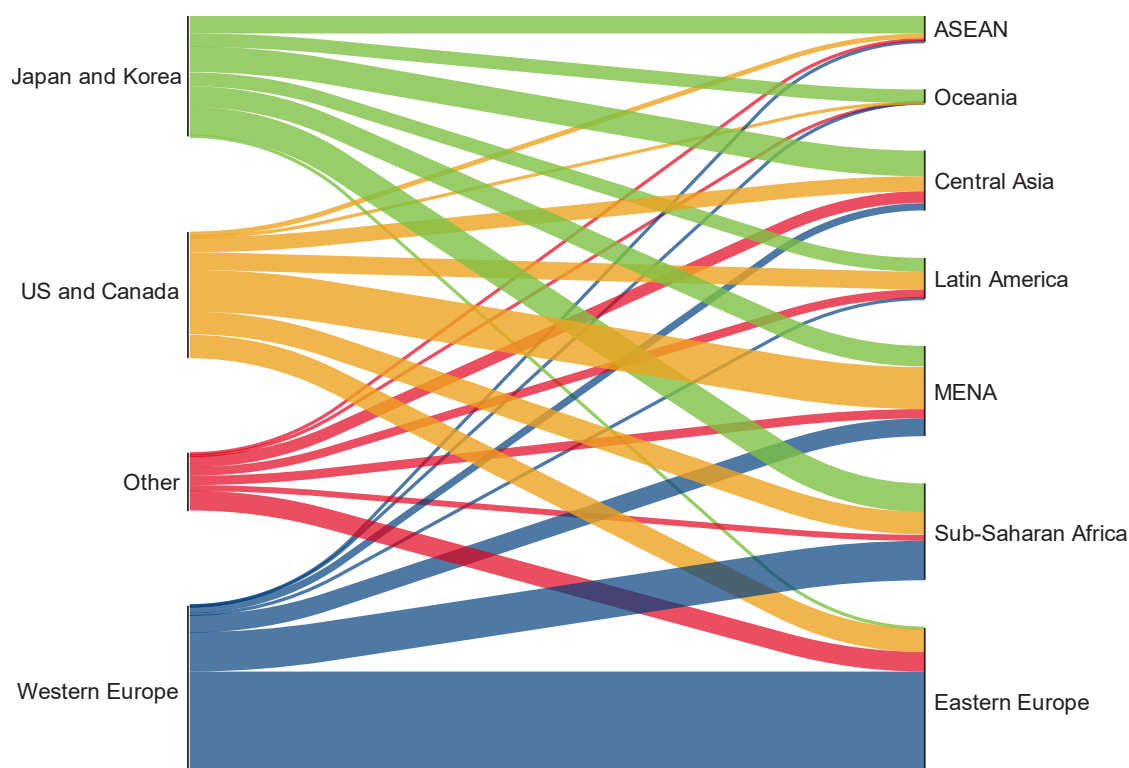
Source: WHO, 2023

Over the last decades, intensive efforts have been devoted to developing new vehicle designs, higher crashworthiness and in-vehicle safety features (e.g. ADAS – Advanced Driver Assistance Systems), taking advantage of the rapid engineering and technological advances. At the same time, the Safe System approach (WHO, 2018) has led to important road infrastructure improvement, with road function better aligned to the road geometry, design of consistent, predictable and self-explaining road environments, as well as ‘forgiving’ roads and roadsides that may reduce the severity of injury in case of a crash. However, these developments have not been deployed across the world region in a systematic and equitable way. In fact, the reality of existing vehicles and existing road networks that have been developed in challenging road environments demonstrates that the regions needing the most urgent interventions in this respect have not receive adequate attention.

The unsafe state of vehicles has been identified as a key crash risk factor in many countries (WHO, 2023b). There are several UN ‘core’ regulations on vehicle safety (WHO, 2023) that, if applied to countries’ manufacturing, production and import standards, would potentially save many lives. These include requiring vehicle manufacturers to meet front and side impact regulations, to include electronic stability control (to prevent over-steering) and to ensure airbags and safety -belts are fitted in all vehicles. It has been well established that without these basic standards the risk of traffic injuries – both to those inside the vehicle and those outside of it – is considerably increased.

In parallel, the global trade in used vehicles plays a significant role in the mobility landscape of many countries, especially LMICs. In Figure 1.4, it can be seen that the biggest used vehicle exporters are Western Europe, USA, Canada, Japan and Korea, i.e. High or Upper middle income regions, while the biggest importers of used vehicles are regions with more LMICs. The influx of older, less safe, and higher-emission vehicles into these regions has long raised serious concerns regarding road safety, environmental impact, and public health (ITF, 2023). The adoption of UN regulations of core vehicle standards in many high income countries, does not prevent the sale of those cars who are deemed sub-standard in those countries (new or second-hand) to be exported to LMICs (Global NCAP, 2020)

Figure 1.4
Flow of Used Vehicles from Exporter to Importer Regions, 2019



*MENA: Middle East & North Africa
Source: OECD/ITF, 2023

The UNRSF project, “Safer and Cleaner Used Vehicles for Africa,” set out to reform the import and export practices surrounding used vehicles by bringing together major exporting regions—including the EU, the United States, and Japan—with African importing countries. The goal of that project was not to restrict the movement of used vehicles but rather to develop harmonized standards and policies governing their transfer. Through this framework, the project aimed to address the low-quality vehicle fleets prevalent in many developing regions, particularly in Africa, where imported used vehicles often lack adequate safety and environmental standards.

The UNECE, collaborating with UNEP, CITA, and FIA, led the Safer and Cleaner Used Vehicles project to establish these minimum safety and environmental standards across Africa. The initiative not only emphasized the regulatory gaps that expose these regions to safety and environmental risks but also sparked regional standardization within ECOWAS and EAC. By coordinating research and engaging stakeholders across import-export regions, the project promoted awareness and initiated an **Informal Working Group under WP.29 (World Forum for Harmonisation of Vehicle Regulations) to create globally relevant technical standards for both new and used vehicles**. These steps intend to uplift vehicle quality, contributing to sustainable development in Africa and potentially influencing other regions with limited vehicle regulations (UNECE, 2022a).

Building on the successes and lessons learned from the first phase of the UNRSF project “Safer and Cleaner Used Vehicles for Africa,” the currently ongoing second phase of the project aims to address these challenges by expanding its focus to the Central and Southern African regions, as well as selected Asian countries. These regions continue to import a critically large share of used vehicles, with a noted absence of higher standards and regulations to ensure their safety and environmental compatibility. Due to this reality, a large share of the used vehicles imported cause more road crashes resulting in fatalities and severe injuries as well as high pollutant and climate emissions. The objective is to put in place regulations, standards, and processes to ensure that importing countries will only receive quality used vehicles; and that exporting countries agree to minimum technical requirements in the transfer of used vehicles.

By addressing the gaps in standards and regulations for the importation of used vehicles, this project seeks to deliver substantial benefits in terms of road safety, environmental protection, and economic development. The outcomes of this report will provide a critical foundation for informed decision-making and policy development, ultimately contributing to safer roads and cleaner environments in the targeted regions.

1.2. Objectives and timeline

The objective of this Report is the development of recommendations based on the analysis of the policy impacts on trade of used vehicles, and the analysis of road safety data linked to used vehicles globally. This initiative aligns with the Global Plan of Action and the Inland Transport Committee (ITC) Recommendations for Enhancing National Road Safety Systems, which emphasize the need for coordinated global efforts to improve vehicle safety standards and environmental sustainability. The report will include a review of the impact of safer and cleaner vehicle policies that have been implemented in LMICs; and a review of road safety accidents and mortality data globally linked to used vehicles. The specific activities that were undertaken towards this end are:

- a. Data collection of vehicle regulations (including vehicle emission standards)
- b. Data collection of road safety mortality and injuries, as well as used vehicles traded.
- c. Consultations stakeholders (including importers and exporters).
- d. Analysis of data and development of reports.

More specifically, the work was conducted in coordination with the World Forum for the Harmonisation of Vehicles (WP.29), including the Exporters, Importing countries (South and Central Africa importing countries, Asian importing countries (Cambodia and Mongolia), Sub-regional bodies (Southern Africa Development Community (SADC) and Economic Community of Central African States (ECCAS), as well as the UNEP (Authors of the Global Trade in Used Vehicles Reports), and other international organizations and associations.

The outcome of this Work are summarised in the present report including:

- i. The analysis of the impact of safer and cleaner vehicle policies, including vehicle emission standards, that have been implemented in LMICs;
- ii. The analysis of road safety data (mortality and injuries) and used vehicles globally, with a case study for each region

1.3. Structure of the report

In Chapter 2 of the present report, a systematic review is presented on the existing regulations and standards in place for the imported used vehicles in LMICs globally. First, the international and regional regulations and frameworks are described, following by analyses on the regional level and subsequently on the country level. Finally, a synthesis of results is carried out, in which the main features of existing regulations and standards are showcased i.e. year of adoption, technical standards included, environmental standards included and their specific thresholds in each case, e.g. vehicle age, emission level etc.

In Chapter 3, an analysis framework is developed for the next stages of the project. First, a review of the state of the art on the effects of vehicle characteristics on road safety is presented. Second, the analysis methods that will be implemented in the next stages of the project are described, together with the data and information that is needed in each case. Third, the case studies to be carried out are outlined, together with the criteria for the candidate countries to be selected.

In Chapter 4 of the present report, the road safety and other relevant data on used vehicles are presented. The results of data collection on the selected indicators is presented based on the needs and criteria set out in Chapter 3. Further data needs are identified and potential implications for the analysis are discussed. The stakeholder consultation is described and the findings are presented, with respect to the attitudes and perceptions of the stakeholders on the safety issues of imported used vehicles and their effective policy approaches.

In Chapter 5 of this report, the findings of the macroscopic analysis are presented, by means of a descriptive analysis and a models' development. Two models are developed associating mortality rates with the rates of used vehicles import, one model for all countries, and one model for LMICs. The parameter estimates of the models are discussed, and their Marginal Effects around the Mean are estimated, giving the expected change in mortality rates resulting from unit increases in other variables.

In Chapter 6, six case studies are conducted to estimate the effect of used vehicle policies on road safety. The case studies are carried out based on before-and-after analysis for six countries: Malaysia, Pakistan, Egypt, Romania, Georgia and Nigeria. The estimated safety effects are assessed in terms of their statistical significance. Eventually, a Weighted Mean Effect analysis allows to summarise the results of the case studies and calculate a global safety effect of used vehicle policies on road safety. The findings are also discussed in terms of their limitations and needs for further analysis.

In Chapter 7, the conclusions of the work are presented, together with the resulting policy recommendations, as well as the recommendations for further scientific research.

2. REVIEW OF THE STATE OF THE ART ON USED VEHICLES POLICIES AND RELATED STANDARDS IN LMICs

2.1. Review of international used vehicle policies and related standards

The United Nations Environment Programme (UNEP) has investigated the overall impact of used vehicles in the environment globally. UNEP has focused mainly on Light Duty Vehicles (LDVs) starting from 2020, with updates on 2021 and 2024. The first iteration of the UNEP reports (UNEP, 2020) focused on LDVs and noted that despite the huge volumes of used LDV vehicles (14 million between 2015 and 2018), in there are no regional or global agreements regulating the trade and movement of used vehicles. There were 146 countries studied as part of that report, and various key outcomes can be summarized in terms of restrictions placed on LDVs. Specifically, 66 countries were found to restrict the age of imported vehicles, in varying degrees ranging from 3 to 15 years. This approach was found to facilitate easy enforcement.

Regarding environmental safeguards, 28 countries had adopted vehicle emission standards, whereas 100 out of 146 countries had not imposed any type of emission standard at all by the time of that report. The last 18 countries had outright banned used vehicle imports. The report also noted that fiscal regulations were found to be effective towards controlling used vehicle imports, and cited flat-out age-based taxation, progressive excise taxation based on greenhouse gas emissions or engine size, and possible exemptions for more desired vehicle types, such as hybrid electric and electric vehicles. Certain countries were found to implement selective bans, such as prohibiting diesel vehicles above a specific age from entering city centers. Additionally, some countries have introduced mandatory labeling of used vehicles to inform consumers about a vehicle's fuel consumption and emissions. The report has determined that 81 out of 146 countries have policies on used LDV regulations that can be characterized as 'weak' or 'very weak', while the policies of 47 countries can be characterized as 'good' or 'very good'. Only 10 out of 54 African countries had a 'good' or 'very good' LDV regulatory framework characterization (in particular: Algeria, Chad, Côte d'Ivoire, Gabon, Lesotho, Mauritania, Mauritius, Morocco, Rwanda, Tunisia).

Overall, the report notes that **the disparity in policy measures between exporting and importing markets has resulted in a global trade of used vehicles that are often obsolete, unsafe, and highly polluting.** While some traded vehicles meet advanced standards and provide affordable access to environmentally friendly technology, many lack basic environmental requirements, contributing significantly to air pollution and climate emissions in recipient countries. Stricter import regulations, combined with appropriate fuel quality, can lead to the adoption of cleaner, more efficient vehicle technologies, improving both environmental impact and road safety. Some countries have introduced incentives for importing used hybrid and electric vehicles, promoting a shift to cleaner fleets, and others that previously banned all used vehicles are now allowing the import of these greener options.

The second iteration of the UNEP reports (UNEP, 2021) provides an update and reaffirmation of the findings of the previous iteration. Specifically, the related document denotes EU, Japan, USA and ROK as the main exporters of used LDVs, 66% of which are imported by developing countries. The previous assessment of used LDV regulations was found to improve, with countries of regulations that are characterized as 'good' or 'very good' rising from 47 to 62. Apart from the cited ECOWAS project, which was examined in the previous, advances in Peru, Cambodia and Ethiopia are also mentioned (outlined in the following section).

The third iteration of the UNEP reports (UNEP, 2024) provides the most recent update providing data of used LDV exports for the period 2015-2022 from the EU, USA, Japan and ROK. Regarding region-wise imports, Africa imported the highest share of used LDVs at 33%, followed by the EECCA (Eastern Europe, Caucasus and Central Asia) region at 24%, Asia-Pacific at 16%, the Middle East at 15%, and LAC at 12%. While there was a temporary dip in export numbers during the Covid-19 pandemic, numbers from 2022 have suggested a full rebound to pre-pandemic levels. Another noteworthy remark is that the EC, which is a major used LDV exporting region, is proposing a new regulation on end-of-life vehicles that, if enacted, will

significantly enhance the quality of used vehicles exported from the region. The regulation will ban the export of end-of-life vehicles, require a valid roadworthiness certificate for exported vehicles, assist importing countries with compliance to their regulations, and create a system for enforcement and information exchange. UNEP is also actively engaging with other major exporting countries to encourage similar measures to regulate the quality of used vehicle exports.

Box 1.

The UNECE / UNEP project “Safer and Cleaner Used Vehicles for Africa”

The UNECE / UNEP, along with CITA, and FIA aimed to address the influx of low-quality used vehicles into developing countries, particularly in Africa (UNECE, 2022a). UNECE’s efforts included studying the global trade of used light-duty vehicles (LDVs) and identifying the major exporters—the EU, Japan, the U.S., and South Korea—as well as the primary importers, with African nations receiving the largest share. The UNECE/UNEP project highlighted critical issues such as pollutant emissions, vehicle safety, energy consumption, and operational costs associated with used vehicles in developing countries. With most developing nations lacking regulations on vehicle imports, UNECE emphasized the need for standardized, enforceable policies to ensure that imported vehicles meet minimum safety and environmental standards, thus reducing air pollution and enhancing road safety.

To drive regulatory improvements, the UNECE/UNEP project facilitated coordination between exporting and importing nations, resulting in significant policy shifts in Africa and other developing regions. Through studies, UNECE/UNEP demonstrated the impact of unregulated imports on roadworthiness and emissions, emphasizing that poor-quality vehicles continue to circulate in countries with weak or no standards. Case studies on Kenya, the Netherlands, and Mauritius illustrated the benefits of stricter import policies, such as age limits and emissions standards, in improving vehicle fleets’ safety and environmental performance. Ultimately, the UNECE/UNEP initiative led to the development of regional vehicle standards within the ECOWAS and EAC blocs, as well as establishing a framework for global technical standards under WP.29, intended to enhance sustainable transportation across Africa and beyond.

Source: UNECE, 2022a

Meanwhile, **the UN Global Plan of Action for Road Safety 2021-2030 explicitly states its Target 5** as follows: “By 2030, 100% of new (defined as produced, sold or imported) and used vehicles meet high quality safety standards, such as the recommended priority UN Regulations, Global Technical Regulations, or equivalent recognized national performance requirements.” (UN, 2020). Therefore, it is anticipated that all nations will gradually move towards adoption of these standards. However, it is also evident that different nations, and especially LMICs, will have to move to these standards from different starting points.

More specifically, **the core UN vehicle safety regulations** that should enter into force (adoption of related UN standards or equivalent national law) are shown in Box.2 (WHO, 2023).

Box 2.

The UN Core Vehicle Regulations

a. Front and Side Impact Protection

This regulation ensures vehicles provide sufficient protection to occupants in the event of a frontal or side collision, reducing the risk of injury or death by requiring vehicles to meet specific crash-test performance criteria.

b. Safety-Belt and Safety-Belt Anchorages

Vehicles must be equipped with safety belts and strong, reliable anchor points that meet safety standards, ensuring safety belts remain secure during crashes to protect occupants from being ejected or thrown inside the vehicle.

c. Electronic Stability Control

This requirement mandates vehicles be fitted with electronic stability control systems, which help drivers maintain control during sharp turns or slippery conditions, preventing skidding and reducing the risk of rollovers and loss of control crashes.

d. Pedestrian Protection

Vehicles are required to include features that minimize injury to pedestrians in the event of a collision, such as impact-absorbing bumpers and hoods, designed to reduce the severity of injuries to pedestrians and other vulnerable road users.

e. Braking Systems

Vehicles must be equipped with advanced braking systems that comply with specific performance criteria, ensuring effective stopping power and including features like anti-lock brakes (ABS) to prevent wheel lockup during emergency braking.

f. Periodic Technical Inspection (PTI) /Assessment

Regular inspections of vehicles are required to ensure ongoing compliance with safety and emissions standards, focusing on critical areas such as brakes, lights, tires, and exhaust systems, to ensure roadworthiness over time.

Source: UNECE, 2022a (WP.29)

These standards aim to ensure that legislation exists that: (i) specifies requirements and standards for vehicle equipment, and (ii) ensures vehicle safety inspections or assessments. However, the above standards for core safety equipment in vehicles is absent in most countries, most notably in low- and middle-income countries. Just over half of countries (88) currently have legislation that specifies the requirements and standards for safety-belts and safety-belt anchorages and only around a third of all countries have legislation on other vehicle core safety elements, including front and side impact protection, electronic stability control, pedestrian protection, and braking systems. Currently, only 35 countries have four-wheeled vehicle legislation mandating all five core areas of safety equipment.

Moreover, a pertinent ITF report examined the transition to electrification and its relation to the global used vehicle trade. Among other findings, it is highlighted that emerging economies (LMICs) will rely on imported second-hand electric vehicles (EVs) to decarbonize, but developed countries should avoid restricting these exports, as it could hinder the transition of LMICs. On the other hand, the shift to electric vehicles in developed countries is leading to a growing export of unwanted combustion-engine vehicles to poorer nations, which could delay their own adoption of EVs. Despite this, electric vehicle exports to emerging economies will remain low for the foreseeable future, due to factors like infrastructure challenges and the retention of valuable end-of-use batteries in exporting countries. Enhanced traceability of used car exports is essential to ensure environmental standards are met, while clear roadworthiness criteria should govern the export of used vehicles to reduce pollution and support sustainable transport strategies in emerging economies (ITF, 2023).

The Euro 1-7 vehicle emissions standards are regulations set by the European Union to limit the levels of harmful exhaust emissions from vehicles. The UN Regulatory Framework offers an equivalent legislative framework in UN Regulation Numbers 83, 154 and 168. Starting on 1992 with Euro 1¹, to the Euro 7 iteration expected on 2025 (EU, 2023), these standards have evolved over time, becoming stricter with each new iteration, to reduce air pollution and improve environmental and public health. Overall, the vehicle market trade is predominantly unregulated in LMICs, and in excess of certification limits (e.g. Newman et al., 2024). However, it is widely accepted that the adoption of standards stricter than those of Euro 6 will largely eliminate real-world NOx emissions from diesel engines (Anenberg et al., 2017); the oncoming Euro 7 standards could very well play this stricter standard role. However, **a barrier for the adoption of high environmental standards in LMICs is the lack of high-quality fuels** in these countries, making it impossible to adopt standards of EURO 5 or higher; the use of low-quality fuel in catalytic converters may result in irreversible damage of the engine. Therefore EURO 4 is the current policy target for LMICs.

Further specifications regarding a fruitful direction are provided by the ITC Recommendations (UNECE, 2022b), which offer a detailed framework for national road safety systems, linking the five key pillars of road safety—management, safe users, safe vehicles, safe roads, and effective post-crash response—with critical action areas such as legislation, enforcement, education, and technology, as well as international support. For each pillar, the Recommendations outline possible actions, designate the appropriate responsible authorities, highlight the importance of national coordination, and suggest ways to apply relevant UN road safety legal instruments. **Among other guidelines, the ITC Recommendations promote actions that are very pertinent for used vehicle imports:**

¹ Within this report, Euro emission standards will be denoted with Arabic numerals only. For example, Euro 5/V is denoted as Euro 5.

- i. A Monitoring Action, regarding the calculation and sharing of indicators linked to actions of national actions, such as indicators regarding the proportion of new or imported (used) vehicles meeting required technical standards
- ii. An Enforcement Action, regarding the undertaking of import/export control on new and used vehicles

The following sections will analyze and scope **the status of current starting points that the nations will have to cover in order to align with the strategy of the UN on used vehicles.**

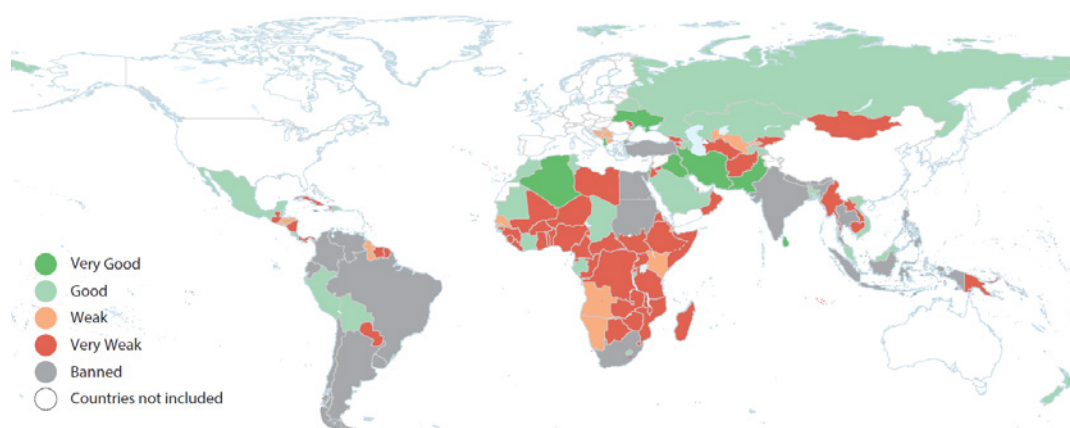
2.2. Review of regional regulations on used vehicles

2.2.1. LMICs

Overall, the vehicle market trade is predominantly unregulated in LMICs, and in excess of certification limits (e.g. Newman et al., 2024). However, it is widely accepted that the adoption of standards stricter than those of Euro 6/VI will largely eliminate real-world NOx emissions from diesel engines (Anenberg et al., 2017); the oncoming Euro 7/VII standards could very well play this stricter standard role.

A very pertinent starting point is the UNEP map for LMICs displaying the strength of regulations in effect regarding the Used Light Duty Vehicle Regulatory Environment (UNEP, 2024), shown in Figure 2.1.

Figure 2.1
Used Light Duty Vehicle Regulatory Environment, 2023



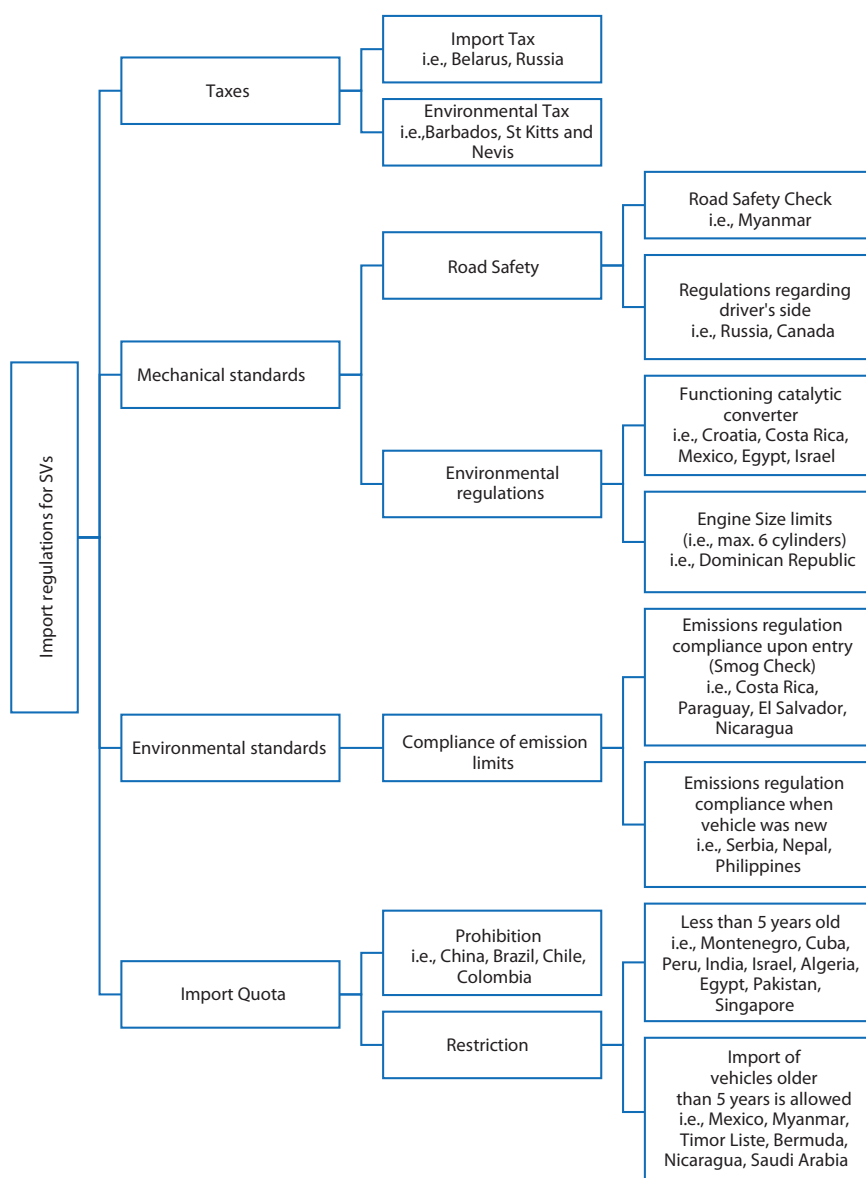
Source: UNEP, 2024

It is evident that there are overarching trends in the established regulations regarding used LDV imports in proximal territories. Specifically, West Africa (mainly ECOWAS countries) have started adopting more rigorous frameworks, where Central and South Africa appear to be lagging considerably in that regard. The Middle East is more fractured, ranging between very good or very weak frameworks, while India and Southeast Asia tend to favour a predominantly overall banning policy, which is common with Latin America too. Undoubtedly, the policy approach of proximal neighbours plays only a part of the overall approach adopted by each country. The following sections endeavour to examine each region at a closer level.

In parallel, it is worth noting the different regulation types that have been devised globally, which are summarized succinctly in a categorization chart by Macias et al. (2013), depicting the overall picture at that time, are shown in Figure 2.2.

Before delving into specific countries, it is fruitful to highlight some of the initiatives that are taken on a continental level, as sub-actions of the UN and other organizations have targeted Africa, Asia or Latin America and the Caribbean explicitly.

Figure 2.2

Import Regulations for Second-hand Passenger Vehicles, 2013

Source: Macias et al., 2013

2.2.2. Africa

Africa is a massive importer of used vehicles, with numbers such as approximately 1.5 million used vehicles imported to Africa every year being provided in related documents (UNEP, 2020), while comprising only 3% of the new vehicle market. The UNRSF project “Safer and Cleaner Used Vehicles for Africa” has core aims that are related to the objectives of the present report. Specifically, the CITA report of Activity 1 of the UNRSF “Safer and Cleaner Used Vehicles for Africa” project had a scoping nature which examined good practices in overseas used vehicle imports and exports. The vehicle export route between New Zealand and Japan was established as the most developed route (CITA, 2021a), while additional findings of the present report will also be examined in the following.

Activity 2 of the project was even more crucial for the present report, as it set out to establish a used vehicle inspection and monitoring framework and implementation compliance systems (CITA, 2021b). Within that activity, it was determined that the entire process of establishing an inspection and monitoring framework for used vehicles and the related compliance system was at a quite initial identification stage, at the time of publication of the report. CITA underlined the need to identify countries that would be highly interested in such systems and securing their commitment to such initiatives. The report promoted a regional approach overall, due to the multifaceted reality and general diversity of the interested countries that are present in the region. The organization stresses that all the supporting material and overall knowledge to establish regional international regulatory frameworks is already present, including inspection schemes, facilities and testing equipment, training and certification processes, anti-fraud controls, international rules, regulations and legal framework knowledge, data platforms and liaising with equipment suppliers, among others.

CITA also mentions three international agreement documents, which are not necessarily African-specific but can serve as guidelines for establishing future inspection programs, which are:

- a. WP.29's Resolution R.E.6 "on the administrative and technical provisions required for carrying out the technical inspections according to the technical prescriptions specified in Rules annexed to the 1997 Agreement"
- b. The African FDARS 1355-2 First Edition 2021, with Parts: "Part 4: Roadworthiness — Requirements for vehicle examiners" and "Part 5: Roadworthiness — Requirements for testing equipment"
- c. The European Union Directive 2014/45/EU (and its Annexes): "Periodic roadworthiness tests for motor vehicles and their trailers" (and its Annexes), aiming to improve road safety by setting minimum requirements for periodic roadworthiness tests of vehicles and trailers in the EU.

Box 3.

The ECOWAS Agreements on Used Vehicle Imports

A notable initiative is the initiative of the **Economic Commission of West African States** (ECOWAS) which was established in 2020. The ECOWAS coalition has established agreements on the importation of only Euro 4/IV standard vehicles by 2021 and placing an age limit of 5 years on Light Duty Vehicles (LDV) and maximum 10 years on Heavy Duty Vehicles (HDV), to be implemented over a period of 10 years (UNRSF, 2020). Moreover, from the above ECOWAS initiative, two key documents were located for regulations regarding specific countries, discussed in the following country-specific sections. Related anticipatory research from the Netherlands (ILT, 2020) reveals that more than 80% of the used vehicles currently exported to West African countries will soon be rejected due to the stricter environmental regulations being enforced by ECOWAS.

Source: UNRSF, 2020

One of the scientific studies closest to this topic is that of Ayetor et al. (2021). The authors initiated their approach by linking the immense percentage of used vehicle fleets found in Africa, cited to be as high as 85%, with a 57% increase of mortality pollution in the thirty years prior to the study, given that road transport has a crucial share of the overall production of pollutants. The authors examined annual international trading data for vehicles and note that policies restricting used vehicle sales in Africa do not generate space for new vehicle production or new vehicle sales, but rather lead to overall vehicle fleet shrinkage. It is also noted that in cases such as Egypt, South Africa and Sudan, vehicles are continuously being imported despite legislative bans.

A similar conclusion is drawn by Boateng and Klopp (2022), who note that administrative corruption and citizen misinformation in Africa exacerbate the problems caused by the presence of inadequate vehicles. Although policymakers attempt to mitigate this issue through outright bans, such vehicles are often rerouted through black markets, causing detrimental effects in safety and emissions to manifest regardless. The authors analyse the flow of used vehicles within a "push-pull" framework, highlighting several key factors that sustain the large-scale importation of used cars into Africa. These include the demand for affordable vehicles, stringent environmental policies in wealthier nations that make used vehicles available for export, limited consumer purchasing power for newer, safer, and less-polluting models, and the weak enforcement of vehicle emissions regulations on the continent. The study discusses the overall policy approach in Ghana, used as a policy case study, which disfavours public transport and sustains the large-scale used vehicle imports in the country, linking it directly with higher emissions and road crash casualties.

Lastly, it should be underlined that the Automobile Association of South Africa (AASA) supports a policy proposal, tentatively suggested by the UN and aimed at improving the quality and safety of imported and exported used vehicles. This policy includes the development of a certified quality label to be displayed on all used vehicles involved in international trade, ensuring compliance with a minimum of Euro 4 or equivalent emission standards, and restricting the import or export of vehicles older than five years. Additionally, it calls for a ban on the trade of zero-rated and one-star NCAP vehicles, as well as vehicles with compromised structural integrity due to crashes. The policy emphasizes that key safety and environmental features must be in place and fully functional, and advocates for incentives to promote the importation of low- to zero-carbon and electric vehicles. Finally, it encourages providing maintenance and aftermarket support for cleaner vehicle technologies (AASA, 2020).

2.2.3. Asia

The Asia-Pacific markets constitute the main markets of used Hybrid electric vehicles (HEV), plug-in hybrid electric vehicles (PHEV) and battery-powered electric vehicles (BEV), with 52% of the total share, imported primarily from Japan and the USA (UNEP, 2021). Furthermore, it is quite likely that Asian LMICs will receive the highest effect of an influx of Chinese vehicles, compared to other LMICs, as China is emerging as a new major exporter in the region, and is projected to reach over 8 million vehicles yearly, a number comparable to the total of OECD countries (ITF, 2023). It is also worth noting that the first wave of domestically produced electric vehicles (EVs) of China is presently reaching the replacement phase, leading to an anticipated increase in EV exports (Chinese Ministry of Commerce, 2021).

Earlier policy documents have strongly connected the need for new vehicle standards to be closely aligned with fuel quality requirements, as advanced technologies can be hindered or rendered less effective by certain fuel characteristics, such as the presence of lead in gasoline or high sulphur content in gasoline or diesel (Asian Development Bank, 2003).

2.2.4. Latin America and the Caribbean

The Inter-American Development Bank (IDB) has examined an array of scenarios considering the adoption of UN regulations and consumer information systems, especially focusing on the UN Global Plan for the Decade of Action for Road Safety 2011-2020. In response to regional requests to address vehicle safety as a crucial aspect of road safety, the IDB has implemented the Regional Public Goods (RPG) initiative, specifically the "Support for the implementation of vehicle regulation standards in Latin America and the Caribbean." This initiative is driven by the need to improve vehicle safety in light of poor safety assessments for passenger cars and the high fatality rates in the Latin America and the Caribbean (LAC) region. The dedicated report (IDB, 2019) outlines, among other findings, that once vehicle regulations be standardized in the LAC region and outside the region, trade changes will also follow. Therefore, it is surmised that if all economies were to adopt the proposed UN regulations and/or equivalent standards, vehicle-importing countries in Central and South America would likely decrease imports of second-hand or non-compliant vehicles, prompting a reorganization of their import systems. This shift could lead these countries to increase imports from regional manufacturing nations, boosting exports from these producers to the newly adjusted markets within the region. Based on earlier European research (Vanherle & Vergeer, 2016), an anticipated impact is that the evolving dynamics of the LAC used car market will influence how the costs and benefits of new legislation are distributed among vehicle owners throughout the lifespan of the used vehicles.

Regarding the adaptation of individual countries of LAC, the three vehicle safety standards with the upmost priority are the following (Wallbank et al., 2019):

- i. Minimum standards for crashworthiness, i.e. regulations that help to protect occupants in front and side impact crashes
- ii. Electronic Stability Control (ESC) for crash avoidance
- iii. Pedestrian protection measures to improve safety for Vulnerable Road Users (VRUs) including pedestrians and bicyclists.

When benchmarking road safety performance in Latin America, the International Transport Forum (ITF, 2017) states that the overarching globally applicable UNECE WP.29 regulations, aiming at harmonizing vehicle regulations, are relevant to this region, including new cars and second-hand cars. The report also pinpoints safety belts as a key mitigating road safety measure, and calls for a combined approach to mitigate wearing rates of safety belts which includes legislation on new and imported/second-hand vehicles.

Moreover, it is critical to note that research of public policy acceptance regarding automobile restriction policies in four Latin American cities (Bogotá, Mexico City, Santiago de Chile, and Quito) has highlighted several key influential factors. A related study highlights the significance of perceived effectiveness, fairness, and personal impacts in shaping public acceptance of urban policies (Boso et al., 2023). Participants generally acknowledge the necessity of these policies for improving traffic and air quality, despite some negative personal impacts. The research also emphasizes the role of local context, particularly in Latin American cities, where high population density, regulatory issues, and infrastructure deficits can lead to lower public support and increased frustration with such measures.

It is worth noting that a LAC-oriented report (Kreuzer & Wilmsmeier, 2014) mentions car and truck scrappage programs, which can be effective, if well-designed. Such programs offer incentives (figures as up to \$4,500 are reported for trading in energy-inefficient vehicles with fuel economy of 18 mpg or less), and their ex-post evaluation studies on consumer welfare and fleet composition (e.g., Busse et al., 2012) suggest that the overall outcomes were positive and did not raise prices in the market of used vehicles.

2.3. Specific LMIC national regulations on used vehicles imports

In the following sections, a detailed review of existing regulations and standards on used vehicles in LMICs is presented. The review seeks to compile existing information from various international reports (e.g. UNEP, FIA, ITF, WHO), and update the information where needed. For that purpose, **the following review method was adopted for each country**, in order of priority:

- Review within existing international reports
- Review within existing scientific literature (i.e. published journal and conference papers)
- Review within national websites (ministries, vehicle authorities etc.)
- Review within other 'grey' sources (opinion articles, blogs, visitor information sites etc.)

While every effort was made to cross-validate the most recent and credible information on the status of country regulations on used vehicles, in a few cases unclear or inconsistent information was found. In this case, we accepted or rejected certain source of information based on expert knowledge. In cases where a clear decision was not possible, a disclaimer on the inconsistency of the found information is explicitly added. For practical reasons, the countries are listed alphabetically.

Afghanistan

Afghanistan imposes no age restriction on the importation of used vehicles only foreigners that own a work permit or working visa & diplomats are allowed to import a vehicle².

Albania

In order to register a used car in Albania for the first time, it needs to meet specific emission criteria based on age. Specifically, EURO 4 emission standards or higher are required for used cars that have been registered before in another country. Euro 3 emission standards or higher are required for used vehicles such as buses and passenger vehicles (ATP, ATV, APV) of categories M2 and N1 that are up to 15 years old before registration in Albania and were registered before in another country³.

Algeria

Algeria imposes a ban on used vehicle imports older than 3 years old⁴.

Angola

Light motor vehicles imported into the country should not be more than 3 years old, while Heavy vehicles should not be more than 5 years old from the date of production. A Pre-Shipment Inspection (PSI) in the country of origin is required. Vehicles imported by Diplomats are not subjected to PSI as they have cleared the vehicle with a Diplomatic Franchise at the destination⁵.

² <https://blog.japanesecartrade.com/73-afghanistan-import-regulation-for-japan-used-cars/>

³ <https://invest-in-albania.org/what-vehicles-can-you-import-to-albania/>

⁴ <https://www.autocango.com/blog-detail/2024-guide-to-import-used-cars-from-china-to-algeria>

⁵ <https://blog.japanesecartrade.com/34-japan-used-car-import-duty-regulation-in-angola/>

Antigua and Barbuda

Antigua and Barbuda impose no age restriction on the importation of used vehicles as long as the vehicles are right-hand driven. A Waiver of Environmental Levy is determined based on the age of vehicle.⁶

Argentina

Overall, Argentina bans used vehicle imports (UNEP, 2017). Exceptions are as follows:

Diplomats can import used cars duty-free with a Diplomatic Franchise issued by the Ministry of Foreign Affairs. Returning Argentine citizens are permitted to bring in one used car and/or one used motorcycle, with a limit of two vehicles per family. Argentine citizens who have lived abroad for more than two years and possess a Residence Certificate issued by the Argentine Consulate in the country of origin may import one car duty-free. Foreign nationals with permanent resident status can import used vehicles under the same conditions as returning Argentine citizens. Individuals with a temporary visa may import used cars under temporary admission without paying duties or taxes. The car is allowed to stay in Argentina for the duration of the visa, and this period can be extended if the visa is renewed. However, the vehicle must be removed from Argentina before the visa expires and must retain its original license plates.

Argentine citizens who have lived abroad for over two years and hold a Residence Certificate issued by the Argentine Consulate in the country of origin can import one car duty-free, provided the car's value does not exceed USD \$15,000. If the car is valued above this amount, duties and taxes will apply. The Residence Certificate must indicate the duration the owner has lived abroad and is applicable to Argentine citizens with a permanent residence visa^{7,8}.

Armenia

As of January 1, 2015, all vehicles produced or imported into the country must adhere to the standards outlined in the following UNECE regulations (5-the ecological class): UNECE rules N 49-04 of the B levels (01.10.1995) UNECE rules N 24-03, addendum 1 (only for diesel engines) (UNEP, 2017). Armenia's import taxation favours smaller cars, but also incentivizes older/used cars⁹. Vehicles are subject to customs duties of 32% of the cost, insurance and freight value¹⁰.

Azerbaijan

The importation of second-hand vehicles older than 10 years is prohibited. Additionally, all imported cars must be equipped with ABS (Anti-lock Braking System) and at least one airbag. Imported vehicles are also required to meet Euro 4 emission standards or higher¹¹.

Bangladesh

Vehicles older than 4 years are not permitted for importation into the country. Used vehicles can only be imported from their country of origin, with no allowance for importation from third countries¹².

As of July 2019, the importation of used vehicles is allowed under Euro 3 emission standards nationwide, and Euro 4 emission standards in Dhaka and Chittagong. A roadmap for further emission standard upgrades has been established, though the timeline for future adoption is yet to be confirmed (UNEP, 2017).

Belarus

In Belarus, there is no specific age limit for imported used cars, apart from vehicles older than five years which are subject to higher tariffs. Used cars manufactured before 2007 must comply with Euro 4 emission standards. Additionally, all imported vehicles must be equipped with ABS, at least one front airbag, and Isofix connectors for child safety seats. For electric vehicles imported by individuals, Belarus has implemented a tax-free policy, in effect until December 2025, as part of its effort to promote environmentally friendly transportation¹³.

⁶ <https://customs.gov.ag/page/vehicle-importation>

⁷ <https://blog.japanesecartrade.com/92-argentina-import-regulation-for-japan-used-cars/>

⁸ <https://www.iamovers.org/ResourcesPublications/CountryGuides.aspx?ItemNumber=3455#>

⁹ <http://www.amcham.am/business/doing-business2/tax-guide.html>

¹⁰ <https://blog.japanesecartrade.com/535-armenia-import-regulation-for-japan-used-cars/>

¹¹ <https://www.autocango.com/blog-detail/azerbaijan-import-regulations-used-cars-from-china>

¹² <https://blog.japanesecartrade.com/75-bangladesh-import-regulation-for-japan-used-cars/>

¹³ <https://www.autocango.com/blog-detail/belarus-import-regulations-used-cars-from-china>

Belize

Imported vehicles must be less than 5 years old and have an indication of vehicle inspection system (UNEP, 2017). In addition, Belize imposes an environmental tax equal to 3% of the vehicle's value¹⁴.

Benin

Recent on-line sources indicate that there is no age restriction on used vehicles imported to Benin. Thus, Beninese can import used vehicle of any age. Taxes on car will depend on age, power, etc. Pre-Shipment Inspection on used vehicles is required prior to their importation into the country¹⁵. However, according to UNEP, there is an import age restriction of 10 years for LDVs and 13 years for tourism vehicles. There is a requirement for control emissions but is currently unspecified (UNEP, 2017).

Bhutan

The import of used vehicles is prohibited, except for officials of Foreign Diplomatic Missions, International Organizations, and experts, who may bring their used vehicles when transferring to Bhutan, provided the vehicles are re-exported upon repatriation. Bhutanese nationals temporarily stationed abroad are allowed to import used vehicles into Bhutan only if certain conditions are met. First, the individual must have worked in a third country for a minimum of two years. Additionally, the vehicle must have been purchased and registered in the person's name in their country of residence at least one year before its importation into Bhutan. The vehicle must be new, not second-hand, and all applicable duties and sales taxes must be paid. No vehicles from third countries can be imported into Bhutan without an import license from the Royal Government of Bhutan. Third-country vehicles, including gifts and prizes, are subject to Customs duty and Sales tax^{16,17}.

Bolivia

Vehicles imported into the country must be no more than 1 year old. Details about the vehicle should be sent to the destination prior to the arrival of the vehicle¹⁸. Documentation supports that diesel vehicles with engines smaller than 4,000 cubic centimetres are prohibited, as are all vehicles that use liquefied petroleum gas in an effort to limit both government fuel subsidies and pollution (UNEP, 2017).

Bosnia and Herzegovina

Vehicles can be imported into Bosnia and Herzegovina with at least EURO 4 standards¹⁹.

Botswana

There are no age Restrictions for the importation of used vehicles in Botswana²⁰, however the vehicle must have driven a maximum of 100,000km²¹.

Brazil

Import of used vehicles is largely banned in Brazil (UNEP, 2017). Brazilian legislation allows the import of used vehicles if they are over 30 years old and classified as collector's items. Typically, these imports are made by individuals who are members of a club affiliated with the Brazilian Federation of Vintage Vehicles (FBVA)²².

Burkina Faso

Burkina Faso poses no age restriction on importation of used vehicles²³.

¹⁴ <https://www.expatsfocus.com/belize/guide/belize-buying-or-importing-a-car>

¹⁵ <https://blog.japanesecartrade.com/41-benin-import-regulation-for-japan-used-cars/>

¹⁶ <https://www.simbashipping.com/shipping-destinations/bhutan/shipping-rules/#>

¹⁷ <https://www.moice.gov.bt/wp-content/uploads/2023/03/Rules-and-Procedures-for-Imports-from-Third-Countries-2002s.pdf>

¹⁸ <https://blog.japanesecartrade.com/95-bolivia-import-regulation-for-japan-used-cars/>

¹⁹ <https://www.otc-global.com/wp-content/uploads/2017/05/Bosnia-and-Herzegovina-Country-Guide.pdf>

²⁰ <https://blog.japanesecartrade.com/39-botswana-import-regulation-for-japan-used-cars/>

²¹ https://www.fiaregion1.com/wp-content/uploads/2020/06/FIA-Report_Final-V3_HR.pdf

²² <https://blog.japanesecartrade.com/931-japan-used-car-import-duty-regulation-in-brazil/>

²³ <https://blog.japanesecartrade.com/42-burkina-faso-import-regulation-for-japan-used-cars/>

Burundi

There is no age restriction on the importation of used vehicles in Burundi. Additionally, Burundians are exempt from any inspection requirements, allowing them to import cars without needing an inspection certificate from authorities such as Japan Export Vehicle Inspection Center (JEVIC), Japan Auto Appraisal Institute (JAAI), or Intertek²⁴.

Cabo Verde

To be imported in the country, each vehicle must have its own Original Bill of Lading (separate from the household goods OBL) stating the weight of the vehicle (in kilograms), the engine and chassis serial numbers, the engine CC, and the make, model and year of manufacture²⁵.

Cambodia

In Cambodia, all vehicles must be left-hand drive, and right-hand drive vehicles face import restrictions. The importation of used cars manufactured before 2000 is discouraged by stricter annual inspection requirements. Additionally, vehicles must meet at least Euro 2 emission standards to pass inspection²⁶. Cambodia adopted Euro 4 vehicle emission standards in 2021 for used LDVs, to be implemented by end of the year (UNEP, 2021).

Box 4.

Country on the Spotlight: Cameroon Commits on the Improvement of vehicle Safety

Cameroon poses a 7-year age limit in vehicle imports (AASA, 2020). In addition, import duties are based on engine size: capacity less than 2000cc warrants a 58% of Cost, Insurance and Freight (CIF), while more than 2000 cc warrants a 77% CIF²⁷.

Moreover, considerable research focus has been placed in the country. The “Assessment of Vehicle Inspection Systems” (AVIS) project (GRSF & CITA, 2020) in Cameroon was initiated through collaboration between the World Bank’s Global Road Safety Facility (GRSF) and CITA (International Motor Vehicle Inspection Committee) to enhance vehicle inspections. The project aims to identify and improve vehicle inspection systems to enhance safety and travel efficiency across three African countries, with Cameroon confirming its participation via its Ministry of Transport. Following several preparatory videoconferences, a mission in Yaoundé in October 2019 engaged stakeholders and collected data to assess the current situation. Meetings were held with public and private entities, including the Minister of Transport, Department of Road Transport, police, gendarmerie, and vehicle inspection center operators.

The AVIS project considered that the current state of vehicle inspection systems in Cameroon reveals several significant challenges, including non-compliance with inspection center specifications, insufficient security for vehicle inspection certificates and stickers, a lack of roadside checks, and the absence of sanctions or corrective measures following audits. The legal framework is inadequate, with undefined procedures and unclear requirements for equipment and skills, leading to a non-uniform system that negatively impacts the standardization and effectiveness of vehicle inspections. Despite these issues, there is a strong political will to improve vehicle inspection and enhance road safety, as evidenced by ongoing projects to secure physical documents and implement administrative monitoring through a unified computer network.

Sources: AASA, 2020; GRSF & CITA, 2020

Central African Republic

Vehicles using Liquefied Petroleum Gas (LPG) can be imported subject to conversion to gasoline, and a Vehicle Identification Number (VIN) sticker is required²⁸.

²⁴ <https://blog.japanesecartrade.com/40-burundi-import-regulation-for-japan-used-cars/>

²⁵ <https://www.worldbaggage.com/kb/cape-verde/2-customs-regulations-for-your-trip-to-cape-verde/>

²⁶ <https://www.autocango.com/blog-detail/cambodia-regulation-import-used-cars-from-china>

²⁷ <https://blog.japanesecartrade.com/43-japan-used-car-import-duty-regulation-in-cameroon/>

²⁸ <https://plc.auction/delivery/cf>

Chad

All used vehicles are exempt from duty, but those older than three years are prohibited from importation²⁹. Vehicle inspection is required upon import (UNEP, 2017).

China (People's Republic of)

Vehicles imported into the country must be no older than 10 years. Those manufactured before January 1, 2005, must pass a rigorous inspection before registration. All vehicles registered in Beijing for the first time must comply with EURO 4 standards, and diesel vehicles are prohibited. The Value-Added Tax (VAT) rate is 17%, with a consumption tax applied based on the vehicle's engine displacement³⁰.

Colombia

The Andean Automotive Agreement of 1993 prohibits the permanent importation of used vehicles into Colombia, with exceptions for classic and vintage collectors' cars, as well as diplomats' vehicles. An Anti-Pollution Control Certificate is required, confirming that all parts and accessories are functional and environmentally compliant. This certificate must be legalized by Colombian authorities at the point of origin.

Comoros

In the Comoros, only left-hand drive vehicles are allowed for import, and there are no age restrictions on imported used vehicles³¹.

Costa Rica

There are no age limits on vehicle imports. Imported vehicles must pass an Environmental Protection Agency (EPA) Smog Test, which must be signed off by the local consulate. Additionally, all imported vehicles are required to have catalytic converters (UNEP, 2017). Vehicles must have a VIN sticker displaying the VIN, engine number, and chassis number. Only left-hand drive cars are permitted, and altering the VIN is strictly prohibited³².

Côte d'Ivoire

In Côte d'Ivoire, the age restrictions for vehicle imports are as follows: cars (tourism vehicles) and taxis must be no more than 5 years old, while minibuses with 9 to 34 seats are limited to 7 years. Buses with more than 34 seats have a 10-year age limit. For trucks, those up to 5 tons are restricted to 7 years, while trucks between 5 and 10 tons, as well as those over 10 tons, are limited to 10 years³³. A fine of FCFA 150,000 is imposed on vehicles older than 10 years and an additional FCFA 10,000 for every year (UNEP, 2017).

Cuba

Until 2013, Cubans were prohibited from buying new and used motorcycles, cars, trucks, and minibuses. In 2013, the purchase of both new and second-hand vehicles was allowed, but only through the government and with a 100% tax. Prior to this, individuals needed personal approval from the vice president, commonly referred to as the "letter." In 2011, the government permitted the sale of used cars between Cubans but retained the requirement for vice-presidential approval. Additionally, the purchase of new cars was still prohibited at that time³⁴. Imported used vehicles must be less than four years old (UNEP, 2017).

Democratic Republic of the Congo

Imported second-hand cars or buses with fewer than 10 seats must be less than 10 years old. Motor vehicles older than 6 months from the date of first registration to the transport document date are exempt from pre-shipment inspection, provided they meet the age requirements for importing used vehicles³⁵.

²⁹ <https://www.icricinternational.org/wp-content/uploads/countries/chad/custom%20laws.pdf>

³⁰ <https://blog.japanesecartrade.com/77-china-import-regulation-for-japan-used-cars/>

³¹ <https://blog.japanesecartrade.com/504-comoros-import-regulation-for-japan-used-cars/>

³² <https://plc.auction/delivery/cr>

³³ <https://blog.japanesecartrade.com/503-cote-divoire-import-regulation-for-japan-used-cars/>

³⁴ <https://www.cubatrade.org/blog/2024/9/6/fueeh9sazh1qup881sw8oc6elqdi8b>

³⁵ <https://www.cma-cgm.com/news/114/import-of-used-vehicles-in-democratic-republic-of-the-congo?cat=rulesandregulations>

Djibouti

Imported cars must be no more than 8 years old, with the date of manufacture indicated on the VIN sticker. Only left-hand drive vehicles are allowed for import. Additionally, LPG vehicles are not permitted to operate on LPG fuel³⁶.

Dominica

In Dominica, vehicles more than 5 years old are not allowed to be imported in the country³⁷.

Dominican Republic

The import of automobiles and light trucks (under five tons) older than 5 years is prohibited (UNEP, 2017). Each person is allowed to import only one vehicle, which must have been in their possession for at least one year prior to importation. All imported vehicles are restricted to an engine size of six cylinders. Luxury vehicles cannot be imported, and motorcycles are subject to taxes and duties. The importer must be present in the Dominican Republic for customs clearance. A new resident can import a car with little or no duty if it has been owned for at least two years and is less than five years old³⁸. Used trucks should not be older than 15 years³⁹.

Ecuador

In Ecuador, importation of used motor vehicles is prohibited, except for diplomatic shipments⁴⁰.

Egypt

Used vehicle imports are banned in Egypt. Passenger vehicles can only be imported within one year of their manufacture date, while electric vehicles may be imported up to three years after manufacture. There are small exceptions for returning Egyptian expatriates (UNEP, 2017). Vehicles must be from the same year as the importation, and the importer must be the original owner. All diesel vehicles are prohibited. Prior authorization is required before importing any vehicle⁴¹.

El Salvador

In general, vehicles older than five years are not permitted for import⁴².

Equatorial Guinea

The permissible age for importing vehicles is up to 10 years for all types. A car import permit from the Ministry of Mines and Energy is required⁴³.

Eritrea

Eritrea poses an age limit of 10 years for used vehicle imports (AASA, 2020).

Eswatini

Import permits are rejected for applications of vehicles older than 15 years⁴⁴.

Ethiopia

Ethiopia introduced in 2019 higher taxation for used LDVs (up to 500%) to discourage import of older used LDVs (UNEP, 2021). Up until recently, there were no age restrictions for importing used vehicles, and no pre-shipment inspection was required⁴⁵. However, as of January 29, 2024, Ethiopia has become the first country to completely ban the import of internal combustion

³⁶ <https://plc.auction/delivery/dj>

³⁷ <https://blog.japanesecartrade.com/97-dominican-import-regulation-for-japan-used-cars/>

³⁸ <https://blog.japanesecartrade.com/98-dominican-republic-import-regulation-for-japan-used-cars/>

³⁹ <https://plc.auction/delivery/do>

⁴⁰ https://lacmassoc.org/uploads/regulaciones/z_regulations_10_eng_ecuador.pdf

⁴¹ <https://blog.japanesecartrade.com/45-egypt-import-regulation-for-japan-used-cars/>

⁴² <https://www.tendata.com/blogs/import/4694.html>

⁴³ <https://plc.auction/delivery/gq>

⁴⁴ <https://www.gov.sz/index.php/component/content/article/327-import-permits?Itemid=31>

⁴⁵ <https://blog.japanesecartrade.com/46-ethiopia-import-regulation-for-japan-used-cars/>

engine vehicles. The Ministry of Transport and Logistics will no longer allow fossil-fuelled vehicles to enter the country due to the high cost of importing gasoline and diesel⁴⁶.

Fiji

In Fiji Islands, used petrol and diesel vehicles that are less than 5 years old must comply with Euro 4 standards, which is the current acceptable level for pollutant emissions across all vehicle categories. Additionally, used special purpose vehicles, as well as those powered by Compressed Natural Gas (CNG), Liquefied Petroleum Gas (LPG), and solar energy, must be less than 8 years old and meet Euro 4 standards to be eligible for importation⁴⁷.

Gabon

Importation of used vehicles above 3 years old is prohibited in the country. All used vehicles imported into Gabon are subject to a tax of 53.4% of the CIF (Cost, Insurance, and Freight) value. Customs authorities in Gabon will assess the vehicle's value, and additional taxes may also apply⁴⁸.

Gambia

Overall, the import of used vehicles is restricted through taxation, with higher taxes applied to older vehicles. Additionally, roadworthiness regulations must adhere to (UNEP, 2017). Vehicles older than 8 years cannot be imported. A vehicle inspection is required before importation into Gambia, and the importer must provide an original or certified copy of the vehicle inspection certificate from the country of origin⁴⁹.

Georgia

There are no age restrictions on the importation of used vehicles, and no pre-shipment inspection is required⁵⁰. In Georgia, imported cars are subject to two types of taxation based on engine volume and age. An excise tax is applied according to the vehicle's age, and an import tax of 0.05 GEL per cubic centimetre of engine displacement is charged. Additionally, an extra 5% of this amount is applied for each year of the vehicle's use (UNEP, 2017).

Ghana

The scientific study of Boateng & Klopp (2022) outlines the course of used vehicle adoption in Ghana in considerable detail. In particular, Ghana enacted the Customs, Excise and Preventive Service (Management) (Amendment) Act, 1998 (Act 552), in June 1998, which prohibited the import of vehicles older than 10 years. Four years later, this Act was replaced by the Customs, Excise and Preventive Service (Management) (Amendment) Act, 2002 (Act 614), which imposed penalties on vehicles exceeding the 10-year age limit (Ghana's EPA, 2017; Kombat & Wätzold, 2019). In a move to promote local manufacturing, Ghana's parliament passed the Customs (Amendment) Act, 2020 (Act 1014) more recently, which bans the importation of all used vehicles over 10 years old, as well as any salvaged vehicles, regardless of age. The passage of this Act was met with strong opposition from mechanics, sprayers, and other garage workers whose livelihoods depend on used vehicles. Despite the protests, the President signed the Act into law, it passed in April 2020 and came into effect in October 2020 (Dontoh & Dzawu, 2020; Larnyoh, 2020).

Moreover, the age of a motor vehicle for importation purposes is determined from the year it was first manufactured. Vehicle age plays a significant role in duty valuation. Customs determines the vehicle's age using a model year reckoning index, which typically provides a rounded year (e.g., 1990 or 2000) or a dealer chart matrix⁵¹.

Grenada

In Grenada, there are no age restrictions on the importation of used vehicles, and no pre-shipment inspection is required⁵².

⁴⁶ <https://research.hktdc.com/en/article/MTYxNzc4NzU5NQ>

⁴⁷ <https://www.frcs.org.fj/wp-content/uploads/2019/06/Import-of-second-hand-vehicles.pdf>

⁴⁸ <https://blog.japanesecartrade.com/47-gabon-import-regulation-for-japan-used-cars/>

⁴⁹ <https://blog.japanesecartrade.com/70-gambia-import-regulation-for-japan-used-cars/>

⁵⁰ <https://blog.japanesecartrade.com/538-georgia-import-regulation-for-japan-used-cars/>

⁵¹ <https://blog.japanesecartrade.com/48-ghana-import-regulation-for-japan-used-cars/>

⁵² <https://blog.japanesecartrade.com/99-grenada-import-regulation-for-japan-used-cars/>

Guatemala

In Guatemala, the importation of left-hand drive vehicles is allowed. However, customs authorities will not permit the import of used vehicles older than 10 years from the date of manufacture. Additionally, one must obtain documentation that establishes the value of the used vehicle before importing it into the country⁵³.

Guinea

Guinea poses an age limit of 10 years for used vehicle imports (AASA, 2020). In 2021, the limit was specialized: for cars and mini-vans up to 3.5 Tons: maximum eight years, for buses for passengers with 7-seaters or more: maximum ten years and for large vans, trucks, and other motor vehicles above 3.5 tons: maximum twelve years⁵⁴.

Guinea-Bissau

Vehicles no older than 5 years can be imported, but a tax of 58% of the CIF (Cost, Insurance, and Freight) value will apply. Importation is subject to approval by a local agent⁵⁵.

Guyana

In Guyana, used vehicles must be no more than 5 years old for importation. Small trucks up to 3 tons must be less than 4 years old. There is no mandatory inspection required for second-hand vehicles being imported⁵⁶.

Haiti

Vehicles older than 3 years cannot be imported in Haiti. Importing a used car can be complex and time-consuming due to extensive paperwork and numerous regulations⁵⁷.

Honduras

As of 2024, imports of motor vehicles older than ten years, except for classic collectible cars, and passenger buses older than thirteen years, are prohibited. Additionally, the import of refurbished and right-hand drive vehicles is not allowed⁵⁸.

India

Importing used vehicles into India is limited only through Customs Port at Mumbai, Chennai, and Kolkata. The vehicles must be at least five years roadworthy, with certification from a government-approved agency. Euro 3 and Euro 4 standards apply in eleven major cities, with Euro 6 standards set for April 2020 (UNEP, 2017).

To import used cars, they must be no older than three years, right-hand drive, and meet new-car standards for headlights and speedometers. Importing is restricted to specific naval docks, and the vehicle must have been loaned, leased, registered, or sold. It must also be roadworthy and certified for at least five years. Cars with engine capacities between 1,000cc and 2,500cc cannot be imported⁵⁹.

Indonesia

Indonesia has mostly prohibited used car imports (UNEP, 2017). Only diplomatic personnel with PP-8 status (i.e. original letter of approval from the Indonesian Ministry of Foreign Affairs) can import vehicles. Vehicles will be free of duty and taxes⁶⁰.

Iran

Vehicles older than three years, diesel engines, and those with right-hand steering wheels are prohibited from importation into the country⁶¹.

⁵³ <https://blog.japanesecartrade.com/928-japan-used-car-import-duty-regulation-in-guatemala/>

⁵⁴ <https://carbuyersbroker.com/importing-used-cars-to-guinea>

⁵⁵ <https://www.fidi.org/sites/default/files/public/2024-01/GUINEA-BISSAU%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

⁵⁶ <https://aaajapan.com/information/car-import-rules-from-japan/guyana>

⁵⁷ <https://www.oneroofautos.com/import-policy/haiti>

⁵⁸ <https://www.trade.gov/country-commercial-guides/honduras-prohibited-restricted-imports>

⁵⁹ <https://www.bajajallianz.com/blog/motor-insurance-articles/guide-to-import-foreign-cars-to-india.html>

⁶⁰ <https://www.simbashipping.com/shipping-destinations/indonesia/shipping-rules/>

⁶¹ https://www.ramadbk.com/iran_import_regulations.php

Iraq

Civilian vehicles must be less than 2 years old for import in the country⁶².

Jamaica

A new set of regulations came into effect in November 2019⁶³. Specifically, individuals can import up to two motor vehicles every three years, excluding motor cars, if they are Jamaican residents, children of Jamaican parents, spouses of Jamaicans, returning residents, contract officers, or holders of work permits. Businesses are categorized into two groups: non-dealer companies, which can import up to three vehicles every three years, and certified motor vehicle dealers, who can import any number of vehicles for resale.

The age restrictions for imported vehicles are as follows: Motorcycles must be no older than five years, motor cars no older than six years, light commercial vehicles between six and ten years old, and motor vehicles for returning residents or special purposes up to ten years old. Commercial vehicles can be between twelve and thirty years old, depending on the type, while heavy-duty commercial equipment must be no older than thirty years.

Jordan

In Jordan, imports of used automobiles have no age limitation, but personal vehicles must be no older than ten years⁶⁴.

Kazakhstan

Vehicles older than five years cannot be imported, and engine volume must not exceed 3 litres⁶⁵.

Box 5.

Country on the Spotlight: Kenya keeps its Vehicle Fleet Young and Clean

In UNEP (2020), it is outlined that in Kenya, over 95% of the light-duty vehicles added to the fleet are imported used vehicles, primarily from Japan. Kenya enforces an 8-year age limit on these imports, resulting in most vehicles being around 7 years old. This regulation keeps Kenya's vehicle fleet relatively young and clean compared to neighbouring countries like Uganda and Rwanda, which also import used vehicles from Japan. It is mentioned that Uganda, with a 15-year age limit, and Rwanda, with no age limit, have significantly older fleets, leading to average fuel consumption and CO2 emissions that are about 25% higher than in Kenya.

Moreover, vehicles must undergo a roadworthiness inspection by an agent appointed by Kenya Bureau of Standards in the country of export and must be right-hand drive⁶⁶.

Source: UNEP, 2020

Kiribati

There is no age restriction for importing used vehicles, and no pre-shipment inspection is required⁶⁷.

Kosovo

Vehicles older than 10 years cannot be permanently imported into Kosovo⁶⁸.

⁶² <https://blog.japanesecartrade.com/540-iraq-import-regulation-for-japan-used-cars/>

⁶³ https://www.tradeboard.gov.jm/ttbl/MV_General.php

⁶⁴ <https://blog.japanesecartrade.com/541-jordan-import-regulation-for-japan-used-cars/>

⁶⁵ <https://blog.japanesecartrade.com/556-kazakhstan-import-regulation-for-japan-used-cars/>

⁶⁶ <http://www.kenyaembassyberlin.de/Guidelines-for-Importation-of.49.0.html>

⁶⁷ <https://blog.japanesecartrade.com/518-kiribati-import-regulation-for-japan-used-cars/>

⁶⁸ <https://www.fidi.org/sites/default/files/public/2024-01/KOSOVO%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

Kyrgyzstan

Kyrgyzstan's customs procedures for importing used vehicles from China have been made more straightforward as of 2024, requiring less effort than before. However, importers must still follow the emissions and safety requirements set by the Eurasian Economic Union. For environmental and safety purposes, cars older than 10 years are prohibited, and imports must meet Euro 2 emission standards or higher. Additionally, left-hand drive vehicles are favoured in order to comply with local traffic laws⁶⁹.

Lao People's Democratic Republic (Laos)

In Laos, there is no age restriction or pre-shipment inspection requirement for importing used vehicles. Left-hand drive vehicles are allowed, but right-hand drive vehicles are prohibited⁷⁰.

Lebanon

Cars older than eight years cannot be imported into Lebanon. Shipping used cars is not recommended, and right-hand drive vehicles are not accepted⁷¹.

Lesotho

All second-hand vehicles imported into Lesotho must undergo an inspection before entry. While customs duties on used vehicles are not applicable, VAT is charged at the standard rate on the cost, CIF duties. If the owner changes permanent residence and has owned the vehicle for at least 12 months, the importation of the vehicle is favoured, but only one vehicle per family is allowed⁷². The vehicle must also be kept for at least 2 years before it can be sold. Additionally, vehicles older than 8 years cannot be imported (AASA, 2020).

Liberia

Only vehicles that are less than 10 years old and left-hand drive cars can be imported into Liberia⁷³.

Libya

The age of used cars imported should not be more than five years at time of importation. Heavy vehicles, forklifts, tractors and trucks of age older than seven years are not allowed to be imported in the country⁷⁴. Moreover, Libya has established a 10-year age restriction for all light and heavy-duty vehicles (ILT, 2020; Boateng & Klopp, 2022).

Madagascar

There is no age restriction for importing used vehicles, but cars and trucks running on liquefied gas are unavailable in the country. Imported vehicles must have a VIN sticker and a left-hand steering wheel⁷⁵.

Malawi

Malawi imposes no age limitation on imported used vehicles, and no specific roadworthiness inspection is required. Only right-hand drive vehicles are permitted, with exceptions for special cases like ambulances and fire engines⁷⁶.

Malaysia

In Malaysia, importing used vehicles older than five years is restricted, and vehicles can only be imported for personal use—not for sale or transfer. The vehicle must be exported when employment ends in the country. To import a car, an Approval Permit (AP) from the Ministry of International Trade and Industry (MITI) must be obtained, and the car must be registered under the owner's name for at least three years before applying for the permit⁷⁷.

⁶⁹ <https://www.autocango.com/blog-detail/guide-importing-used-cars-china-kyrgyzstan-2024>

⁷⁰ <https://blog.japanesecartrade.com/520-laos-import-regulation-for-japan-used-cars/>

⁷¹ <https://www.fidi.org/sites/default/files/public/2021-08/LEBANON%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

⁷² <https://www.carsotho.com/en/posts/importing-cars-to-lesotho>

⁷³ <https://www.fidi.org/sites/default/files/public/2024-01/LIBERIA%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

⁷⁴ <https://blog.japanesecartrade.com/49-libya-import-regulation-for-japan-used-cars/>

⁷⁵ <https://plc.auction/delivery/mg>

⁷⁶ <https://blog.japanesecartrade.com/51-malawi-import-regulation-for-japan-used-cars/>

⁷⁷ <https://aaajapan.com/information/car-import-rules-from-japan/malaysia>

Maldives

Right-hand drive cars are allowed to be imported in the country. There is no age restriction or pre-shipment inspection requirement for importing used vehicles⁷⁸.

Mali

Left-hand drive cars are allowed to be imported in the country. There is no age restriction or pre-shipment inspection requirement for importing used vehicles⁷⁹.

Marshall Islands

There is no restriction on the age of the vehicle. Only left-hand drive cars can be imported in the Marshall Islands⁸⁰.

Mauritania

Vehicles older than eight years cannot be imported. Import taxes for used vehicles are 50% of the car's value⁸¹.

Mauritius

Mauritius has successfully implemented policies to enhance the quality of imported used vehicles. The country permits only used vehicles that are no older than three years (AASA, 2020) and has introduced a CO₂-based vehicle taxation scheme, along with a verification and inspection process for these vehicles. Recently, Mauritius also introduced fiscal incentives for low and zero-emission vehicles, leading to a significant increase in the importation of used hybrid and electric vehicles (UNEP, 2020).

Box 6.

Country on the Spotlight: Mexico Regulated Vehicle Imports along Major North-American Corridor

Used motor vehicles may not be imported into Mexico if their circulation is restricted or prohibited in their country of origin, if they have been reported as stolen, or when the vehicle does not comply with all applicable physical-mechanical properties and environmental protection requirements. Mexican customs authorities have the authority to work with their foreign counterparts and require information and documentation from motor vehicle importers to ensure that any motor vehicle imports comply with these requirements. Current import regulations for used motor vehicles also provide reduced duty treatment to subject merchandise whose vehicle identification number corresponds to a vehicle manufactured or assembled in Mexico, the United States and/or Canada⁸². Previous regulations (UNEP, 2017) also mentioned that vehicles must be at least five years old but no more than ten years old, have a gasoline engine, and be equipped with a catalytic converter.

Mexico has also received attention as a research case, as the main exporting partner to Mexico is the US through the North American Free Trade Agreement (NAFTA), with the bilateral corridor being cited as the highest in volume worldwide in the past. On December 24, 2008, the Mexican government issued a decree establishing conditions for the importation of second-hand vehicles, which came into force on January 10, 2009. This decree was largely a response to pressure from vehicle production and distribution associations (AMDA and AMIA). While its compliance with NAFTA has been questioned, neither the U.S. nor the Canadian governments have filed any complaints, so Mexico has not been called to formally defend this decision. The decree imposes import duties on certain vehicles over 10 years old, a 1% duty on vehicles between 5 and 9 years old, and differential treatment for vehicles remaining in the border region or exiting it (Macias et al., 2013).

A related study (Jin et al., 2021) has quantified the health and climate benefits of key policies targeting the reduction of air pollutant emissions from HDVs in G20 economies. Within this research, the benefits for Mexico were assessed across five different scenarios. The presently adopted regulation scenario indicates a plateauing of NO_x emissions of above 200 thousand tons by 2030 and a rebounding towards 235 thousand tons by 2040. A scenario concerning the expansion

⁷⁸ <https://blog.japanesecartrade.com/522-maldives-import-regulation-for-japan-used-cars/>

⁷⁹ <https://blog.japanesecartrade.com/506-mali-import-regulation-for-japan-used-cars/>

⁸⁰ <https://plc.auction/delivery/mh>

⁸¹ <https://blog.japanesecartrade.com/53-mauritania-import-regulation-for-japan-used-cars/>

⁸² <https://research.hktdc.com/en/article/NjQyOTY2ODQx>

of existing policies to impose Euro VI-equivalent regulations to imported used vehicles is projected to lead to an additional NO_x reduction of 17% by 2030 and 31% by 2050. With a similar approach, black coal (BC) emissions are estimated to plateau at about 6 thousand tons by 2036 and onwards, with the expansion scenarios lead to further BC reductions of 34% by 2030 and 99% by 2050. The other additional considered scenarios of that paper are not exclusive to used vehicles and thus out of the scope of the present report.

Sources: UN/UNEP, 2017 ; Macias et al., 2013; Jin et al., 2021

Micronesia

There is no age restriction or pre-shipment inspection requirement for importing used vehicles⁸³.

Moldova

Models up to maximum seven years old are acceptable for import. Vehicles must have a VIN sticker and chassis, and only left-hand drive cars are allowed⁸⁴.

Mongolia

There is no age restriction or specific inspection requirement for importing used vehicles into Mongolia. However, cars imported from Japan are checked for radiation. Although Mongolia is a left-hand drive country, right-hand drive vehicles can be imported with a valid import license⁸⁵.

Montenegro

No restrictions exist overall. There are minor variations in the required documentation depending on whether the importer is a returning citizen, foreigner or diplomat⁸⁶.

Morocco

In Morocco, light-duty and heavy-duty vehicles must be less than five years old and from 2010 or later. Since 2015, imported vehicles must meet at least Euro 4 emission standards (ILT, 2020; UNRSF, 2020).

Mozambique

Used cars must be less than five years old (AASA, 2020), while used vans must be less than nine years old (UNEP, 2017).

Myanmar

Only vehicles manufactured in 2015 or later are eligible for import into Myanmar. Vehicles older than five years cannot be imported, and no specific inspection is required for used vehicles⁸⁷.

Namibia

In September 2022, Namibia prohibited the importation of second-hand motor vehicles older than 12 years into the country⁸⁸.

Nauru

There is no age restriction or pre-shipment inspection requirement for importing used vehicles⁸⁹.

Nepal

Nepal allows importation of used vehicles that comply with Euro 3 standards (UNEP, 2017).

⁸³ <https://blog.japanesecartrade.com/524-micronesia-import-regulation-for-japan-used-cars/>

⁸⁴ <https://plc.auction/delivery/md>

⁸⁵ <https://blog.japanesecartrade.com/83-mongolia-import-regulation-for-japan-used-cars/>

⁸⁶ <https://www.fdi.org/sites/default/files/public/2022-02/MONTENEGRO%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

⁸⁷ <https://blog.japanesecartrade.com/525-myanmar-import-regulation-for-japan-used-cars/>

⁸⁸ <https://www.lac.org.na/laws/2022/7902.pdf>

⁸⁹ <https://blog.japanesecartrade.com/527-nauru-import-regulation-for-japan-used-cars/>

Nicaragua

Nicaraguan law prohibits the importation of cars older than ten years, except for returning Nicaraguan citizens who have been abroad for at least one year before their final return⁹⁰.

Niger

There is no age limit for used vehicles originating from within the Southern African Customs Union (SACU) region, but vehicles from outside SACU must be less than five years old (UNEP, 2017; AASA, 2020).

Nigeria

Nigeria has established a ban on imports of vehicles that are more than 8 years old (Ayetor et al., 2021; Boateng & Klopp, 2022).

Niue

Vehicles older than 15 years from the current year are banned from importation to Niue⁹¹.

North Macedonia

Vehicles must be less than 10 years old and owned by the importer at the time of import. All vehicles must have a catalytic system and a Euro 3 emission standard certificate⁹². Taxation upon registration is based on engine size and power, with an additional flat fee and a yearly road tax. From July 2015, Euro 4 standards apply to new light-duty vehicles, and Euro 5 standards to new LDVs imported and registered in North Macedonia (UNEP, 2017).

Pakistan

The import of used cars in Pakistan has been extended to vehicles up to three years old from the date of manufacturing, instead of the previous two years limit for overseas Pakistanis. All vehicles imported must be right hand drive (RHD).⁹³

Panama

There are no restrictions on importing new or used cars and trucks into Panama. The duty tax for importing a vehicle is approximately 30% of the car's value. According to Panamanian customs laws, import taxes apply to all automobiles except those belonging to qualified diplomats, members of international organizations, retirees, or U.S. contractors. These individuals are allowed to import one vehicle duty-free, while other vehicles are subject to import duties⁹⁴.

Papua New Guinea

There is no age restriction on the importation of used vehicles, but cars up to 3.5 tons must be five years old or newer, while cars over 3.5 tons must be 15 years old or newer⁹⁵.

Paraguay

Vehicles that are older than 10 years cannot enter into Paraguay⁹⁶.

Peru

Import duties and excise tax are 69% for used cars, based on their value. Vehicles must be no older than two years or have more than 32,000 kilometres. Used cars older than two years from the model year are prohibited (e.g., a 2020 fabrication with a 2021 model year is considered a 2021 vehicle). Diesel cars are banned, and carbon monoxide emissions must not exceed 9% by volume. Damaged cars are not allowed. Taxes, duties, and fees apply as well. One car can be imported duty-free if its value does not exceed the amount authorized by the Peruvian Ministry of Foreign Affairs and its engine capacity does not exceed 2,000 cc⁹⁷.

⁹⁰ https://lacmassoc.org/uploads/regulaciones/regulations_eng_2023061211:11:01_nicaragua.pdf

⁹¹ <https://niue.tradeportal.org/procedure/44/step/44?l=en>

⁹² <https://www.rainieros.com/forms/customs-regulations-by-country/moving-to-macedonia/>

⁹³ <https://blog.japanesecartrade.com/87-pakistan-import-regulation-for-japan-used-cars/>

⁹⁴ <https://blog.japanesecartrade.com/103-panama-import-regulation-for-japan-used-cars/>

⁹⁵ <https://blog.japanesecartrade.com/85-papua-new-guinea-import-regulation-for-japan-used-cars/>

⁹⁶ <https://www.fidi.org/sites/default/files/public/2023-09/PARAGUAY%20Import%20-%20FIDI%20Customs%20Guide.pdf>

⁹⁷ <https://customs.sirva.com/countries/peru/peru---returning-citizen#:~:text=Import%20duties%20and%20excise%20tax,of%20the%20car%20is%20prohibited>

In addition, Peru adopted Euro 6 emission standards in October 2021 with an implementation date of October 2024 concurrently with 10 ppm Sulphur fuels for used LDVs. The country is mentioned as the first Latin America and the Caribbean (LAC) country to move towards this direction (UNEP, 2021).

Philippines

The Philippine government has implemented specific import policies to ensure the quality and safety of used cars entering the country. Only left-hand drive vehicles are permitted. While used cars of any age can be imported, they must have been registered for at least six months prior to shipment⁹⁸. Gasoline-powered vehicles must have an engine displacement of no more than 2800ml and a weight not exceeding 1500kg. Additionally, all imported vehicles must adhere to the safety and emission standards set by Philippine technical regulations. The policy also covers other vehicle types, including private cars under the No Dollar Importation (NDI) scheme, diplomatic mission vehicles, commercial trucks with a Gross Vehicle Weight (GVW) of 2.5 tons or more, commercial buses with a GVW of 6 tons or more, special purpose vehicles (SPVs), and motorcycles⁹⁹. Moreover, limited importation of used passenger cars is permitted through Special Economic Zones or Ports, with Euro 4 emission standards required (UNEP, 2017).

Rwanda

Importation of left-hand drive vehicles is allowed in the country. Right-hand drive vehicles are allowed in the country only for two-weeks. There is no age restriction on importation of used vehicle. There is no requirement of an inspection before the shipment of used vehicle¹⁰⁰.

Saint Helena

No age restriction or inspection requirement on importation of used vehicle. Left Hand Drive vehicles are allowed to be imported in the country¹⁰¹.

Saint Vincent and the Grenadines

No age restriction or inspection requirement on importation of used vehicle. Left Hand Drive vehicles are allowed to be imported in the country¹⁰².

Samoa

Vehicles must be no more than 8 years old from the year of manufacture, with cars from 2010 and onwards eligible for import. All used vehicles and machinery must be inspected upon arrival and undergo a water blasting treatment. Importers should ensure that vehicles and machinery are thoroughly cleaned of all contamination before shipping¹⁰³.

São Tomé and Príncipe

No age restriction on importation of used vehicle. There is no requirement of the inspection before the shipment of used vehicle¹⁰⁴.

Senegal

Only vehicle less than 8 years old can be imported¹⁰⁵.

Serbia

Euro 5 standards apply to new light-duty vehicles (LDVs) from 2011, while Euro 3 standards apply to used vehicles from 2005 (UNEP, 2017).

⁹⁸ <https://blog.japanesecartrade.com/86-philippines-liberia-import-regulation-for-japan-used-cars/>

⁹⁹ <https://www.autocango.com/blog-detail/philippines-import-regulations-chinese-used-cars>

¹⁰⁰ <https://blog.japanesecartrade.com/57-rwanda-import-regulation-for-japan-used-cars/>

¹⁰¹ <https://blog.japanesecartrade.com/508-st-helena-import-regulation-for-japan-used-cars/>

¹⁰² <https://blog.japanesecartrade.com/922-japan-used-car-import-duty-regulation-in-saint-vincent/>

¹⁰³ <https://blog.japanesecartrade.com/530-samoa-import-regulation-for-japan-used-cars/>

¹⁰⁴ <https://www.otc-global.com/wp-content/uploads/2017/05/Sao-Tome-and-Principe-Country-Guide.pdf>

¹⁰⁵ <https://www.fidi.org/sites/default/files/public/2023-11/SENEGAL%20Import%20%E2%80%93%20FIDI%20Customs%20Guide.pdf>

Seychelles

All imported used motor vehicles require an import permit and must be less than five years old (UNEP, 2017). For returning graduates or residents, the used vehicle must be right-hand drive, and registered in the owner's name for at least 12 months^{106, 107}.

Sierra Leone

Sierra Leone has no age limitation or compulsory inspection for imported used vehicles. Only left-hand drive vehicles are allowed¹⁰⁸.

Solomon Islands

There is no age restriction or pre-shipment inspection requirement for importing used vehicles. Right-hand drives are allowed¹⁰⁹.

Somalia

Returning nationals having lived abroad for more than 2 years are allowed to import one used motor vehicle duty and tax free provided the vehicle was registered in the shipper's name in the country of origin. Left Hand Drive vehicles are allowed to be imported in the country¹¹⁰.

Temporary foreign residents are allowed to import one used motor vehicle duty and tax free within the first six months of their arrival in Somalia (date of issue of work permit) provided that the vehicle was registered in the shipper's name.

South Africa

Used vehicle imports are banned, except for permits issued to returning South African nationals with permanent residence. This restriction aims to protect the local motor vehicle manufacturing industry, with permits granted only under specific conditions¹¹¹.

South Sudan

South Sudan imposes age restrictions on imported vehicles, which must be typically at most eight years from the year of manufacture. The year of manufacture must also be indicated on the vehicle's documents¹¹². Pre-Shipment Inspection is required for vehicles from Japan, the United Kingdom, Singapore, the United Arab Emirates, and South Africa. Odometer tampering is prohibited. Both left-hand and right-hand drive vehicles can be imported, although driving is on the right side of the road. Buses must be left-hand drive only¹¹³.

Sri Lanka

The age limits for importing used vehicles are as follows: used cars must be under 2 years old, while vans and dual-purpose vehicles should be no more than 5 years old. Buses, motor cars, SUVs, all-terrain vehicles, ambulances, and motorbikes must be under 3 years old. Single cabs, double cabs, lorries/trucks (Gross Vehicle Weight (GVW) less than 5 Metric Tons (MT)), and refrigerated trucks (GVW less than 5 MT) must be under 4 years old. A compulsory inspection must be passed before export, and the original inspection certificate is required for customs clearance and LCA (Letter of Credit Authorization) upon arrival. Importation of only Right-Hand Drive (RHD) vehicles is allowed in Sri Lanka¹¹⁴.

Sudan

Sudan has imposed a ban on used vehicle imports, except for immigrants, vintage and racing cars, vehicles adapted for physically disabled, and donated vehicles for welfare organizations (UNEP, 2017). Only diplomats can import second-hand cars into the country, provided their diplomatic mission in Sudan grants a diplomatic exemption. Used vehicles cannot be imported through non-diplomatic shipments¹¹⁵.

¹⁰⁶ <https://www.simbashipping.com/shipping-destinations/seychelles/rules-and-regulations/>

¹⁰⁷ <https://blog.japanesecartrade.com/60-seychelles-import-regulation-for-japan-used-cars/>

¹⁰⁸ <https://www.oneroofautos.com/import-policy/sierra-leone>

¹⁰⁹ <https://blog.japanesecartrade.com/529-solomon-islands-import-regulation-for-japan-used-cars/>

¹¹⁰ <https://blog.japanesecartrade.com/511-somalia-import-regulation-for-japan-used-cars/>

¹¹¹ <https://www.gov.za/services/services-organisations/import/importing-second-hand-or-used-vehicle>

¹¹² <https://www.carsjuba.com/en/blog/import-regulations-for-japan-used-cars-in-south-sudan>

¹¹³ <https://blog.japanesecartrade.com/64-south-sudan-import-regulation-for-japan-used-cars/>

¹¹⁴ <https://blog.japanesecartrade.com/89-sri-lanka-import-regulation-for-japan-used-cars/>

¹¹⁵ <https://www.icricinternational.org/wp-content/uploads/countries/sudan/custom%20laws.pdf>

Suriname

In UNEP (2017) it is indicated that used cars may be 5 years old or younger. However, exporter's sources suggest that used cars must be no more than 8 years old, while used buses must be no more than 15 years old from the date of manufacture. There is no age restriction for used trucks. Right-hand drive (RHD) vehicles are permitted, and no specific inspection is required for importing used vehicles into Suriname¹¹⁶.

Syrian Arab Republic

Syria has imposed a ban on used vehicle imports. Only foreigners and diplomats may import motor vehicles. The shipper must be present for customs clearance, and automobiles must be less than 2 years old. Diesel engine vehicles are not allowed¹¹⁷.

Tajikistan

Since 2022, Tajikistan has prohibited the sale of transport vehicles that do not meet Euro 4 emission standards to reduce air pollution. Additionally, cars manufactured before 2013 are banned to decrease the number of old, heavily polluting vehicles on the roads¹¹⁸.

Tanzania

In Tanzania, there is no age limit for importing used cars. However, used motor vehicles aged 8 years or older are subject to additional excise duties: 25% for non-utility vehicles and 5% for utility vehicles. Starting July 1st, the age restriction has been reduced to 8 years instead of 10 years, with dumping fees applicable for vehicles exceeding this limit. This rule applies to light cars and trucks but does not affect machinery. The old age fees have been in effect since July 1, 2014. The age restriction does not apply to cargo in transit, machinery, or earth-moving equipment such as graders, bulldozers, and agricultural equipment¹¹⁹.

Thailand

There is no age restriction for importing used vehicles, and no pre-shipment inspection is required. An importer is allowed to import only one used or second-hand vehicle for personal use¹²⁰. Importation of used vehicles that do not meet Euro 4 standards is prohibited. Discussions are ongoing regarding the implementation timeframe for Euro 5/6 standards (UNEP, 2017).

Timor-Leste

Importing light passenger and mixed vehicles more than 5 years old from the date of manufacture is prohibited in Timor-Leste, except for specified cases. No inspection is required for importing a vehicle¹²¹.

Box 7.

Country on the Spotlight: Togo Focuses on PTI Capacity Building

Used vehicles imported in the country should pass safety and emission regulations which remain unspecified. Importation of left-hand drive vehicles is allowed in the country¹²². An earlier AVIS project in Togo (GRSF & CITA, 2017) was initiated through collaboration between the World Bank's Global Road Safety Facility (GRSF) and CITA (International Motor Vehicle Inspection Committee) aimed at enhancing vehicle technical inspection systems. The primary objective was to evaluate the existing vehicle inspection and importation frameworks and develop strategies for their improvement.

¹¹⁶ <https://blog.japanesecartrade.com/106-suriname-import-regulation-for-japan-used-cars/>

¹¹⁷ <https://s3.amazonaws.com/rdcms-iam/files/production/public/newimages/member/shippers/syria.pdf>

¹¹⁸ <https://www.autocango.com/blog-detail/tajikistan-regulation-import-used-cars-from-china>

¹¹⁹ <https://blog.japanesecartrade.com/65-tanzania-import-regulation-for-japan-used-cars/>

¹²⁰ <https://blog.japanesecartrade.com/90-thailand-import-regulation-for-japan-used-cars/>

¹²¹ <https://blog.japanesecartrade.com/515-east-timor-import-regulation-for-japan-used-cars/>

¹²² <https://blog.japanesecartrade.com/66-togo-import-regulation-for-japan-used-cars/>

The periodic technical inspection system in Togo, managed by SOTOPLA-CEVA under the Direction des Transports Routiers et Ferroviaires (DTRF), is structured but ineffective, as evidenced by an extremely low rejection rate of 0.64% despite the visibly poor condition of the vehicle fleet. The inspection process suffers from inadequate supervision, insufficient inspection stations, and underutilized equipment, particularly in pollution control. The lack of technical requirements for vehicle registration, coupled with minimal oversight and ineffective data management, further undermines the system's efficacy in ensuring vehicle safety and environmental compliance. Additionally, the network capacity is inadequate for the size of the fleet, and heavy trucks are often inspected without proper facilities, highlighting the need for significant improvements in the inspection framework.

The analysis of Togo's vehicle inspection system led to several recommendations aimed at enhancing road transportation reliability and traffic safety. These recommendations are categorized into three areas: improving entry requirements for vehicles, building administrative capacity within the DTRF to manage inspections, and upgrading existing inspection stations. For vehicles entering Togo, it was suggested to introduce technical requirements based on international standards like the UNECE 1958 Agreement.

Source: GRSF & CITA, 2017

Tonga

There is no age restriction for importing vehicles into Tonga, but only right-hand drive vehicles are allowed. Vehicle inspection is not required for importation¹²³.

Trinidad and Tobago

Trinidad and Tobago have set age restrictions on used vehicles, and specifically 4 years for gasoline and 3 years for diesel light duty vehicles (UNEP, 2017).

Tunisia

Importation of liquefied gas vehicles is prohibited. All types of vehicles (cars, trucks, cargo, and passenger vehicles) must have a VIN sticker and be no more than 5 years old¹²⁴.

Turkey

Used car imports require permission from the Ministry of Trade and are subject to regular inspections at authorized testing centers. TUVTURK is the sole authorized company for these inspections and performance tests. There is no age limit for imported used cars in Turkey, allowing for a broad selection of vehicles¹²⁵. Imported vehicles must also comply with the current EU standards, namely Euro 5 for LDVs from 2010, Euro 4 for HDVs effective from January 2015¹²⁶.

Turkmenistan

Importing vehicles over five years old from the date of production is prohibited. The country also bans damaged or wrecked vehicles, high-speed sports cars, cars with engine displacements over 3.5 litres, passenger cars with fewer than two airbags, and motorcycles exceeding 120 kilometres per hour. Additionally, only white, silver, and other light-coloured cars can be imported, while black cars are prohibited¹²⁷.

Uganda

Starting July 4, 2023, Uganda has prohibited the importation of used vehicles older than 15 years. The Uganda National Bureau of Standards (UNBS) has set a compulsory "Road Worthiness Inspection (RWI)" requirement following standards adopted in 2017 to ensure that imported used cars meet specific standards¹²⁸.

¹²³ <https://blog.japanesecartrade.com/533-tonga-import-regulation-for-japan-used-cars/>

¹²⁴ <https://plc.auction/delivery/tn>

¹²⁵ <https://www.autocango.com/blog-detail/turkey-import-regulations-used-cars-china>

¹²⁶ <https://dieselnet.com/standards/tr/>

¹²⁷ <https://www.autocango.com/blog-detail/turkmenistan-regulation-import-used-cars-from-china>

¹²⁸ <https://blog.japanesecartrade.com/67-uganda-import-regulation-for-japan-used-cars/>

Ukraine

Starting January 1, 2013, the import of LDV vehicles below Euro 3 standards is banned. Euro 5 LDVs were introduced on January 1, 2016, and Euro 6 LDVs became mandatory from January 1, 2018. Excise taxes are lower for vehicles with smaller engine volumes and lower CO₂ emissions (UNEP, 2017). Moreover, vehicles older than 7 years may be subject to high taxation, while only left-hand drive cars are allowed to be imported¹²⁹.

Uzbekistan

Import policies regulate the age and quality of used cars based on commercial use. Beginning January 1, 2024, personal importation of cars for commercial use was prohibited. Individuals are allowed to import only one car per year for personal use. Corporate entities are exempt from compulsory certification for vehicles imported for their own use, provided they are not intended for sale within the country¹³⁰.

Vanuatu

There is no age restriction for importing used vehicles, and no inspection is required before shipment¹³¹.

Venezuela

Importation of used vehicles is banned in Venezuela (UNEP, 2017).

Vietnam

Imported vehicles in Vietnam must be no more than 5 years old, while right hand drive vehicles are allowed to be imported in the country¹³². The vehicle must have been registered in the owner's home country for at least six months and have a minimum mileage of 10,000 km¹³³. The country has adopted Euro 5 emission standards for LDVs¹³⁴.

West Bank and Gaza Strip

Imports of motor vehicles older than 3 years, according to Article 3, Paragraph 11 (a) of the Paris Protocol is prohibited¹³⁵.

Yemen

Only vehicles less than 4 years old may be imported. Diplomats and individuals authorized by a Yemeni Ministry can import vehicles duty-free; otherwise, customs duties must be paid¹³⁶.

Zambia

There is no age restriction or inspection requirement before shipping used vehicles. Used vehicles to be imported into Zambia must be certified for roadworthiness as per the Zambian Standard ZS 560 from Zambia Compulsory Standards Agency certified authorities^{137, 138}.

Zimbabwe

In Zimbabwe, used vehicles must be no more than 10 years old for importation. Commercial vehicles like tractors, haulage trucks, earth-moving equipment, and specialized vehicles for mining and construction are exempt from this age restriction. Used cars older than 10 years require an import license from the Ministry of Industry and Commerce, while commercial and specialized vehicles used in agriculture and mining do not. Starting June 1, 2022, EAA inspection is mandatory for all imported vehicles¹³⁹.

¹²⁹ <https://plc.auction/delivery/ua>

¹³⁰ <https://www.autocango.com/blog-detail/uzbekistan-regulation-import-used-cars-from-china>

¹³¹ <https://blog.japanesecartrade.com/534-vanuatu-import-regulation-for-japan-used-cars/>

¹³² <https://blog.japanesecartrade.com/91-vietnam-import-regulation-for-japan-used-cars/>

¹³³ <https://www.vico.com.hk/resource/news/tips-from-vico-how-to-import-cars-into-vietnam>

¹³⁴ <https://www.transportpolicy.net/standard/vietnam-light-duty-emissions/>

¹³⁵ <http://www.pipa.ps/page.php?id=1bb75ay1816410Y1bb75a>

¹³⁶ <https://www.icricinternational.org/wp-content/uploads/countries/yemen/custom%20laws.pdf>

¹³⁷ <https://www.zcsa.org.zm/index.php/roadworthiness-inspection-of-imported-used-vehicles/>

¹³⁸ <https://blog.japanesecartrade.com/68-zambia-import-regulation-for-japan-used-cars/>

¹³⁹ <https://blog.japanesecartrade.com/69-zimbabwe-import-regulation-for-japan-used-cars/>

2.4. Summary

2.4.1. Overview of used vehicles regulations and standards in LMICs

Table 2.1 provides an updated overview of existing regulations on used vehicles in LMICs, building upon the resources cited in the previous sections. Moreover, information for each LMIC is appended, regarding their adherence to UN standards and regulations as per the indicators of the global status report on road safety 2023 (WHO, 2023a). For that correspondence, the abbreviations are as follows:

- UN_PTI (WHO variable CP_44_FN): On PTI country adheres to corresponding UN or equivalent international safety regulation
- UN_FrontSide (WHO variable CP_47_FN): On front and side impact protection country adheres to corresponding UN or equivalent international safety regulation
- UN_SafetyBelts (WHO variable CP_131_FN): On safety belts and safety belt anchorages country adheres to corresponding UN or equivalent international safety regulation
- UN_ESC (WHO variable CP_49_FN): On ESC country adheres to corresponding UN or equivalent international safety regulation
- UN_Pedestrians (WHO variable CP_51_FN): On pedestrian protection country adheres to corresponding UN or equivalent international safety regulation
- UN_ABS (WHO variable CP_53_FN): On ABS country adheres to corresponding UN or equivalent international safety regulation

From Table 2.1, it is evident that **the regulation landscape of used vehicle imports is very fragmented across LMICs**. There are considerable differences in whether imports are permitted (and for which societal or population groups), in vehicle ages that are permitted for import, in environmental standards, left- or right-hand driving configurations, and in inspection requirements (pre-shipment or upon arrival), among other factors.

In earlier periods, polluting older vehicles of developed nations were exported to developing nations, which had even more polluting vehicles in the first place. Therefore trade was a venue to decrease average emissions in both exporting and importing countries which was, reasonably, considered as beneficial. However, the magnitude of scale and composition effects for used vehicles was noted to remain critical, among additional considerations, such as the fact that increased retirement age in the more loosely regulated importing country leads to higher total emissions per vehicle (Davis & Kahn, 2010). Nowadays, a more holistic approach is required, given the highly connected and globalised environment. **The ethical and responsible way forward is to pursue beneficial shortcuts, (also known as ‘leapfrogging’ effects), where developing nations such as LMICs can rapidly benefit from the technological attainments of developed nations**, without having to take every single step of the journey of emission reductions or road crash reductions.

It is crucial to note that, overt and rapid electrification of vehicle fleets in developed countries exacerbate the number of internal combustion vehicles exported in LMICs, and suppresses their own electrification efforts. Direct exports of used electric vehicles will remain negligible for the foreseeable future in LMICs, especially as there are considerable end-of-life and second-life applications for vehicle batteries in the countries of original vehicle use (ITF, 2023).

Regarding the facilitation of information exchange during the vehicle exporting process, the CITA report of Activity 1 of the UNRSF “Safer and Cleaner Used Vehicles for Africa” project mentioned that, from all analysed platforms, EUCARIS is found to be the most preferred platform on a broader international use to share registration data (CITA, 2021a). EUCARIS is a data exchange platform within Europe that facilitates the sharing of information related to vehicles, vehicle owners, and drivers, with plans to expand its reach beyond European borders. The platform can provide recipient countries with valuable information to assess whether imported vehicles meet their specific requirements, such as age and emission class. The platform’s ability to share the history of roadworthiness in exporting countries should be of particular interest to all nations involved in vehicle importation.

Apart from broad and encompassing, this strategy should be highly adaptable to countries of varying cultures, origins and present states as well. This is reflected upon the selection of particular case studies, which will provide a proof-of-concept towards acquiring the crucial data needed and conducting road safety-oriented analysis, in order to move towards high-end frameworks for the handling and regulation of used vehicle imports. The aforementioned 'leapfrogging' effects should be pursued whenever possible, without neglecting the ethical aspects of transport, promoting fair and equitable access to vehicles to every societal stratum and preventing exclusionary strategies and regulations. By embracing the needs of underprivileged societal groups, and integrating them into a more sustainable and less financially demanding strategy, the black-market trade will be significantly limited and undercut as well. This will lead to (i) fewer adverse effects by inappropriate vehicles, (ii) reduce 'dumping ground' uses in LMICs and (iii) increased accuracy of reported data for the available fleets, increasing the effectiveness of all road safety policies and countermeasures.

The road safety impacts of used vehicle imports in LMICs remain largely unexamined, with a noticeable lack of research and data on this issue. While significant focus has been placed on the environmental benefits of reducing emissions from older vehicles, the safety aspects of these imports are often neglected. Imported used cars may not meet contemporary safety standards, posing potential hazards to drivers, VRUs and all road users overall (Newman et al., 2024). This is not a purely academic issue either. As LMICs concentrate on environmental goals, the critical road safety concerns associated with these vehicles tend to be overshadowed in the policymaker agendas as well, as evidenced by gaps in their regulatory and policy approach.

Even for a purely conservationist approach, however, **safety should not be neglected as it is extremely supplementary to environmental approach towards sustainability**, which is highly affected by governance quality (Wang, 2019). Peripheral problems, such as the lack of regulations and the offline nature of transactions, can lead to a decline in used vehicle quality. In turn, these gaps have been cited as detrimental towards the quality of the exported used vehicles, and also towards the perception of their quality by potential buyers, ultimately damaging the exporting market (Kim, 2024).

Table 2.1
Summary of existing regulations on used vehicle imports in LMICs

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Afghanistan	--	Permitted only for foreigners	None		None		✓					
Albania	2022	Permitted	<15 years		EURO 3 / 4		✓	✓	✓	✓	✓	✓
Algeria	2022	Regulated	<3 years									
Angola	--	Regulated	<3 years			Pre-shipment inspection						
Antigua and Barbuda	--	Regulated	None			Environmental levy (higher for vehicles > 1 year old)	✓					
Argentina	2017	Banned										
Armenia	2017	Permitted	None		UNECE			✓	✓	✓	✓	✓
Azerbaijan	2023		<10 years	ABS, Airbag	EURO 4					✓	✓	✓
Bangladesh	2017		<4 years		EURO 3 / 4							
Belarus	2023		None	ABS, Airbag, CRS Isofix	EURO 4	Higher tariffs for >5 years old	✓	✓	✓	✓	✓	✓
Belize	2021		<5 years									
Benin	2024					Pre-shipment inspection						
Bhutan	2023	Permitted only for foreigners and diplomats										
Bolivia	2017		<1 year	No liquified petroleum gas								
Bosnia and Herzegovina	2016				EURO 4				✓	✓	✓	✓

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Botswana	2018	Permitted	None	Max 100,000 Km								
Brazil	2017	Banned										
Burkina Faso	--	Permitted	None									
Burundi	--	Permitted	None									
Cabo Verde	2023											
Cambodia	2024		Made after year 2000	LHD	EURO 2							
Cameroon	2018	Permitted	<7 years			Taxation as per engine capacity						
Central African Republic	--											
Chad	2017		<3 years			Inspection upon import						
China (People's Republic of)	--		<10 years	No diesel	EURO 4	Inspection for vehicles made before 2005				✓	✓	
Colombia	1993	Permitted only for collector's cars and diplomats										
Comoros	--		None	LHD								
Costa Rica	2023		None	LHD, Catalytic converters								
Côte d'Ivoire	2018		< 5 years for cars, < 10 years for buses and HGV									
Cuba	2017		<4 years									
Democratic Republic of the Congo	2013		<10 years			Pre-shipment inspection						
Djibouti	2021		<8 years	LHD								

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Dominica	2017		<5 years			Environmental levy						
Dominican Republic	2017		<5 years									
Ecuador	2003	Permitted only for diplomats										
Egypt	2017	Mostly banned	<1 year	No diesel				✓	✓	✓	✓	✓
El Salvador	2023		<5 years									
Equatorial Guinea	--		<10 years									
Eritrea	2018		<10 years									
Eswatini	2023		<15 years									
Ethiopia	2024	Banned non-electric vehicles										
Fiji	2019		<8 years for special vehicles		EURO 4							
Gabon	2017		<3 years									
Gambia	2017		<8 years	Roadworthiness regulations		Pre-shipment inspection						
Georgia	2017		None			2 types of taxation based on engine volume and age		✓	✓	✓	✓	✓
Ghana	2017		<10 years			No salvaged vehicles allowed						
Grenada	--											
Guatemala	--		<10 years	LHD								
Guinea	2018		<8 years for cars, <10 years for buses and <12 years for HGV									
Guinea-Bissau	2023		<5 years									

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Guyana	2024		<5 years for cars, <4 years for HGV									
Haiti	--		<3 years									
Honduras	2024		<10 years for cars, <13 for buses with exceptions	RHD								
India	2024	Cars with engines between 1,000cc- 2,500cc are banned	<3 years	RHD, roadworthiness regulations	EURO 3/4 and 6 in specific regions					✓	✓	
Indonesia	2017	Permitted only for diplomats				PP-8 status diplomatic personnel only						
Iran	2023	Banned diesel engines	<3 years	RHD								
Iraq	--		<2 years									
Jamaica	2019	Permission limits for citizens and businesses	<5 years for PTWs, <6 years for cars, <6-10 years for LDV									
Jordan	--		<10 years for personal vehicles									
Kazakhstan	--		<5 years	Engine volume max. 3 litres			✓			✓	✓	✓
Kenya	2005		<8 years			Roadworthiness inspection by a KEBS agent						
Kiribati	--		None									

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_Frontside	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Kosovo	2023		<10 years									
Kyrgyzstan	--		<10 years	LHD	EURO 2							
Lao People's Democratic Republic	--			LHD								
Lebanon	2021		<8 years	LHD								
Lesotho	2018		<8 years			Vehicle inspection upon entry						
Liberia	2023		<10 years	LHD								
Libya	2019		<5 years for cars, <10 years for LDVs and HDVs, <7 years for heavy specialized vehicles									
Madagascar	--	Cars and trucks running on liquefied gas are unavailable in the country		LHD								
Malawi	--			RHD								
Malaysia	2017		<5 years			Imports for personal use, not for sale or transfer		✓	✓	✓	✓	✓
Maldives	--			RHD								
Mali	--			LHD								

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_Frontside	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Marshall Islands	--			LHD								
Mauritania	--		<8 years									
Mauritius	2018		<3 years			Fiscal incentives for low and zero-emission vehicles						
Mexico	2017, 2024	Imports prohibited if vehicle circulation restricted in country of origin	>5 years and <10 years									
Micronesia	--											
Moldova	--		<7 years	LHD								
Mongolia	--			Left-hand drive, licence for right-hand drive								
Montenegro	--								✓	✓	✓	✓
Morocco	2022		<5 years for LDVs and HDVs		EURO 4							
Mozambique	2018		<5 years for cars, <9 years for vans									
Myanmar	2018		<5 years, only vehicles after 2015									
Namibia	2022		<12 years	LHD								
Nauru	--											

2. REVIEW OF THE STATE OF THE ART ON USED VEHICLES POLICIES AND RELATED STANDARDS IN LMICs

Used vehicles regulations and standards								UN core vehicle regulations				
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Nepal	2017				EURO 3							
Nicaragua	2022		<10years LDV			Exceptions for returning citizens						
Niger	2018		<5 years if coming from outside SACU									
Nigeria	--		<8 years				✓	✓	✓	✓	✓	✓
Niue	2020		<15 years									
North Macedonia	2017		<10 years	Catalytic system	EURO 3, (4 for S vehicles, 5 for LDVs)				✓	✓	✓	✓
Pakistan	2006		<3 years	RHD				✓	✓	✓	✓	✓
Panama	--					Some diplomats can import one vehicle duty-free						
Papua New Guinea	--		<5 years for vehicles<3.5 tons, <15 years for vehicles>3.5 tons									
Paraguay	2023		<10 years									
Peru	2024		<2 years	Max 32,000 Km	EURO 6, Diesel cars banned, carbon monoxide emissions max 9% by volume	No damaged vehicles allowed						

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Philippines	2017			LHD	EURO 4, Gasoline engines <2800ml and weight <1500kg	Pre-shipment registration 6 months prior						✓
Rwanda	--			LHD								
Saint Helena	--			LHD								
Saint Vincent and the Grenadines	--											
Samoa	--		<8 years, only vehicles after 2010			Vehicle inspection and water blasting upon entry						
São Tomé and Príncipe	2016											
Senegal	2023		<8 years									
Serbia	2017				EURO 5 (new LDVs), EURO 3 (cars from 2005)				✓	✓	✓	✓
Seychelles	2017		<5 years	RHD		Import permit required						
Sierra Leone	--			LHD								
Solomon Islands	--			RHD								
Somalia	--			LHD								
South Africa	2002	Banned, specific exceptions for expats								✓	✓	✓

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
South Sudan	2017		<8 years	LHD (Buses)		EAA Pre-Shipment Inspection is required						
Sri Lanka	2016		<2 years for cars, <5 years for LDVs, various other limits									
Sudan	2012	Banned, specific exceptions for immigrants, diplomats, collectors										
Suriname	2017		<8 years for cars, <15 years for buses	RHD								
Syrian Arab Republic	2013	Banned, specific exceptions for diplomats and foreigners	<2 years		Diesel engines not allowed							
Tajikistan	2022		Only vehicles after 2013		EURO 4							
Tanzania	2024					Vehicles aged >8 years incur additional duties						

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Thailand	2017	Only one used or second-hand vehicle for personal use			EURO 4					✓	✓	
Timor-Leste	--		<5 years for light passenger and mixed vehicles									
Togo	--			LHD								
Tonga	--			RHD								
Trinidad and Tobago			<4 years for gasoline, <3 years for diesel LDVs									
Tunisia	--		<5 years		Liquefied gas vehicles are prohibited		✓			✓	✓	✓
Turkey	2024				Euro 5 for LDVs, Euro 6 for HDVs	Regular inspections at authorized testing centres		✓	✓	✓	✓	✓
Turkmenistan	2024		<5 years	No cars with <2 airbags and PTWs >120 km/h	No engine displacements >3.5 litres	No damaged vehicles allowed						
Uganda	2023		<15 years	EAA compulsory Road Worthiness Inspection (RWI)			✓	✓	✓	✓	✓	✓

Used vehicles regulations and standards							UN core vehicle regulations					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Ukraine	2017				Euro 6 for LDVs	Lower taxes for vehicles with lower engine volumes and CO2 emissions						
Uzbekistan	2024	Importation of cars for commercial use is prohibited										
Vanuatu	--											
Venezuela	2017	Banned										
Vietnam	2017		<5 years	RHD	Euro 5 for LDVs	Minimum mileage of 10,000 km						
West Bank and Gaza Strip	2015		<3 years									
Yemen	2013		<4 years									
Zambia	--			Roadworthiness certificate								
Zimbabwe	2022		<10 years for personal use vehicles	EAA compulsory Road Worthiness Inspection (RWI)								

Source: Table developed by authors with regulation data from WHO, 2023

2.4.2. Implications and needs for road safety analysis

From the examination of all relevant reports and online resources, it is evident that the regulation landscape and the overall conceptual approach regarding used vehicle imports remains largely fragmented as of 2024. It stands to reason that higher degrees of international cooperation towards the standardization and adoption of import rules and regulations will lead to considerable ease of the entire process, as well as saving of resources, due to economies of scale in all its stages (identification, inspection, transport, authentication etc.). Since the end goals are common across countries, a smart, anticipatory strategy will have to be established, which will take into account rapid technological developments such as electrification. In order to be truly futureproof, developments in cooperative, connected and automated mobility (CCAM) and how the automated configuration of each vehicle will operate in the importing and exporting countries, will also have to be considered as well.

This reality ties in with **the present status quo which typically lays all the weight of the policy to the receiving country**, also known as regulating the demand side (Macias et al., 2013). In comparison, there is a considerable gap in regulating the supply side, and even larger gaps in coordination of both sides, which should yield benefits larger than the sum of the two parts.

As per the aforementioned, there are gaps in the overall vehicle registration and recording process. Overall, these should be solved by a standardised and unified approach, at least at a continental level. Total vehicular import bans have been cited to reduced imported vehicle numbers, reported as high as 76% (Coffin et al., 2016), but more recent research suggests that high vehicle numbers are rerouted through black markets and as such are undocumented (Boateng & Klopp, 2022). Therefore realistic approaches on the treatment of imported used vehicles are urgently needed.

It is noted that road safety is not directly among the used vehicle import criteria; indirectly, vehicle age and certain technical requirements serve road safety purposes, however the impact of used vehicles on road safety is currently unknown. There are no dedicated studies associating the number of imported vehicles with the number of road crashes in a country, or the introduction of a relevant regulation on road safety outcomes. It appears that road safety goals are neglected over the environmental goals, not only in policy efforts, but also in policy impact assessments. **The present report aims to fill this gap by explicitly looking in the road safety implications of used vehicles policies** (or lack thereof).

3. ANALYSIS FRAMEWORK

3.1. State of the art on road safety and vehicle characteristics

Vehicle state is a well-known risk factor in road safety. Apart from the baseline vehicle characteristics (mass, speed, protection offered to the occupants), **vehicle technical features and age / year of first registration have been associated with crash outcomes** by earlier studies. In a longitudinal analysis of data between 1985-2001, it was found that the risk of driver death increases each year vehicles age, also probably due to changes in vehicle use patterns. In that study, after separating out the vehicle design effects and making adjustments for the vehicle age effects, an adjusted picture emerged: without the vehicle design changes, the historical decline in driver fatality risk would have ended in 1993, with risk climbing ever since. The contribution of vehicle design and equipment standards, road safety technologies and crashworthiness has been paramount to the decline in traffic fatalities observed over the last decades in most of the developed countries (Farmer & Lund, 2006).

However, **the quantification of the impact of vehicle age and technical standards has received little attention in the international literature** (Høye, 2019). In many cases, the impact of vehicle age is 'hidden' in the impact of other known risk factors such as driver age, safety-belt use, vehicle technical inspection and others, for many of which the literature is abundant.

Vehicle age and year of registration

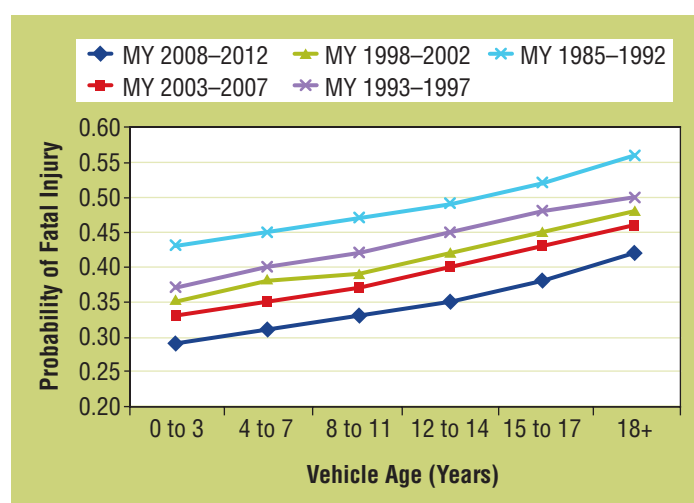
Only a few recent studies can be found on the topic, and all of them come from industrialised countries. For example, Santolino et al. (2022) examined the impact of vehicle age on occupants' injury severity and found that the expected severity of the crash increases linearly up to a vehicle age of 18 and remains constant thereafter at the highest level of body injury severity. A relevant study using UK data from the motorway's road network in the period 2014-2016 found that crashes of older vehicles are more likely to occur due to skidding, and skidding and overturning. The results supported the assumption that crash types vary across the different vehicle age groups, describing the inter-relation between road environment features and different vehicle age groups (Török, 2020)

A study from Canada assessed the safety issues arising from Canada a rapid increase in used vehicle imports between 2000-2005. It was found that the fatality, serious injury and crash rates rise rapidly with increasing vehicle age, except for vehicles 0 to 2 years old. For example, the fatality rate involving occupants of vehicles >15 years and older is approximately three times as high as that of vehicles 3-5 years old (Lécuyer et al, 2006).

Another study from the USA (NHTSA, 2013) modelled the impact of vehicle age on fatality risk, and found that, a driver of a vehicle 4 - 7 years old is 10% more likely to be fatally injured than a driver of a vehicle that was >3 years old; respectively, a driver of a vehicle 8 -11 years old is 19% more likely, and a driver of a vehicle 12-14 years old is 32% more likely. The model estimate (controlling for a number of other variables e.g. driver age and gender, vehicle size, crash type, speed limit etc. set in their reference value) is highlighted in Figure 3.1.

Figure 3.1

Model Estimates of the Probabilities of Car Driver Being Fatally Injured, by Vehicle Age And Vehicle Model Year

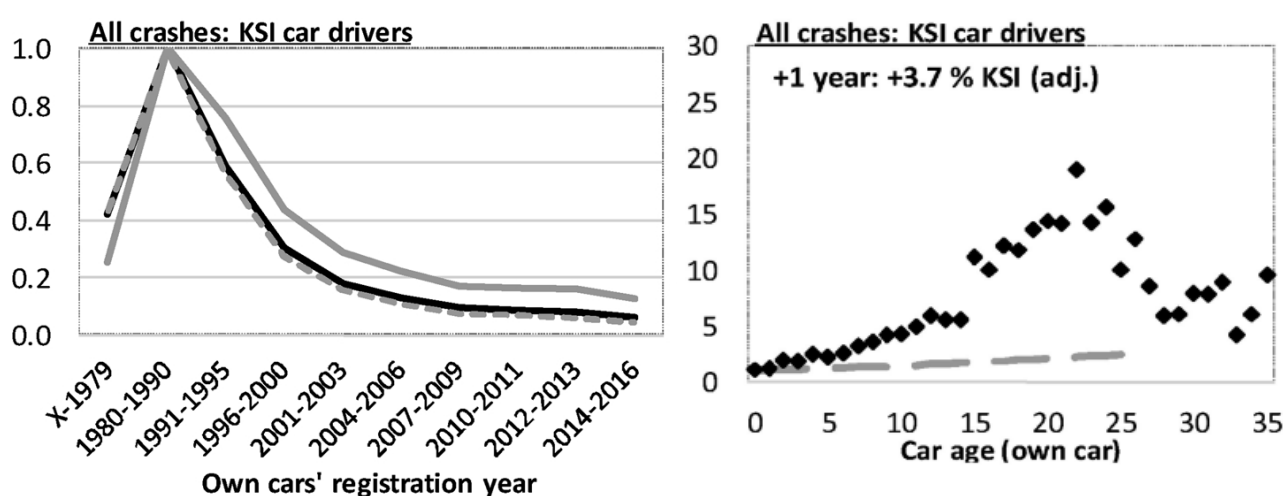


Source: NHTSA, 2013

These findings also showcase an important aspect: A large part of the effect of car age is probably due to effects of registration year which is strongly related to car age. Therefore, in any statistical analysis it is important to account for the relationship between the two variables in relation to the year of the crash. An example from Norway (Høye, 2019) highlights this interaction as follows: according to Poisson model estimates, controlling for vehicle kilometres of travel, there were more KSI (Killed or Seriously Injured) drivers in older cars than in newer cars, namely about 15 times as many KSI in 20-year-old cars than in new cars, without adjusting for the year of registration. Once year of registration is controlled for, the effect becomes smaller, however even in this case the number of KSI drivers was found to increase by 3.7% per year in all crashes (see Figure 3.2). The effect is considerably larger in single vehicle crashes (+5.3% per year).

Figure 3.2

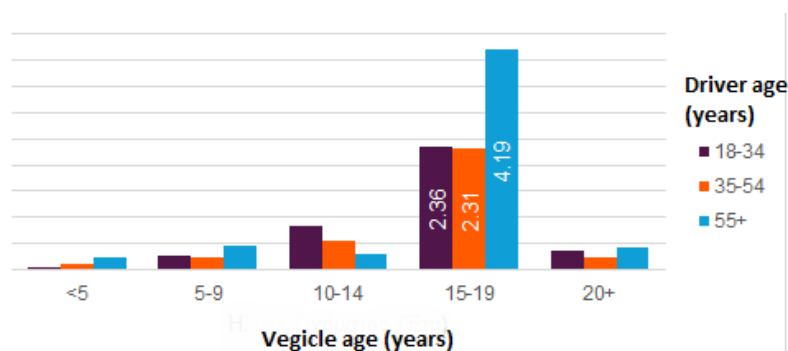
Relative Number of KSI Drivers per Year of Vehicle Registration [Left Panel] and Number of Expected Casualties per Vehicle Age [Right Panel], Norway



Source: Høye, 2019

The Norwegian study also highlights the importance of taking into account the effect of traffic risk exposure, i.e. the amount of vehicle kilometres of travel. **Older vehicles tend to travel less mileage than newer vehicles, often because they are owned by older drivers who have smaller traffic participation and exposure.** This leads to a disproportional estimate of risk, when dividing the number of crashes to the number of vehicle-kilometres travelled by older people. While this is a known ‘artefact’, the risk of driving older vehicles remains much higher. Data from Greece suggest that the actual fatality risk of passenger car drivers increases almost exponentially with vehicle age, while people older than 55 years have ~10 times higher fatality risk when driving vehicles older than 15 years (see Figure 3.3). Moreover, in Greece it has been estimated that older HGV (Heavy Goods Vehicles) vehicles have a 4.5 times higher probability of crash involvement compared to newer vehicles, while crashes involving older HGV have a four times higher probability of resulting in a fatality (Christoforou et al., 2010).

Figure 3.3
Fatality Risk per Vehicle Age Group, Car Drivers, Greece, 2022



Source: National Technical University of Athens

Overall, there appears to be a clear association between vehicle age and adverse road **safety** outcomes, either in terms of crash probability (especially single vehicle crashes, ran-off road etc.) or in terms of fatality and injury, **there are also relationships with other properties of the cars, as well as with the drivers’ age, gender and behaviour.** For example, older vehicles are expected to have lower level of crashworthiness and fewer active safety features. Moreover, older cars are more often driven by either young and inexperienced, or elderly people, by males, and by drivers who are drunk, speeding, and/or not wearing safety belts (Høy, 2019). In the following sections, we summarize the impact of these key vehicle features and driver behaviours on road safety.

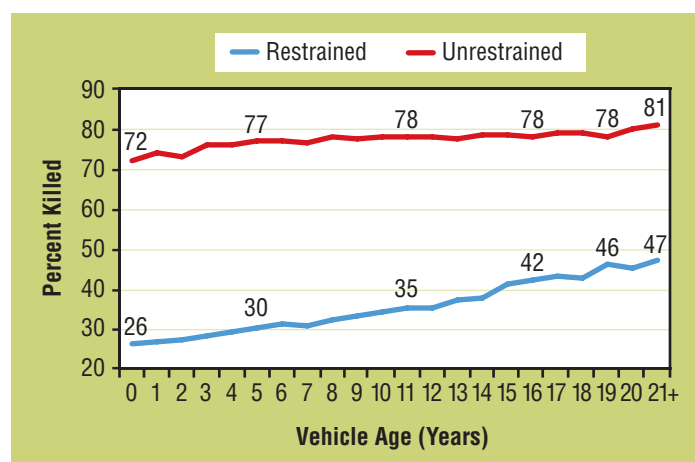
Presence and use of restraint systems

It has been estimated (e.g. Høy, 2016) that **the risk of having a fatal crash is more than 8 times higher for unrestrained drivers** compared with drivers wearing a safety belt. This difference is explained by the fact that, in addition to the severity of the impact suffered by the human body), not using safety belts correlates with other risk factors such as drink-driving, speeding, night-time driving, and previous traffic offences. ETSC (2017) estimated that, if 99% of car occupants wore their safety belts, ~900 deaths per year could be avoided in the EU.

The age of the vehicle is also related to the presence and use of restraint systems. On the one hand, older vehicles are less likely to dispose proper restraint systems as per the UN standards. On the other hand, the lack or non-use of safety belts exposes the older car occupants to much higher risk of fatality or severe injury. According to NHTSA (2013), for drivers wearing a safety belt, the risk of being killed rises from 25% (in a new vehicle) to 35% in vehicles >10 years old and 45% in vehicles more than 20 years old. At the same time, an unrestrained driver has a ‘baseline’ fatality risk of ~70% in a new vehicle, which rises to >80% for vehicles older than 20 years (see Figure 3.4).

Figure 3.4

Percentage of Drivers Killed by Vehicle Age and Restraint Use, USA 2005–2011



Source: NHTSA, 2013

However, **the safety belt wearing rates in LMICs are in many cases alarmingly low**, ranging from as low as 20-25% in Jordan and Malawi, to 90-95% in Brazil, China and Moldova (see Table 3.1). The average safety belt wearing rate for drivers in those LMICs for which data is available is 60% with a standard deviation of 24%; this reveals that the safety belt wearing rates in LMICs are very low compared, for instance, to the EU average of 92% (standard deviation of 9%).

Table 3.1

Percentage of Drivers Wearing a Safety Belt in LMICs , 2021

Country	%	Country	%
Argentina	55.0	Moldova	98.0
Benin	30.0	Mali	50.0
Brazil	97.6	Montenegro	79.2
China	94.7	Mongolia	29.9
Côte d'Ivoire	30.0	Malawi	24.0
Colombia	66.9	Nicaragua	92.0
Costa Rica	70.6	Peru	58.2
Ghana	55.6	Serbia	86.3
Jordan	20.3	Thailand	35.7
Kenya	43.6	Tunisia	51.4
Sri Lanka	75.0	Türkiye	50.1
Morocco	64.3	South Africa	81.0

Source: WHO, 2023

Vehicle active and passive safety technologies

In addition to the above basic vehicle factors affecting crash and fatality risk, **an important role is attributed to the various active and passive safety systems** and Advanced Driver Assistance Systems (ADAS) that aim to prevent deviations from a safe situation, and recovery in case these deviations occur. The UN core vehicle safety standards require a set of design standards and technologies to be available in vehicles, including ABS, ESC and crashworthiness / crash protection. A full review of the safety impacts of these features is beyond the scope of the present report, and we provide below an overview of related findings from the literature:

- **Frontal impact protection** (airbags – EuroNCAP configuration) may significantly reduce injury severity in case of crash, an effect found both in crashworthiness testing and in real-world crashes (Martin, 2017). Side airbags have been found to reduce the probability of head injury by 75%, although minor injuries to upper limbs may be induced, according to some studies (Jänsch & O’Connell, 2017).
- **Pedestrian protection**, either as active technologies (i.e. pedestrian airbags) or as changes in vehicle shape are considered beneficial for crash outcomes, although based on a limited number of studies¹⁴⁰.
- **ADAS** are all associated with safety benefits: e.g. Intelligent Speed Adaptation (ISA) associated with 15-30% crash reductions, Lane Departure Warning / Lane Keeping assist with 25-35% crash reductions, and Autonomous Emergency Braking (AEB) with 35-44% crash reductions. ESC technologies are found to have significant road safety benefits both on crash occurrence and on crash severity in numerous studies (Phan, 2017).

The above review of the state of the art on vehicle-related safety impacts suggests that **the presence and use of restraint systems, as well as the presence of active and passive vehicle systems and technologies, are equally important determinants with the age of the vehicle** with respect to road safety outcomes in case of a crash.

Therefore, within the prospective regulations for the technical standards of imported used vehicles, the adherence to UN core vehicle regulations should be promoted as well. In fact, it has been found that, in countries with no vehicle regulations in place, exporters / industry may put on the market substandard vehicles (new or used / second hand) (Global NCAP, 2020).

3.2. Analytic methods and planning

In this section, the proposed analyses aimed to address the objective of this project are described, namely to identify the impacts of used vehicles regulations and policies on road safety outcomes (fatalities and injuries). The selection of methods takes into account their theoretical properties in line with the properties of the variables of interest. In each case, the data and information that is needed to perform the analysis are outlined.

3.2.1. Macroscopic analysis (global level)

The aim of this type of analysis is to establish a **macroscopic relationship between the trade of used vehicles and the level of road safety** around the globe, with focus on LMICs. The questions to be investigated are as follows:

Is the number of used vehicles imports in the car fleet associated with road safety performance?

The following data will be collected for all LMICs and the following indicators will be estimated:

- Mortality rate (fatalities per 100,000 population) per country (year 2021 or last available)
- Passenger car ownership (registered passenger cars per 100,000 population) per country (year 2021 or last available)
- Imported used car ownership (number of imported cars per 100,000 population) per country (year 2021 or last available)
- Regulation on the imports of used vehicles (no regulation, some regulation, complete ban) per country (year 2021 or last available)

¹⁴⁰ <https://www.roadsafety-dss.eu/#/references?topic=COUNTERMEASURE&taxonomy=6159>

Adherence to UN core requirements for vehicle safety (yes / no) per country (year 2021 or last available), more specifically (see also chapter 2.1):

- Safety belt standards
- Front and side collision impact standards
- Braking standards
- Pedestrian protection standards
- Lateral control protection standards
- Vehicle Technical Inspection standards

The analysis methods that will be implemented at this stage in order to answer the above questions include:

- Descriptive analysis:** this will associate mortality rates with the used car ownership and the share of used vehicles in the fleet for different groups of countries as regards the presence of regulations in the imports of used vehicles and the adherence to UN Vehicle standards.
- Statistical model:** the application of statistical modelling such as count-based **Generalized Linear Models (GLMs)** provides insights on which variables have a statistically significant influence on the values of the Mortality Rate of each country. The analysis will be conducted separately to create two models: (i) including LMICs only and (ii) including all countries,

In this analysis, a particular category of GLMs, namely Negative Binomial regression is used to model integer values (counts) of Mortality Rates across the various countries, using a dispersion parameter to handle the extra variability (see section 5.1.2).

The **Negative Binomial Model** is formulated as follows:

$$y = \log(p) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where p the expected integer of Mortality Rate values, β_0 is the intercept, $\beta_1, \dots, \beta_n$ are the coefficients, and $X_1, \dots, X_n$ are the predictor variables.

The logarithmic link is used to ensure non-negative predicted values for count data. The independent predictor variables X_1 used in each regression must not be mutually correlated, which would bias the results through multicollinearity. Multicollinearity occurs when two or more predictor variables in a model are highly correlated, meaning they provide redundant information. Variance Inflation Factor (VIF) is a useful statistical measure.

Moreover, **Marginal Effects to the Mean (MEM)** are used for increased interpretability of results. MEMs measure **the change in the expected count of the dependent variable in response to a change in an explanatory variable, while holding other variables constant**. Marginal effects help interpret how predictors influence the outcome variable's mean in a non-linear model like NB, where relationships are not as straightforward as in linear models. The estimation of VIF and MEM is detailed in Appendix A.

The outcome of this analysis will be set of model coefficients and their marginal effects between crash mortality rates and the amount of imported used vehicles in the fleet (i) across LMICs and (ii) across the world.

3.2.2. Cross-sectional before-and-after analysis (country / regional level)

While the analysis of the effect of policy interventions over time can yield useful insights, it is not always advisable to yield the actual estimate of the safety effect of the policy from this type of analysis. Even if the intervention variable is found to be statistically significant, it only reflects the impact of the intervention on the *temporal* development, and it would be difficult to conclude that the size of its estimated parameter corresponds to the actual safety effect of the treatment, in terms of e.g. percentage reduction in fatalities.

In practice it is difficult to detect the change due to one measure in the development of the national victim total. A number of factors, which will be detailed in the following paragraphs, challenge the validity of country-specific analyses.

The analysis method that will be used to estimate the safety effect of an intervention (i.e. policy or other treatment implemented) is a cross-sectional **before-and-after study**. This is a quasi-experimental study, in which the road safety outcomes following a policy or intervention in a certain country are compared to the road safety outcomes in a group of similar countries where no relevant intervention took place. It is therefore possible to assess the real safety impact of the policy, not only by comparing the before-and-after situation in one country, but by taking into account of what would have been the result if the policy did not take place.

The aim of this type of analysis is to quantify the safety effect of policy interventions on the imports of used vehicles in terms of percentage reduction in fatalities. It uses a 'comparison group' of countries where relevant policies were not put in force, in order to estimate the relative improvement not only compared to the 'before' state, but also compared to a 'do nothing' state. The question to be investigated is as follows:

What is the actual safety impact of a policy intervention on used vehicles on road safety?

The following data are required in order to implement this type of analysis:

- Selection of a country of interest (test country)
 - Year of policy intervention on used vehicles (occurring at least five years earlier than the last available year of road safety data)
 - Number of fatalities per year (at least 5 years before the intervention)
 - Number of fatalities per year (at least 5 years after the intervention)
- Selection of a group of countries (comparison group) with similar socioeconomic and transport characteristics, similar road safety culture etc. For each country:
 - Number of fatalities per year (at least 5 years before the test country intervention)
 - Number of fatalities per year (at least 5 years after the test country intervention)

The rationale behind the quasi-experimental before-and-after studies is that there are confounding factors which influence the number of road crashes and casualties and, therefore, should be accounted for in the estimation of a real safety effect of the treatment. The threats to validity when simply comparing crash outcomes 'before' and 'after' an intervention are (Hauer, 1997):

- **Regression to the mean:** Road crashes have a random behaviour. Hence, in some periods, the values measured on given points of the network can be greater (or less) than the average values expected for those points. In the measurements made after the treatments, an effect of decrease of crashes is sometimes registered, independent of the treatments, simply due to the random fluctuation in crash numbers. While this randomness is more prevalent in local crash data (e.g. road segment-based crash counts), it is still important to account for it even when working with national data.
- **Uncontrolled environment:** Road crashes occur in a setting, which, unlike a laboratory, is not 'controlled'. Therefore, for some types of road crashes, some medium-long term trends can be observed due to various safety features of vehicles or a change in driver habits. If a decreasing road crashes trend existed already in the previous years, the reduction of road crashes after a treatment would probably have occurred even without the treatment¹⁴¹.
- **Other external factors:** These can also affect the number of road crashes where a treatment took place; for instance, a reduction or an increase in traffic flows may bring about a variation in the number of road crashes, independent of the treatment. Other co-incident factors mostly concern the parallel implementation of other measures like lowering the speed limit or increasing control activities.

As mentioned above, in order to properly quantify the effects of a treatment, a simple before-and-after comparison is not sufficient, as it is necessary to compare the situation with the treatment ("**after**") to the situation that would have existed if the treatment were not applied. The latter presents a corrected value of a previously observed ("**before**") situation. Determining

¹⁴¹ This type of bias can be controlled by time series analysis such as ARIMA.

what would have occurred in a site / country without the treatment is a critical part of the entire process and is performed in two steps: first, the determination of the correct “before” value (of the effect), which accounts for the regression-to-the-mean, and second, the determination of the correct “after” value without the treatment, accounting for the uncontrolled environment.

Two basic approaches are possible for estimating the global effect on the sample of sites / countries: (i) using a comparison group or (ii) using a multivariable model. Because the later requires the availability of detailed data and the development of statistical models, we focus on the first approach. Using a **comparison group**, assumes that changes in the safety effect in the comparison group forecast accurately the changes that would have occurred at the treatment site in the absence of treatment. The evaluation of the treatment effect is performed by means of an odds-ratio. In this case, **the safety effect of the treatment at site i is estimated as:**

$$\theta_i = \frac{\frac{\mu_A}{\mu'_B}}{\frac{C_A}{C'_B}}$$

Where:

- θ_i is the estimated effect
- μ_A - the number of road crashes / fatalities observed at the treatment area (site i) in the “after” period
- μ'_B - the number of road crashes / fatalities at the treatment area (site i) in the “before” period
- C_A - the number of road crashes / fatalities observed at the control group area in the “after” period
- C'_B - the number of road crashes / fatalities at the control group area in the “before” period

The applicable value of the safety effect, i.e. **the best estimate of crash / fatality reduction associated with the treatment (in percent), is calculated as $(1-\theta_i)*100$** . In case there are multiple treatment sites, each with their own estimated safety effect θ_i , it is possible to calculate the Weighted Mean effect for all sites, as well as its confidence interval. The formulation of these calculations is provided in Appendix B.

The outcome of this analysis will be a statistically validated estimate of the safety effect (% reduction in fatalities) of used vehicles policies in the test countries, together with a global estimate (if possible).

3.3. Set-up of the case studies

As indicated by the review of scientific knowledge, as well as the existing policies and standards relevant to road safety of used vehicles (presented in Chapters 2 & 3.1), in order to analyse the impacts of policies on used vehicles, **it is important to ensure that not only imported vehicles do not exceed certain age thresholds, but also that the minimum required vehicle technical and safety standards are adopted** by a country. The synthesis of the state of the art on the impact of vehicle features on road safety (see Chapter 3.1) confirms this hypothesis. The import of ‘younger’ vehicles alone by no means guarantees improved safety, if the imported vehicles do not adhere to UN standards.

Therefore, **the criteria for selecting countries for the case studies** are as follows:

- i. Country has implemented a policy on used vehicles imports at least 5 years earlier than the year of its last available road safety data
- ii. Country adheres to (most of) the core UN requirements for vehicle safety
- iii. UN regions are adequately represented
- iv. Availability of time series of road fatality data at least 10 years before and 5 years after the policy introduction
- v. Availability of time series of used vehicle imports data at least 10 years before and 5 years after the policy introduction

3. ANALYSIS FRAMEWORK

- vi. Availability of time series of registered vehicles data at least 10 years before and 5 years after the policy introduction
- vii. Availability of information on other road safety policies in the country

From the detailed review of Chapter 2, **a number of countries are candidates** for the present case studies according to criteria (i), (ii), (iii). Their features are detailed in the following Table 3.2. It is noted that further analysis, including a more in-depth look on the available data, is required to conclude on the most pertinent countries for the case studies.

Table 3.2
Used Vehicle Policies and Standards, Adherence to Core UN Vehicle Safety standards for Case-study Candidate Countries

Used vehicles regulations and standards							UN core vehicle standards					
Country	Year	Type	Age limit	Technical standards	Environmental standards	Remarks	UN_PTI	UN_FrontSide	UN_SafetyBelts	UN_ESC	UN_Pedestrian	UN_ABS
Albania	2022	Regulated	<15 years		EURO 3 / 4		✓	✓	✓	✓	✓	✓
Egypt	2017	Mostly banned	<1 year	No diesel				✓	✓	✓	✓	✓
Georgia	2017	Regulated	None			2 types of taxation based on engine volume and age		✓	✓	✓	✓	✓
Kazakhstan	Unknown	Regulated	<5 years	Engine volume max. 3 litres			✓			✓	✓	✓
Malaysia	2017	Regulated	<5 years			Imports for personal use, not for sale or transfer		✓	✓	✓	✓	✓
Nigeria	2020	Regulated	<8 years				✓	✓	✓	✓	✓	✓
North Macedonia	2017	Regulated	<10 years	Catalytic system	EURO 3, (4 for S vehicles, 5 for LDVs)				✓	✓	✓	✓
Pakistan	2006	Regulated	<3 years	RHD				✓	✓	✓	✓	✓
Serbia	2017	Regulated			EURO 5 (new LDVs), EURO 3 (cars from 2005)				✓	✓	✓	✓
South Africa	2002	Banned, specific exceptions for expats								✓	✓	✓

Source: Table developed by authors with regulation data from WHO, 2023

The final countries selected for case studies, because their policies and available data were considered sufficiently robust, are as follows:

- **Egypt**
- **Georgia**: selected with reservation; regulation (vehicle age and size) was introduced through taxation only.
- **Malaysia**
- **Pakistan**
- **Romania**: selected as a new country; originally not included in the present review, it can be considered for a case study as a country that joined the EU on 2007, and can serve as a useful case study of adopting UN vehicle regulations.
- **Nigeria**.

The remaining countries were rejected for a variety of reasons:

- Albania: rejected, regulation was introduced in 2022 and no effect can be identified
- Kazakhstan: rejected; year of regulation is unknown.
- North Macedonia: rejected; very small country, unlikely to be representative of policy impacts.
- Serbia: rejected; regulation was introduced only on environmental standards, no age limit in the imported used vehicles.
- South Africa: rejected; regulation was introduced in 2002, data is not available before 2000.

4. DATA COLLECTION

This chapter summarises the data collection that has been carried out in this project, as per the analysis requirements outlined in Chapter 3.

4.1. Road Safety data

Road safety data were collected from the WHO Global Status Reports on Road Safety (WHO, 2023) for the period 2010-2021. In these reports, data are available every 3 years, namely 2010, 2013, 2016, 2019 and 2021. There are three road mortality indicators published:

- **The number of country-reported fatalities:** this is the estimate provided by the country representatives, usually based on data collected by the Traffic Police, according to the definition of fatality as “a person killed within 30 days from the crash”
- **The number of WHO-estimated fatalities:** this is an estimate of the actual number of traffic fatalities, including the persons who have died beyond the 30 days, and correcting for possible under-reporting of fatalities in national databases. In some countries, namely those that have credible data from the health sector, this estimate is made on the basis of death certificates. In countries with no credible vital registration data, the estimate is based on regression models developed by WHO¹⁴².
- **The mortality rate,** calculated as the number of fatalities (WHO-estimated) per 100,000 population.
- In the present study, the second and third indicators from the list above will be used. In Appendix C, the collected data for LMICs are provided. In the next phases of the project, the above data will need to be collected per year for a time span of at least 10 years before and 5 years after the used vehicles policy intervention.

4.2. Used vehicles data

Used vehicles data have been initially collected from the ITF Dashboard¹⁴³ on used vehicles trade across the world, as per the relater report of ITF (2023). The meta-data provide full information about the national (e.g. Ministries, statistical offices, trade associations) and international (e.g. Eurostat) sources used to gather the data, including personal communications with country experts. The dataset includes the following variables (see Appendix C):

- Number of imported vehicles by country per exporting country, per year (1988-2023)
- Number of exported vehicles by country per importing country, per year (1988-2023)

Additional data on vehicle fleet have been collected from the WHO (2023), on the following indicators (see Appendix C):

- Number of registered vehicles (vehicle fleet) in the country (2021 or last available year)
- Number of registered cars and 4-wheeled light vehicles (2021 or last available year)
- Percentage of passenger of powered light vehicles in the fleet (2021 or last available year)

¹⁴² The regression models' estimates have been validated in several countries that have implemented studies to improve their crash fatality estimates, e.g. Thailand, Tunisia and others.

¹⁴³ <https://www.itf-oecd.org/used-vehicles-dashboard>

4.3. Other relevant data

Finally, in order to calculate additional indicators needed for the analysis, namely the vehicle ownership rate and the used vehicle ownership rate, population (2021 or last available year) data were collected from WHO (2023), together with additional sociodemographic and transport safety indicators, that may be used in the analyses, e.g. for the selection of appropriate comparison groups in before-and-after analysis:

- Population density
- Mean age of the population (years)
- Share of population living in urban areas (%)
- Gross Domestic Product (GDP) per capita
- Vehicle ownership (1000 inhabitants)
- PTW ownership (1000 inhabitants)
- Number of registered Powered-Two-Wheelers (PTW)
- Number of registered Heavy Trucks and Buses
- Safety-belt wearing rates (%) for drivers
- Helmet wearing rates (%) for PTW riders
- Road network density (km/ sq.km)
- Share of paved roads (%)
- Percentage of reported fatalities for passenger cars (%)
- Percentage of reported fatalities for PTW (%)
- Percentage of reported fatalities males / females (%)
- Percentage of males in reported fatalities (%)
- Mean annual precipitation (mm/year)
- Mean annual temperature (°C)
- HDI (Human Development Index)
- Hospital beds per population (1000 inhabitants)
- Adherence to one or more of the 7 UN road safety conventions
- Used vehicles regulation (banned, permitted with regulation, unregulated)

4.4. Stakeholders' consultation

A **stakeholders' consultation** took place, focusing: (i) on the collection of in-depth data and information for the case study countries, and (ii) on the exchange of knowledge, experiences and information with other key stakeholders that have been involved in the investigation of the impacts of used vehicle imports in LMICs. A survey was dispatched among the members of the World Forum for the Harmonisation of Vehicle Regulations (WP.29) including country representatives, importers and exporters, regional bodies (e.g. Economic Community of Central African States (ECCAS), Southern Africa Development Community (SADC)) and other relevant stakeholders (e.g. FIA, World Customs Organization, Global-NCAP). The survey questionnaire is available in Appendix F.

4. DATA COLLECTION

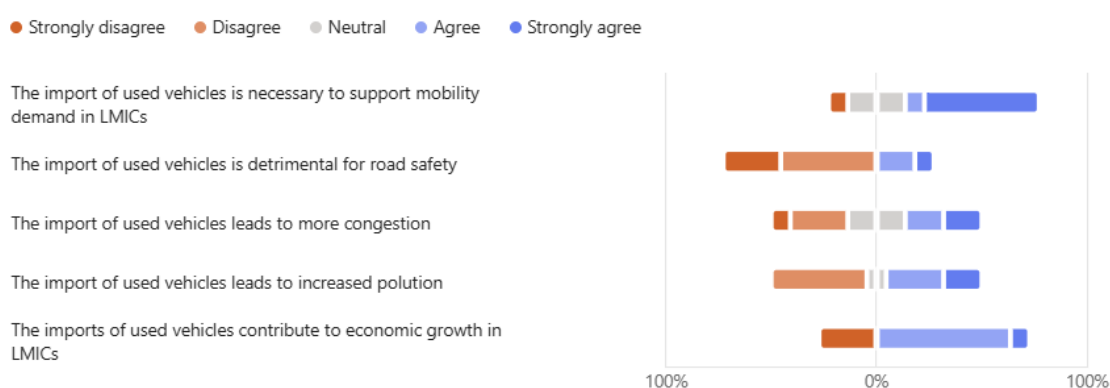
The survey included 4 sections and 10 questions in total, structured as follows:

- Stakeholders' attitudes towards the socioeconomic impacts of imported used vehicles
- Stakeholders' perceptions of the road safety impacts of imported used vehicles and the key vehicle characteristics that affect it.
- The share of imported used vehicles on the roads in different countries, and their respective share among the vehicles involved in road crashes
- Stakeholders' suggestions and recommendations on policies that may improve road safety of imported used vehicles.

A total of 11 responses to the survey were received. The majority of responses came from country representatives with various functions (i.e. land transport agencies, standardisation agencies), while additional individual responses came from regulatory bodies, importing agencies, vehicle industry / manufacturers and research institutions.

The first set of questions included the socioeconomic impacts of imported used vehicles. Participants were asked to indicate how much they agree with a number of statements, on a 5-point Likert scale from 'strongly disagree' to 'strongly agree'. The results are shown in Source. It is interesting to note that, while all stakeholders tend to agree that imports of used vehicles support mobility needs and contribute to economic growth of LMICs, their perception of the traffic risks associated with them is low; **only 27% of respondents agree or strongly agree that imported used vehicles are detrimental for road safety**. On the other hand, the majority of respondents are neutral or agree with the detrimental effect on congestion (64%) and on pollution (55%).

Figure 4.1
Stakeholders' perception of the impacts of imported used vehicles in LMICs



Source: UNECE, 2025

The stakeholders further elaborated on their views, by means of open-text responses. It has been mentioned by more than one participant that, imported vehicles may appear as increasing the number of fatalities merely because they will increase traffic exposure and mileage travelled in the country. Some stakeholders stated that the road safety problem in LMICs is no different between new or used vehicles. At the same time, the road safety problems in LMICs are considered to be affected by a number of factors related to the state of the road infrastructure, drivers' behaviour and lack of traffic compliance, sub-standard traffic regulations and lack of enforcement, drivers' training etc., **making the impact of imported used vehicles difficult to isolate**. Therefore, the imports of used vehicles are one component of a holistic road safety policy (or its deficiencies).

Moreover, it is pointed out that, the safety impacts of imported used vehicles depend upon the quality/condition not only of the imported vehicle, but also of the vehicles they replace. For instance, if the used imported vehicle replaces a machine in worse condition regarding safety and emissions performance, then it will improve health. Quoting one of the participants:

Used vehicles from regulated markets are sometimes better than new vehicles from non-regulated markets.

Several participants put emphasis on the **technical inspection of these imported used vehicles**, and the general framework of vehicle maintenance in the country. Quoting one of the participants:

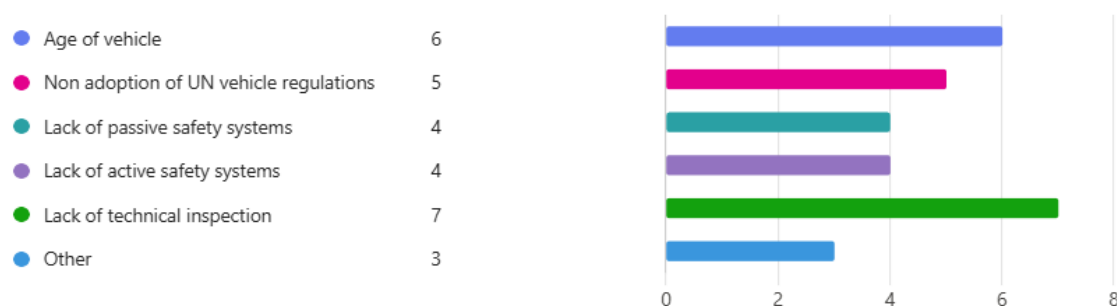
In LMICs, the periodic technical vehicle inspection are not strictly enforced / no enforcement and they are also lacking facility. The used vehicles especially with older ages tend to have bad condition and can increase traffic accident risk.

Vehicle age and/or mileage may directly lead to deterioration of parts and systems related to both safety and emissions, and there is lack of technical inspection from either the export or the import country. In addition, there is often insufficient supply of spare parts and after sale service from the export agency, and lack of monitoring from the local authority on the use of the vehicle, e.g. for overload or illegal retrofitting. It was also mentioned that, even for accidents that are a result of mechanically unsafe vehicles, this is attributed to inability of owner to maintain the vehicle irrespective of whether it was acquired new or imported as second hand. This statement may reflect a safety culture according to which the responsibility for safety lies on the end-user, i.e. the driver. While this statement may not be representative of the views of the majority of stakeholders, it is important to underline that this is not in line with **the Safe System approach, which states that road safety is a shared responsibility between designers, operators, and users of the transport system.**

Figure 4.3 shows the stakeholders' **ranking of several proposed factors / vehicle characteristics that may play a key role in their road safety impacts**. It is shown that 7 out of the 11 respondents indicate the lack of technical inspection, while 6 respondents indicate the age of the vehicle and 5 respondents indicate the non-adoption of UN core vehicle regulations. This finding is in line with the international literature; however it is noted there seems to be more emphasis placed on the importance of PTI rather than any other technical characteristic of the vehicle.

Figure 4.2

Stakeholders' Ranking of the most Important Factors associated with Negative Impacts of Imported Used Vehicles



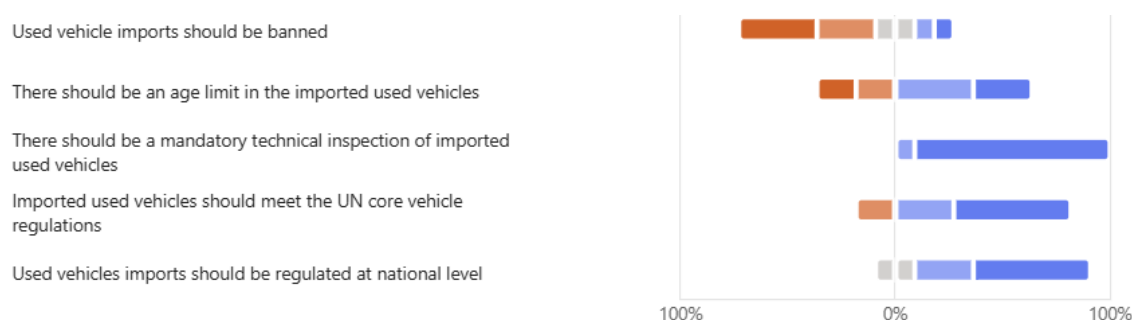
Source UNECE, 2025

As was expected, very little or no information of the share of imported used vehicles on the roads in each country was reported, and this could not be supported by published resources. In order to maintain the anonymity of respondents, specific estimated are not reported here.

Subsequently, participants were asked to indicate how much they agree with a number of statements regarding potential effective policies on used vehicles imports. The summary of responses is provided in Source: . It is interesting to note that only 18% of respondents agree or strongly agree with banning the imports of used vehicles. Moreover, ~64% of respondents agree with introduction of age limits. Notably, **all respondents strongly agree that there should be technical inspections of imported used vehicles**, and almost all stakeholders agree or strongly agree that used vehicle imports should be regulated at national level. Finally, more than 80% of the respondents agree or strongly agree that imported used vehicles should meet the UN core vehicle regulations.

Figure 4.3

Stakeholders' Ranking of the most Important Factors associated with Negative Impacts of Imported Used Vehicles



Source: UNECE, 2025

Further **recommendations for policy actions at the country level**, proposed by the respondents, can be summarised as follows:

- Setting an age limit and a technical inspection of vehicles needs to be accompanied by an enforcement programme on the use of vehicles, i.e. strengthening monitoring of transportation, especially overload, over speed and illegal refitting.
- The development of technical requirements for used vehicles needs to be developed in line with the requirements for new vehicles.
- Implementing a mandatory vehicle inspection at the country of export to ensure that imported used vehicles meet minimum roadworthy standards and comply with Euro 4 emissions standards
- Carrying out periodical technical inspection at the country of import.
- Coordinating between authorities from export and import countries, and setting up assessment and recognition mechanism for export agencies and dealers.
- Establish corresponding testing items and testing method standards to support the requirements and testing of the safety and environmental performance of second-hand cars.
- For countries with home-grown automotive industry, policies should safeguard importers to ensure they get value for money. Local vehicle assemblers and businesses dealing with new vehicles usually may advocate for a ban to importations, but governments should balance the mobility and economic benefits for the whole population.
- Imposing taxes on imported used vehicles which can be eventually invested in improvement of infrastructure.
- Receiving countries must always keep their sovereignty by defining which vehicles are acceptable.

Overall, it is suggested that **there should be more systematic communication and collaboration between exporting and importing countries**, so that eventually a mutually agreed set of technical standards can be identified, upon which technical inspections upon importation can be based. Quoting one of the stakeholders:

Receiving countries have to set up their requirements with the support of countries with vehicle regulatory experience. It is necessary to set up communication between exporting and importing countries' authorities to facilitate information exchange. Involved stakeholders (importers, traders) must be accountable in case of trying to sell non-compliant vehicles.

Specific suggestions for actions from the exporters' side include:

- Perform pre-export inspections to ensure vehicles are safe, sharing of the registration data of vehicle with importer country, providing roadworthiness certification and Vehicle Service Book / PTI certificate.
- Enhance qualifications of export dealers, encourage export dealers to improve their capability of after sale service and supply of spare parts.
- Target and address illegal practices related to used vehicle exportation.
- Help setting up the regulatory framework for the importing countries for regulating these vehicles.
- Exporting Countries should support Infrastructure Development in LMICs, like in vehicle maintenance and setting up inspection centres from the funds they receive from trade of these vehicles.

Overall, a concern was expressed by a number of participants that the safety concerns over these vehicles may undermine the economic development of LMICs by setting barriers to a valuable source of mobility and economic development, without any concrete quantitative evidence. Quoting one of the stakeholders:

Used vehicle as a very valuable resources should be fully used, especially for LMICs [...]. In many developed countries, used vehicles are abundantly traded and become low-income families and beginners preferences. The key point is how to balance the positive and negative aspect of used vehicles. If effective measures be taken by both export and import countries, [...] the negative affects coursed by used vehicles can be limited to extend which is acceptable.

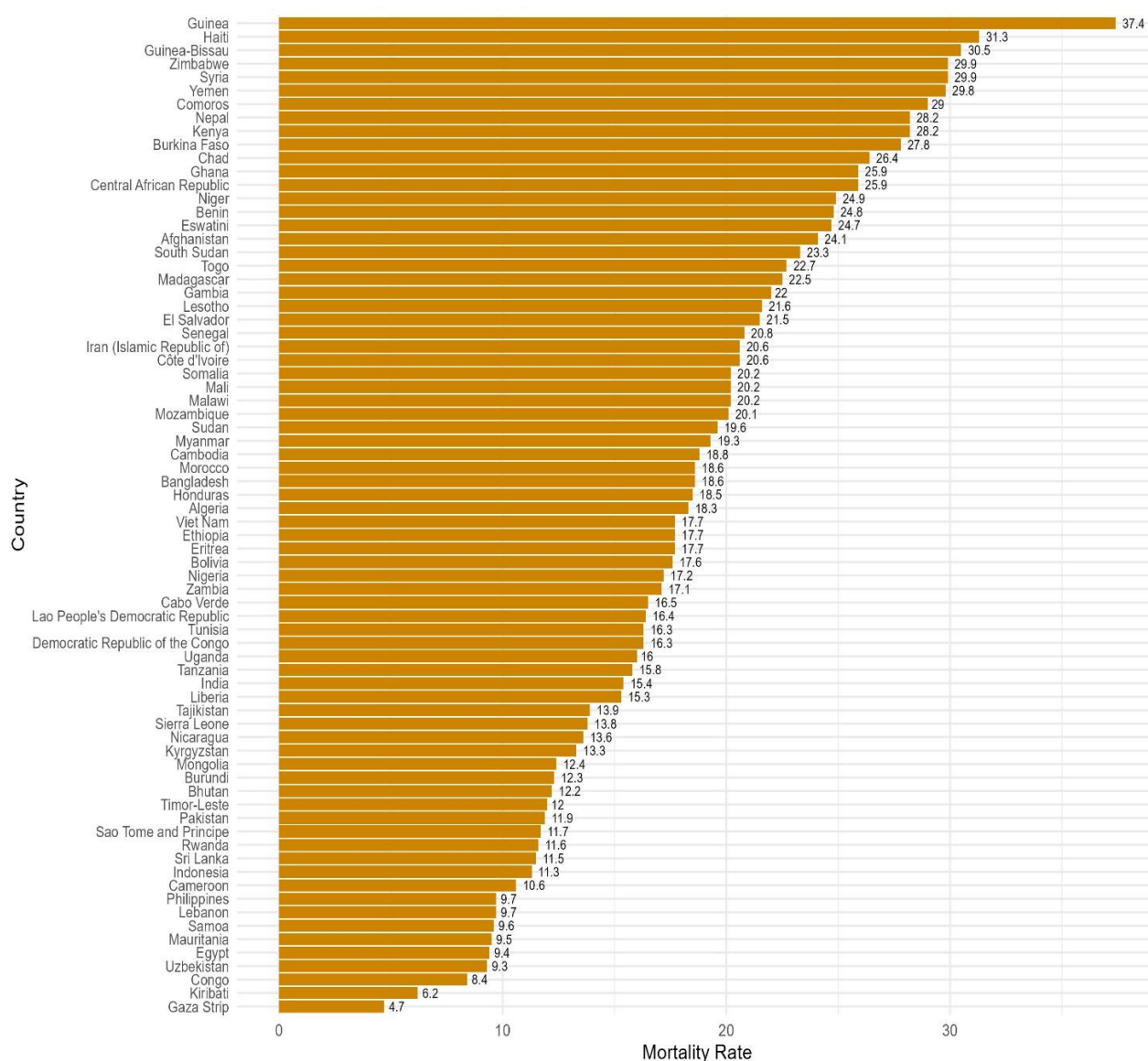
5. RESULTS OF MACROSCOPIC ANALYSIS

5.1. Descriptive analysis

5.1.1. Basic statistics

Initial descriptive analysis was conducted for the collected data to ascertain their basic statistical profile and properties before more in-depth analysis. Figure 5.1 shows the mortality rates of LMICs for year 2021. It is observed that these values range from 4.7 (for the Gaza Strip) to 37.4 (for Guinea).

Figure 5.1
Mortality Rates per Country, 2021



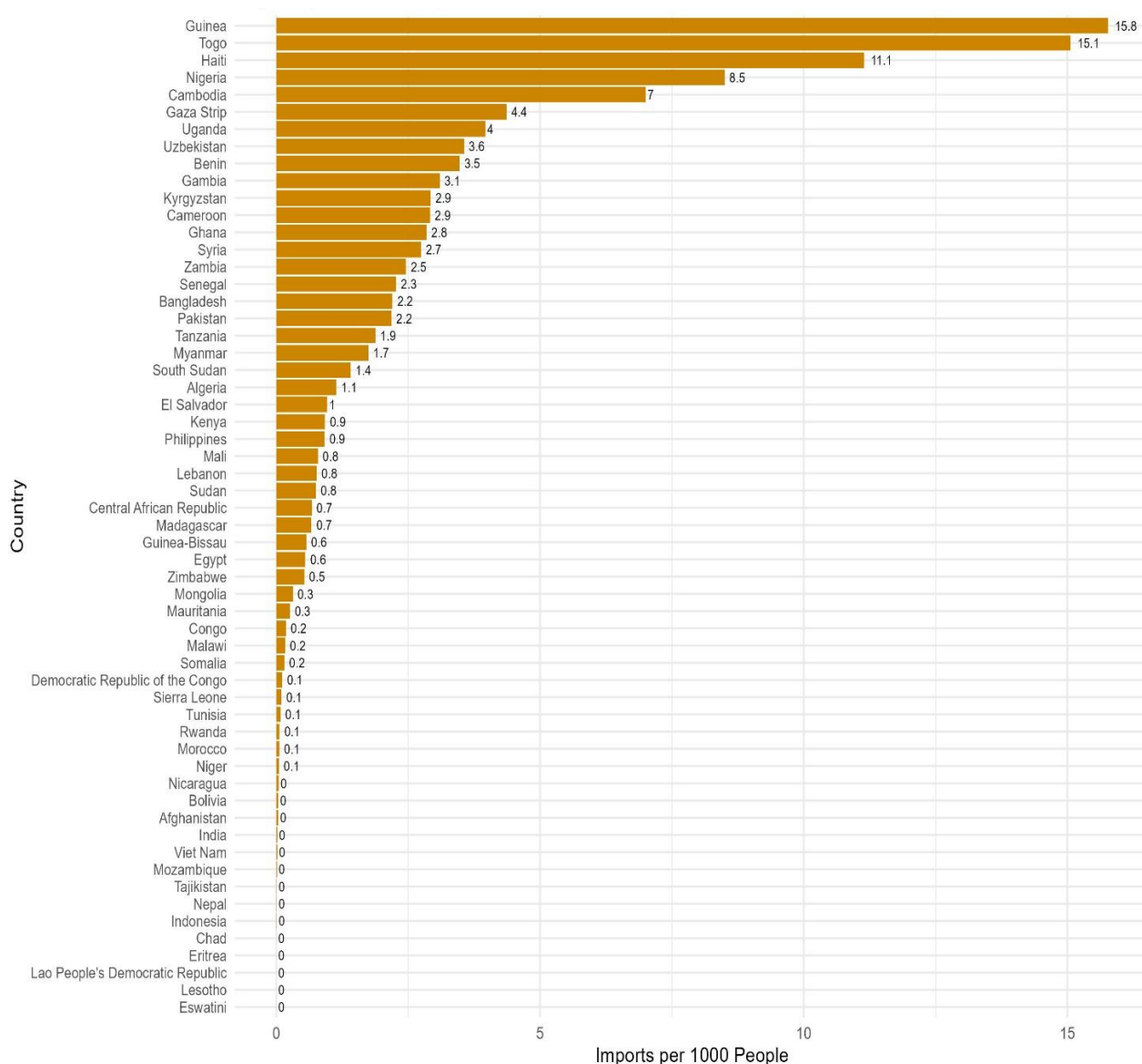
Source: WHO, 2023

In order to explore the distribution of countries with respect to the magnitude of used vehicles imports, a relevant standardisation method is required. The used vehicle imports are expressed as the average imports of the most recent 5 years for which data has been published (ITF, 2024 - namely between 2017-2021), divided by the total population (and multiplied by 1000 inhabitants). Mathematically, this rate is expressed as:

$$AVG_{IMPORTS_{2017-2021}} = \frac{\text{average}(Imports \text{ between years } 2017 - 2021)}{\text{population}} * 1000$$

The results, presented in Figure 5.2 show that several countries import more than XX vehicles per 1000 inhabitants over the last 5 years. Two countries had exceptionally high rates of vehicle imports and were omitted from the graph, namely: Yemen (with 31 imports per 1000 people) and Liberia (with 29.7 imports per 1000 people). However, the vast majority of countries imported less than 5 vehicles respectively.

Figure 5.2
Average Used Vehicles Imports (2017-2021) per Population (1000 Inhabitants)



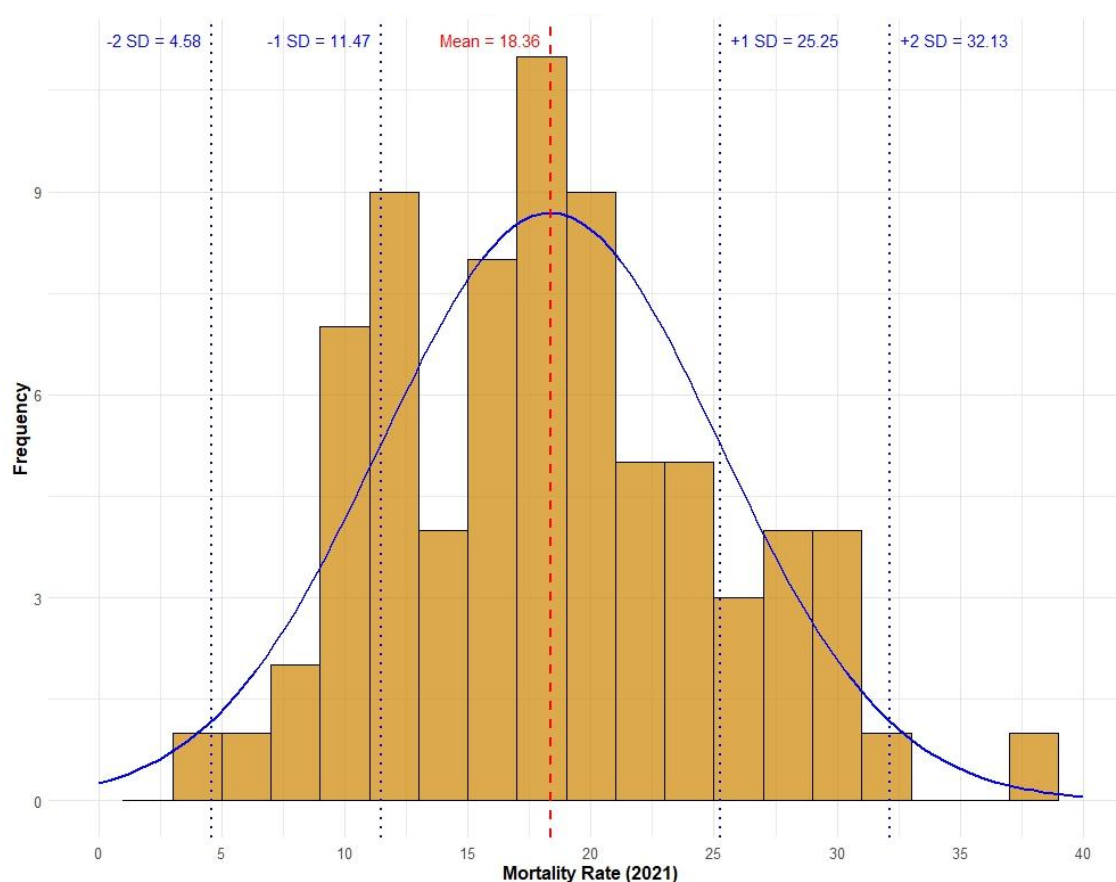
Source: WHO, 2023, & ITF, 2023, and author's independent processing

These results suggest that the imports of used vehicles exhibit a very large dispersion across countries. A descriptive analysis of basic indicators was carried out (see Appendix F), in which mortality rates were plotted against the average used vehicles imports per population over the past 5 years, for all countries in general, and for LMICs in particular. These results confirm that there is a lot of variation in the data, and it is not possible to observe any macroscopic associations. Therefore, the development of a macroscopic statistical model associating mortality rates with vehicle regulations, fleet and imports, appears to be the pertinent option.

5.1.2. Distribution of mortality rates

The distribution of mortality rates of 2021 for LMICs (provided by WHO, 2023) can be examined by creating a related histogram, as shown in Figure 5.3 below. Each bar width represents a value of 2 of mortality rates (also known as categorical bins).

Figure 5.3
Histogram of Mortality Rates (2021) vs. Normal Distribution



Source: WHO, 2023

From Figure 5.3, it can be concluded that:

- The distribution of Mortality rate values of 2021 in LMICs appears to follow a normal distribution pattern for the most part.
- The blue line is a theoretical normal distribution (mean = 18.356, st.dev = 6.889).

This finding denotes that linear models and Generalised Linear Models (GLMs) are appropriate for effective further analysis of mortality rate values. At the same time, the variable does not have large enough values for robust log-linear modelling (namely to create models without numeric stability issues). GLMs are also widely used for modelling relationships in data where the response variable has the form of count data (McCulloch, 2003).

Therefore, by rounding the Mortality Rates for 2021 to integers, count data that refer to mortality are obtained. The starting GLM is typically the Poisson model, which is frequently selected as the most appropriate distribution for the response variable due to the nature of the data (Hilbe, 2014). However, it has been established that overdispersion (the condition in which the variance of the dependent variable is larger than the mean) leads to biased results when examined with a Poisson GLM. Instead, Negative Binomial (NB) GLMs should be implemented to account for overdispersion (Lord & Mannering, 2010).

Therefore, the mean and variance of the Mortality Rate variable were subsequently examined. It was determined that the variance (with a value of 47.459) was significantly larger than the mean (with a value of 18.356) of the Mortality Rate variable (using 2021 values for LMICs). Moreover, when considering all countries, the mean (with a value of 13.520) of the Mortality Rate variable was even smaller than the variance (with a value of 63.540), again using 2021 values. Therefore, the data has considerable overdispersion. In these cases, the Poisson distribution assumptions are violated, and Negative Binomial (NB) Regression GLMs are preferred.

5.2. Models development

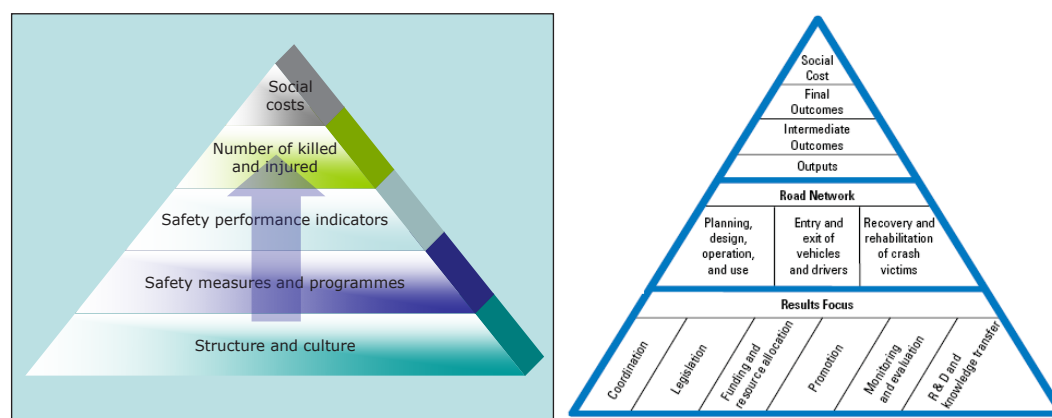
5.2.1. Variables selection: the 'pyramid' road safety model

The selection of variables for macroscopic road safety models has been a challenging scientific question for road safety analysis (Page, 2001; Kopits & Cropper, 2005; Yannis et al., 2011). The level of motorisation of a country (e.g. the vehicle ownership rate) and the traffic exposure of the population (e.g. the amount of vehicle kilometres of travel) are well established as the fundamental predictors of road safety outcomes in a country, region or local road network. It is often of particular interest for road safety analysis to test variables that are related to policy interventions. However, these policy interventions are strongly correlated with one another, and also with the behaviour of road users and the resulting transport demand and exposure.

The 'pyramid' concept of road safety management systems (Koornstra et al., 2002; Wegman et al., 2005) was developed in order to take into account the structure of these correlations and help in the understanding of road safety policy mechanisms. This concept has been tested in numerous applications (e.g. (Bliss & Breene, 2009; Yannis & Papadimitriou, 2013; EC, 2023) (see Figure 5.4).

Figure 5.4

Generic structure of the pyramid of road safety management systems



Sources: adapted from (Koornstra et al, 2002; ERSO, 2021) – left panel, Bliss & Breen, 2009 – right panel

One of the most important applications of this concept has been presented within the **UNECE SafeFITS model for global road safety predictions** (UNECE, 2018; Yannis et al., 2018). In this model, it was established that the following layers constitute the road safety management system:

- **Economy and Management:** This layer includes data concerning the basic characteristics of each country, economic indicators and indicators describing the road safety management structure (i.e. the institutional and socio-economic background of the system, or the 'policy input')

- **Transport demand and exposure:** This layer includes characteristics related to the amount of travel per mode in each country, e.g. share of traffic per mode (passenger cars, powered 2-wheelers, public transport, pedestrians and cyclists etc.), vehicle- and person-kilometres of travel, the time spent in traffic per mode etc. Due to lack of data at international level, this will be often approximated by indicators reflecting the level of motorisation.
- **Road Safety Policies and Measures:** This layer includes indicators regarding legislation and interventions made by authorities covering the 5 UN pillars of the road safety Action Plan: road user, vehicles, roads, post-crash care. (i.e. the ‘policy output’)
- **Road Safety Key Performance Indicators (KPIs):** An intermediate layer is required to properly understand the mechanism of road safety impacts. Any policy or measure will first affect the operational level of road safety, which can be expressed by indicators reflecting the state of the road infrastructure, the state of the vehicle fleet (e.g. vehicle age, safety standards) and – most importantly – the behaviour and compliance of road users (e.g. share of speeding, use of restraint systems, driving under the influence etc.).
- **Fatalities and injuries:** the final outcomes, in terms of fatalities and injuries, reflect the eventual impact of the policies.

The Safe FITS model established that, for a complete road safety performance analysis, variables from all 4 layers of the pyramid, and all 5 pillars of the UN action plan (i.e. management, roads, vehicles, road users and post-impact care) need to be included in a macroscopic model (UNECE, 2018; Yannis et al., 2018). However, it is also possible to focus on one pillar, as long as the 4 layers of the pyramid are accounted for, to represent the whole mechanism of road safety. In the present study (see

Table 5.1), **we use the SafeFITS layers of road safety management with specific focus on the ‘Vehicles’ pillar of the UN road safety action plan.**

Table 5.1
Variables Selection for Macroscopic Analysis focused on Vehicle Characteristics

Vehicle related indicators	
Economy & Management	• GDP per capita • Level of motorization (vehicle ownership, PTW ownership) • Quality of road infrastructure • ...
Transport Demand & Exposure	• Vehicle-kilometres of travel • Level of motorization (vehicle ownership, PTW ownership) • Rate of imported used vehicles • ...
Policies and measures	• Adherence to UN vehicle regulations • Regulation of used vehicle imports • ...
KPIs	• Share of road users using restraint systems (e.g. safety belts) • ...
Final outcomes	• Number of fatalities • Mortality rate

Source: UNECE, 2025

Therefore, a number of vehicle-focused independent / explanatory variables can be considered for model development. The following Tables showcase the descriptive statistics of all considered variables at the level of all countries (not just LMICs). A separate dataset was also created for LMICs, as a subset of the dataset outlined below. The number of missing values per variable is also provided in each Table. For continuous (numerical) variables, results are presented in Table 5.2.

Table 5.2
Continuous variables of the data sample exploited for nb glms across all countries

Variable	Origin	Description	Mean	Min	Max	St. Dev.	Missing
Mortality_rate_2021	WHO	WHO-estimated mortality rate per 100 000 population for 2021	13.52	1.30	37.40	7.97	1
Paved_km	WHO	Total paved kilometres (per year)	682353	19	34446633	4023764	50
Total_registered_veh	WHO	Total registered vehicles	17318493	1	395264200	52418634	34
Total_registered_veh_per 10^8pop	WHO	Total registered vehicles per 100 million people	21.89	0.12	77.31	19.16	34
Registered_cars_4wh	WHO	Cars and 4-wheeled light vehicles	12553191	1	259314638	38346298	74
Powered_2_3_wh	WHO	Powered 2- and 3-wheelers	4538809	62	121602401	15905632	64
Heavy_trucks	WHO	Heavy trucks	658507	19	14288000	2043860	72
Buses	WHO	Buses	141446	0	2196000	357161	72
UN_num	WHO	Adherence to one or more of the 7 UN road safety conventions	2.41	0.00	7.00	2.34	0
Helmet_wearing_rate	WHO	Helmet wearing rate (%Driver)	81.32	0.00	100.00	27.19	102
Safetybelt_percentage	WHO	Safety-belt wearing rate (% Drivers)	76.66	0.00	100.00	19.71	99
Male_percentage	WHO	Percentage of males	76.56	7.10	100.00	14.43	38
Population	WHO	Population	45384453	17003	1425893465	157843621	0
Imports_2011	ITF	Used vehicle imports for the year 2011	27838	3	274663	52539	72
Imports_2010_2016_avg	ITF	Used vehicle imports for the years 2010-2016 (average)	21419	5	246202	37351	73
Imports_2017_2021_avg	ITF	Used vehicle imports for the years 2017-2021 (average)	19746	4	187879	29967	73
Imports_2011_1000pop	ITF/WHO	Used vehicle imports for the year 2011 per 1000 people	18.26	0.00	1389.87	139.73	72
Imports_2010_2016_1000pop	ITF/WHO	Used vehicle imports for the years 2010-2016 (average) per 1000 people	30.16	0.00	2573.11	259.70	73
Imports_2017_2021_1000pop	ITF/WHO	Used vehicle imports for the years 2017-2021 (average) per 1000 people	21.25	0.00	1754.67	177.08	73

Source: WHO, 2023, ITF, 2024, & UNECE, 2025

Indicative observations from the above distributions are the following:

- The average estimated mortality rate per 100,000 population for 2021 was 13.52, with substantial variation (1.3 to 37.4) across countries, indicating different levels of road safety.
- The infrastructure-related variable of paved kilometres showed a broad range, with an average of 682,353 km but reaching up to over 34 million in some countries, which may influence road safety and vehicle usage rates.
- The average number of total registered vehicles was about 17.3 million, with variation in the number of vehicles per capita. Countries ranged from having only one registered vehicle to nearly 400 million, highlighting disparities in vehicle density.

5. RESULTS OF MACROSCOPIC ANALYSIS

- The dataset also includes specific types of vehicles like cars, light vehicles, and powered 2- or 3-wheelers. Notably, powered 2- and 3-wheelers, which are known to display higher crash risks, averaged around 4.5 million per country but varied widely.
- Countries adhered to different levels of UN road safety conventions (average adherence was 2.41 out of 7 for Safety Conventions and Equipment Use), while helmet and safety belt usage varied, with average rates of 81.32% and 76.66%, respectively. The large standard deviations in these variables suggest inconsistency in safety measures and practices across countries.

The overview of continuous variables reflects significant variability in road safety indicators, infrastructure, and vehicle types, which may affect crash and mortality rates both due to used vehicles and overall.

For binary discrete (categorical) variables, results are presented in Table 5.3 (row percentages in parentheses):

Table 5.3
Discrete variables of the data sample exploited for nb glms across all countries

Variable	Origin	Description	Yes/TRUE	No/FALSE	Missing
UN_Hi_Qual_Std	WHO	Presence of high-quality safety standards for used-vehicle imports/exports 2	113 (66.08%)	45 (26.32%)	13 (7.60%)
UN_atleast_1	WHO	Adherence to at least 1 of the 7 UN road safety conventions	128 (74.85%)	43 (25.15%)	0 (0.00%)
UN_atleast_3	WHO	Adherence to at least 3 of the 7 UN road safety conventions	123 (71.93%)	48 (28.07%)	0 (0.00%)
safetybelt_above_70	WHO	Safety-belt wearing percentage above 70% (yes/no)	68 (25.19%)	103 (38.15%)	99 (36.67%)

Source: WHO, 2023, and author's independent processing

Indicative observations from the above distributions are the following:

- A substantial majority of countries (74.85%) adhere to at least one UN road safety convention, with 71.93% meeting at least three, indicating that a more recognition of international safety frameworks has started gaining ground.
- Nearly 37% of data on seatbelt compliance and 7.60% on vehicle standards compliance are missing, suggesting challenges in obtaining consistent safety data across all nations.
- Only 25.19% of countries report a seatbelt usage rate above 70%, revealing significant room for improvement in enforcing or promoting safety belt safety regulations.
- About two-thirds of the countries (66.08%) maintain high-quality safety standards for used-vehicle imports and exports, reflecting a moderate emphasis on vehicle safety in the international trade of used vehicles.
- Consistent adherence rates across the UN conventions categories (around 70%) may indicate similar factors influencing international safety standard adoption.

The above presented variables were selected on the basis of data availability and completeness in international data sources. **All variables were tested for their statistical significance; in the following section, only statistically significant modelling results are presented.**

5.2.2. Model for all countries

The NB GLM for all countries is presented in the following tables. The rounded Mortality Rates of 2021 are set as the independent variable for the NB model. Out of the 171 countries reported by WHO, 1 did not report a mortality rate. Moreover, 34 countries had missing data regarding total vehicle registrations. In addition, 73 countries had missing values of imports based on data from ITF (ITF, 2024). Regarding safety belt percentages, 99 countries displayed missing values. However, to avoid extreme data thinning, a decision was made to impute the missing safety belt percentage data based on the average value of each WHO region. After obtaining the missing values via imputation, safety belt percentages were fully available for modelling across all countries.

Following all preprocessing steps, the final model was estimated with **79 countries with full data entries**.

Table 5.4

Parameter Estimates of the Negative Binomial Model of Mortality Rates – All Countries

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
<i>Intercept</i>	2.784	0.110	25.326	< 0.001	***
<i>Imports_2017_2021_1000pop</i>	0.0002	0.000	1.620	0.105	.
<i>Total_registered_veh_per 10^8pop</i>	0.005	0.003	2.097	0.036	*
<i>safetybelt_above_70</i> _[ref: below 70]	-0.169	0.093	-1.824	0.068	.
<i>UN_atleast_1_TRUE</i> _[ref: false]	0.008	0.104	0.077	0.939	

NB dispersion parameter: Theta: 9.27, Std. Err.: 2.31.

Source: UNECE, 2025

In other words, the Theta parameter helps account for overdispersion in the data—meaning that the variance of mortality rates is higher than what would be expected in a simple Poisson model, where variance equals the mean. Here, Theta is estimated at 9.27, which indicates a moderate level of overdispersion. This is because a higher Theta value in NB models suggests that the overdispersion becomes less pronounced in the data.

The standard error of Theta (2.31) shows some degree of uncertainty in this estimate, implying that Theta's true value could vary around this estimate. However, the value is precise enough to provide reliable insights about the model (since it is about 25% of the Theta value), suggesting that the Negative Binomial approach is appropriate for handling the variability in mortality rates across countries. Therefore, the model accounts for moderate overdispersion, and the standard error shows some uncertainty, but the estimate is still reliable enough for modelling purposes.

Table 5.5

Variance Inflation Factors (VIF) AND Marginal Effects at the Mean – All Countries

Variable	VIF	MEM
<i>Imports_2017_2021_1000pop</i>	1.066	0.004
<i>Total_registered_veh_per 10^8pop</i>	1.018	0.091
<i>safetybelt_above_70</i> _[ref: below 70]	1.020	-2.903
<i>UN_atleast_1_TRUE</i> _[ref: false]	1.051	0.136

SOURCE: UNECE, 2025

As per the previous model, given that all VIF values are very close to 1, and thus very low, it can be ascertained that the independent variables are not mutually correlated. Moreover, there is a significant constant in the model. Therefore the model results are unbiased and can be considered robust for interpretation. In particular:

- The variable of **Imports of Used Vehicles (2017-2021) per 1,000 population has a positive influence on mortality rates**, however the effect is marginally statistically significant at the 90% level. This finding suggests that higher levels of vehicle imports loosely correlate with increased mortality rates, highlighting a potential risk associated with imported used vehicles in all countries.
- **The total registered vehicles per 10⁸ of population (100 million people) has a positive and statistically significant effect on mortality rates** at a higher level, namely 95%. This finding suggests that higher registered vehicle numbers are correlated with increased mortality rates in all countries.
- The categorical variable expressing **Safety belt Usage Above 70% shows a negative effect on mortality rates at the 95% level**. The coefficient implies that high safety belt compliance has a protective effect, likely due to its effectiveness in reducing fatalities in road traffic incidents.

- **The adherence to at least one UN road safety convention does not impact mortality** on a statistically significant level, which may imply that simply meeting one convention might not suffice for substantial safety improvements.

Based on the Marginal Effects of the above model parameters, it was estimated that:

- Each 1 unit increase of average used vehicle imports of the most recent 5 years ago per 1000 population (years 2017-2021) leads to an increase of about 0.004 of the crash mortality rate.
- Each 1 unit increase of total registered vehicles per 10⁸ of population leads to an increase of about 0.091 of the crash mortality rate.
- Safety belt compliance above 70% decreases mortality rates by more than 2.9 on average.

5.2.3. Model for LMICs

The Negative Binomial (NB) GLM for LMICs is presented in the following tables. The rounded Mortality Rates of 2021 are set as the independent variable for the NB model. Out of the 75 LMICs reported by WHO, all reported a mortality rate. However, 16 LMICs had missing data regarding total vehicle registrations. In addition, 14 LMICs had missing values of imports based on data from ITF (ITF, 2024). Regarding safety belt percentages, 47 LMICs displayed missing values. As in the previous model, to avoid extreme data thinning, a decision was made to impute the missing safety belt percentage data based on the average value of each WHO region. After obtaining the missing values via imputation, safety belt percentages were fully available for modelling across all LMICs.

Following all preprocessing steps, the final model was estimated with **44 LMICs with full data entries**.

Table 5.6
Parameter Estimates of the Negative Binomial Model - LMICs

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
<i>Intercept</i>	2.555	0.131	19.463	< 0.001	***
<i>Imports_2017_2021_1000pop</i>	0.024	0.008	2.917	0.004	**
<i>Total_registered_veh_per 10^8pop</i>	0.004	0.003	1.667	0.096	.
<i>safety belt_above_70[ref: below 70]</i>	0.224	0.103	2.184	0.029	*
<i>UN_atleast_1_TRUE[ref: false]</i>	0.121	0.106	1.143	0.253	

NB dispersion parameter: Theta: 19.29, Std. Err.: 8.82.

Source: UNECE, 2025

In the case of LMICs, Theta has a relatively higher value of 19.29, indicating that the model recognizes and adjusts for moderate overdispersion in the data, helping to fit the variability better than a standard Poisson model would. The standard error of Theta is 8.82. While this shows a small increase of uncertainty around the Theta estimate, it remains small enough relative to Theta (since it is about 46% of the Theta value), to provide confidence that the NB model adjustment for overdispersion is reliable. Therefore, the model has accounted for overdispersion, with a Theta value showing moderate dispersion, and the standard error indicates a reasonable degree of uncertainty in estimating that dispersion.

Table 5.7

Variance Inflation Factors (VIF) AND Marginal Effects at the Mean – LMICs

Variable	VIF	MEM
<i>Imports_2017_2021_1000pop</i>	1.068	0.476
<i>Total_registered_veh_per 10^8pop</i>	1.014	0.086
<i>safety_belt_above_70[ref: below 70]</i>	1.023	4.233
<i>UN_atleast_1_TRUE[ref: false]</i>	1.057	2.318

Source: UNECE, 2025

Given that all VIF values are very close to 1, and thus very low, it can be ascertained that the independent variables are not mutually correlated. Low VIF values mean these predictors independently contribute to understanding variations in mortality rates, supporting robust interpretation of the coefficients. Moreover, there is a significant constant in the model. Therefore the model results are unbiased and can be considered robust for interpretation. In particular:

The variable of **Imports of Used Vehicles (2017-2021) per 1,000 population has a positive and statistically significant effect on mortality rates** at the 95% level. This finding suggests that higher levels of vehicle imports correlate with increased mortality rates in LMICs, highlighting a potential risk associated with imported used vehicles in LMICs. It is reminded that, when departing from a LMIC-focused analysis to all-countries analysis (section 5.2.2), the impact expressed by this variable is diluted, likely by socioeconomic factors, e.g. the fact that the exporting countries are also included in the dataset.

- The **total registered vehicles per 10⁸ of population (100 million people) also has a positive and statistically significant effect on mortality rates** at the 90% level. This finding suggests that higher registered vehicle numbers are correlated with increased mortality rates in LMICs.
- The categorical variable expressing **Safety belt Usage Above 70%** shows a positive effect on mortality rates at the 95% level. This counterintuitive finding might result from improved crash fatality reporting in countries with stricter regulations, or much higher exposure to road crashes due to higher motorization in these countries, and not necessarily a causal relationship with crash mortality levels. It is reminded that, when expanding to envelop all countries, the increased and more representative sample of observations serves to clarify the role of this variable (i.e. a negative association between safety belt wearing rates and traffic mortality around the world).
- The **adherence to at least one UN road safety convention** does not impact mortality on a statistically significant level, which may imply that simply meeting one convention might not suffice for substantial safety improvements.

Finally, based on the Marginal Effects of the above model parameters, it was estimated that:

- Each 1 unit increase of average used vehicle imports of the most recent 5 years ago per 1000 population (years 2017-2021) leads to an increase of about 0.476 of the crash mortality rate in LMICs.
- Each 1 unit increase of total registered vehicles per 108 of population leads to an increase of about 0.086 of the crash mortality rate.
- A LMIC having safety belt usage above 70% has appears to increase mortality rates by more than 4.2 on average. This finding is a direct implication of over-reporting safety-belt use in those few countries where there is a reported figure, and it should be considered invalid¹⁴⁴.

¹⁴⁴ The variable is kept in the model for model estimation stability.

6. CROSS SECTIONAL BEFORE-AND-AFTER ANALYSIS – CASE STUDIES

6.1. Case study 1: Malaysia

General background

Malaysia, a country in Southeast Asia, is a federal constitutional monarchy comprising 13 states and three federal territories. It belongs to the Upper Middle income country group, and in the Western Pacific WHO region. Based on 2021 data, Malaysia suffered 4539 fatalities (85% male, 15% female). The country has a total of 2243 paved kilometres, while no data are available for total vehicle registrations (WHO, 2023).

Road safety context and development

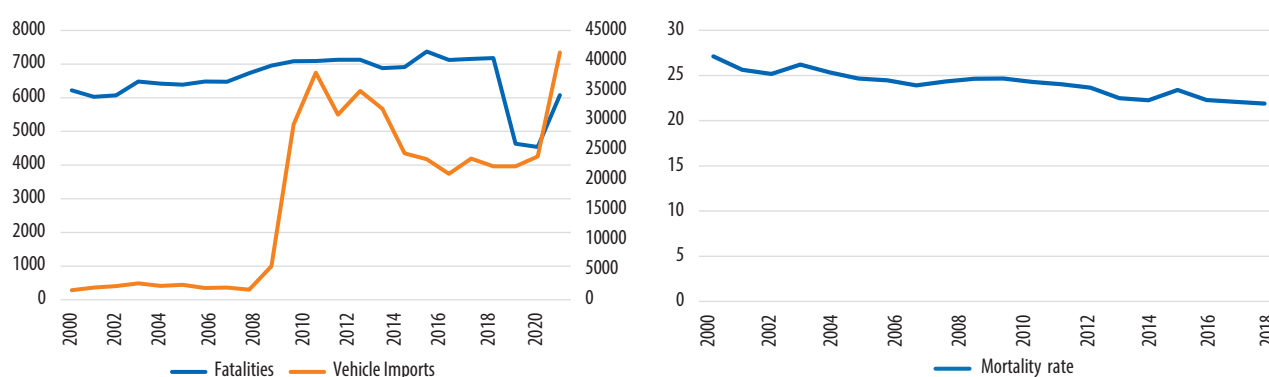
Malaysia has a national law on universal access to emergency care, but does not have a national law guaranteeing free-of-charge access to rehabilitative care for all injured individuals. Moreover, the country has a substantial presence of strategies to promote alternatives to individuals use of powered vehicles (WHO, 2023).

Malaysia has several road safety and transport systems in place. The country has a Civil Registration and Vital Statistics system as of 2021. Furthermore, Malaysia partially monitors the frequency and distribution of journeys by modal type. There are systems in place to track speeding violations and speeding-related injuries and fatalities. Malaysia also addresses driving under the influence of alcohol or drugs, along with related road traffic fatalities and injuries. The country has enforcement for powered two- and three-wheeler helmet use and regulations concerning mobile phone use while driving. However, Malaysia does not have systems in place for monitoring safety belt and child-restraint system use (WHO, 2023).

Figure 6.1 shows the trends of traffic fatalities and imported used vehicles between 2000-2022. It is noted that used vehicles imports were practically initiated in 2008, with a steep increase until 2010. Afterwards, imports showed a decreasing trend, with a steep increase after 2020 (possibly also related to the Covid-19 pandemic). At the same time, traffic fatalities showed a mildly increasing trend, and were obviously affected by the Covid-19 pandemic on 2020-2021.

Figure 6.1

Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Malaysia, 2000-2022



Source: WHO, 2023; ITF, 2020

Used vehicles policies

In Malaysia, importing used vehicles older than five years is restricted since 2017, and vehicles can only be imported for personal use—not for sale or transfer. The vehicle must be exported when employment ends in the country. To import a car, an Approval Permit (AP) from the Ministry of International Trade and Industry (MITI) must be obtained, and the car must be registered under the owner's name for at least three years before applying for the permit¹⁴⁵.

Control group

A number of Southeast Asian countries were considered as a control group for this case study. However, none of them was suitable, because most of the countries applied similar regulation: Myanmar, Thailand, Vietnam and Philippines also regulated vehicle imports on 2017-2018, while Indonesia has a total ban of used vehicles imports. Therefore, it was decided to consider India as the control group, being a country large enough to control for regression-to-the-mean of fatalities counts, and at the same time having adequately similar socioeconomic indicators (see Table 6.1). India regulated used vehicle imports only in 2024, therefore serves as a suitable control group.

Table 6.1

Selection of Control Group on the Basis of Socioeconomic and Geographical Indicators – Malaysia

Indicators	Malaysia	India
<i>Population density</i>	108	488
<i>Share of population living in urban areas (%)</i>	78.7	36.4
<i>GDP per capita</i>	33,434	8,379
<i>Income group</i>	Upper middle income	Lower middle income
<i>Road network density (km/sq.km)</i>	85	204
<i>Share of paved roads (%)</i>	71	67
<i>Vehicle ownership (1000 inhabitants)</i>	572	57
<i>PTW ownership (1000 inhabitants)</i>	447	157
<i>Mean age of the population (years)</i>	31.8	29.8
<i>Mean annual precipitation (mm/year)</i>	2,875	1,083
<i>Mean annual temperature (°C)</i>	26	25
<i>HDI (Human Development Index)</i>	0.807	0.644
<i>Hospital beds per population (1000 inhabitants)</i>	2.0	1.6

Source: WHO, 2023

Before-and-after analysis

The results of before-and-after analysis for the introduction of used vehicle imports regulations in 2017 is presented in Table 6.2). Treatment year is 2017, with a 5-year before and a 5-year after period. It is noted that, for the after period, data for the period 2020-2022 were collected for Malaysia and India from national web resources¹⁴⁶, ¹⁴⁷. A cross-checking was carried out to ensure that the data are comparable with the WHO data of the rest of the study period; in case of discrepancy, the additional national data were corrected through a calculated coefficient to align with the estimates of WHO data.

¹⁴⁵ <https://aaajapan.com/information/car-import-rules-from-japan/malaysia>

¹⁴⁶ <https://www.carz.com.my/2023/06/915874-road-accidents-recorded-from-2021-2022-ministry-of-transport>

¹⁴⁷ https://morth.nic.in/sites/default/files/RA_2022_30_Oct.pdf

Table 6.2
Before-and-After Analysis Results - Malaysia

	Before	After
Number of fatalities	2012-2016	2018-2022
Treatment group	42,519	29,587
Control group	1,251,299	1,033,461
CMF	Lower bound	Upper bound
0.843	0.832	0.853
Safety effect (% reduction)		
15.75	14.68	16.81
Significant effect - positive		

Source: UNECE, 2025

Table 6.2 shows that the policy resulted in a Crash Modification Factor (CMF) of 0.843, corresponding to a 15.75% reduction in fatalities in Malaysia. The upper and lower bounds of the estimate are both positive, which is statistically significant¹⁴⁸ at 95% confidence level.

These findings suggest a positive effect of the policy on road safety. However, they should be considered with some reservation, because the policy was introduced rather recently, and the available data may not reflect the full impact of the change. It is well understood that this type of policies may need a 3–5-year time lag before an actual change in fatalities can be visible. It is recommended that this case study is repeated when data after the treatment are fully available and validated by WHO.

6.2. Case study 2: Pakistan

General background

Pakistan, a country in South Asia, is a federal parliamentary Islamic republic comprising 4 Provinces, 1 Federally Administered Territory and 2 Autonomous Territories. It belongs to the Lower Middle income country group, and in the Eastern Mediterranean Region WHO region. Based on 2021 data, Pakistan suffered 5816 fatalities (gender distribution not available). The country has a total of 506,706 paved kilometres, while total vehicle registrations amount to 34,301,159 (or 14,823.2 per 100,000 people) (WHO, 2023).

Road safety context and development

Pakistan does not have either a national law on universal access to emergency care or a national law guaranteeing free-of-charge access to rehabilitative care for all injured individuals. Moreover, the country lacks strategies to promote alternatives to individuals use of powered vehicles (WHO, 2023).

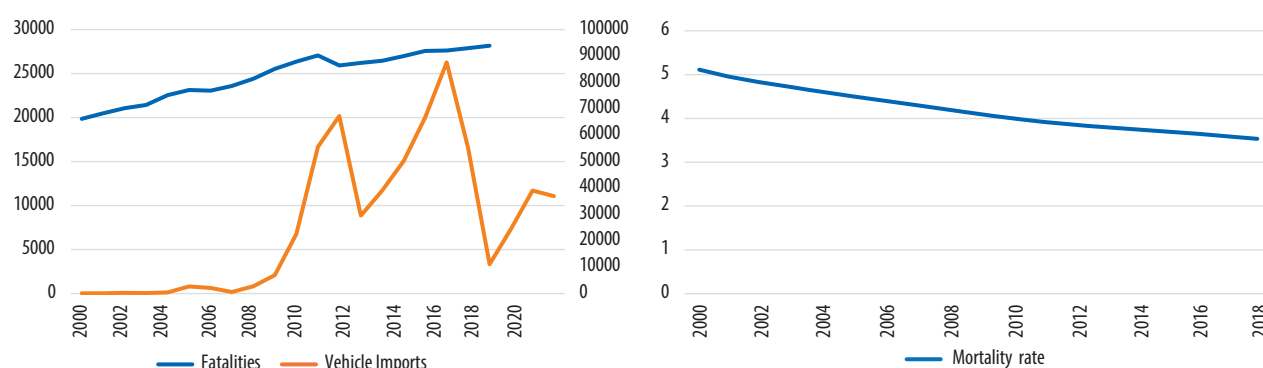
Pakistan has some road safety and transport systems in place. The country has a Civil Registration and Vital Statistics system as of 2021. Furthermore, Pakistan partially monitors the frequency and distribution of journeys by modal type. There are systems in place to track speeding violations and speeding-related injuries and fatalities. However, Pakistan does not have systems in place for monitoring driving under the influence of alcohol or drugs, along with related road traffic fatalities and injuries. The country does not monitor neither the use of safety belt and child-restraint systems, nor powered two- and three-wheeler helmet use nor regulations concerning mobile phone use while driving (WHO, 2023).

¹⁴⁸ For a statistically significant estimate, both the upper and lower bound should be within the positive values range.

Source: WHO, 2023 shows the trends of traffic fatalities and imported used vehicles between 2000-2022. It is noted that used vehicles imports were practically initiated on 2005-2006, and were boosted since 2010-2012, with a steep increase until 2018. The subsequent decrease trend is probably related to the Covid-19 pandemic. At the same time, traffic fatalities showed a mildly increasing trend over the whole examined period. It is interesting to note, however, in the right panel of Source: WHO, 2023, that the mortality rate showed a steadily decreasing trend, as a result of the country's population increasing far more rapidly than the traffic fatalities.

Figure 6.2

Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Pakistan, 2000-2022



Source: WHO, 2023 ; ITF, 2020

Used vehicles policies

In 2006, the import of used cars in Pakistan has been extended to vehicles up to three years old from the date of manufacturing, instead of the previous two years limit for overseas Pakistanis. All vehicles imported must be right hand drive (RHD)¹⁴⁹. It is thereby noted that Pakistan had a regulation for used vehicles before the study year, however the imports of vehicles were very limited before that date. The case study therefore serves to demonstrate the potential impact of early regulation of used vehicles imports.

Control group

India was selected as the control group, being a country large enough to control for regression-to-the-mean of fatalities counts, and at the same time having adequately similar socioeconomic indicators (see Table 6.3). India regulated used vehicle imports only in 2024, therefore serves as a suitable control group.

¹⁴⁹ <https://blog.japanesecartrade.com/87-pakistan-import-regulation-for-japan-used-cars/>

Table 6.3

Selection of Control Group on the Basis of Socioeconomic and Geographical Indicators – Pakistan

Indicators	Pakistan	India
Population density	326	488
Share of population living in urban areas (%)	38.0	36.4
GDP per capita	6,437	8,379
Income group	Lower middle income	Lower middle income
Road network density (km/sq.km)	30	204
Share of paved roads (%)	70	67
Vehicle ownership (1000 inhabitants)	69	57
PTW ownership (1000 inhabitants)	75	157
Mean age of the population (years)	22.9	29.8
Mean annual precipitation (mm/year)	524	1,083
Mean annual temperature (°C)	21.4	25
HDI (Human Development Index)	0.54	0.644
Hospital beds per population (1000 inhabitants)	0.5	1.6

Source: WHO, 2023

Before-and-after analysis

The results of before-and-after analysis for the introduction of used vehicle imports regulations in Pakistan is presented in Table .4. Treatment year is 2005, with a 5-year before and a 5-year after period. All the relevant data were available from WHO.

Table 6.4

Before-and-After Analysis Results – Pakistan

	Before	After
Number of fatalities	2000-2005	2007-2011
Treatment group	128,518	126,969
Control group	1,020,005	1,013,892
CMF	Lower bound	Upper bound
0.994	0.987	1.001
Safety effect (% reduction)		
0.61	-0.08	1.29
<i>Non-significant</i>		

Source: UNECE, 2025

Table 6.4 shows that the policy resulted in a Crash Modification Factor (CMF) of ~1, corresponding to a 0% reduction in fatalities in Pakistan. The upper and lower bounds of the estimate are [-0.08, 1.29], which is non-significant¹⁵⁰ at 95% confidence level.

¹⁵⁰ For a statistically significant estimate, both the upper and lower bound should be within the positive values range.

A separate analysis was conducted by considering a 5-year time lag in the impact of the policy. The results are presented in Table 6.5. In this case, the 'after' period is taken to be 2011-2015. Table 6.5 shows that the policy resulted in a Crash Modification Factor (CMF) of 1, corresponding to a practically 0% reduction in fatalities. The upper and lower bounds of the estimate are [-0.07, 0.61], which is non-significant.

Table 6.5
Before-and-After Analysis Results with 5-Year Time Lag in the Effect of Policy – Pakistan

	Before	After
Number of fatalities	2000-2005	2011-2015
Treatment group	128,518	132,652
Control group	1,020,005	1,052,066
CMF	Lower bound	Upper bound
1.001	0.987	1.001
Safety effect (% reduction)		
	-0.07	0.61
	Non-significant	

Source: UNECE, 2025

These findings suggest that there was no effect of the policy on road safety. It is likely that the influx of imported used vehicles after the regulation year, together with the steep growth of the country's population, resulted in increased level of motorisation and increased traffic exposure in the country. Consequently, it is possible that the age limit policy was successful in preventing a significant worsening of road safety from the increase of used vehicles imports, but could not counterbalance the other transport and traffic consequences that resulted from it.

6.3. Case study 3: Egypt

General background

Egypt, a transcontinental country located mainly in Northern Africa (and partially on Asia) is a unitary semi-presidential republic, divided into 27 administrative regions known as governorates. It belongs to the Lower Middle income country group, and in the Eastern Mediterranean WHO region. Based on 2021 data, Egypt suffered 7101 fatalities (83% male, 17% female). The country has a total of 125,312 paved kilometres, while total vehicle registrations amount to 10,909,456 (or 9984.7 per 100,000 people) (WHO, 2023).

Road safety context and development

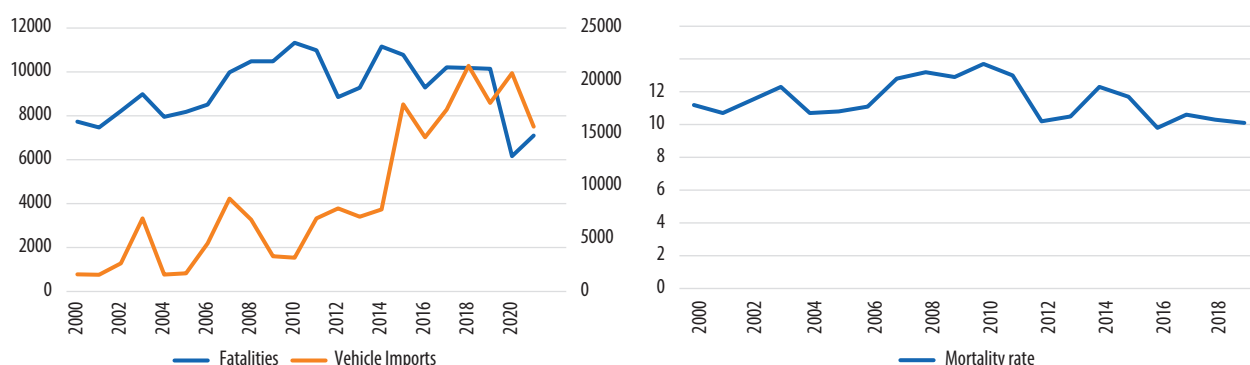
Egypt has a national law on universal access to emergency care, but does not have a national law guaranteeing free-of-charge access to rehabilitative care for all injured individuals. Moreover, the country lacks strategies to promote alternatives to individuals use of powered vehicles (WHO, 2023).

Egypt has several road safety and transport systems in place. The country has a Civil Registration and Vital Statistics system as of 2021. Egypt has systems in place for monitoring driving under the influence of alcohol or drugs, along with related road traffic fatalities and injuries. There are also systems in place to track speeding violations and speeding-related injuries and fatalities. The country monitors powered two- and three-wheeler helmet use and mobile phone use while driving. However, Egypt does not monitor the frequency and distribution of journeys by modal type or the safety belt and child-restraint systems use (WHO, 2023).

Figure 6.3 shows the trends of traffic fatalities and imported used vehicles between 2000-2022. It is noted that used vehicles imports showed an increasing trend over the years, with a lot of fluctuation, and were boosted between 2014-2018 (regulation was introduced on 2017). At the same time, traffic fatalities and the mortality rate showed a fluctuating trend over the whole examined period, with a notable decrease only during the Covid-19 pandemic.

Figure 6.3

Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Egypt, 2000-2022



Source: WHO, 2023; ITF, 2020

Used vehicles policies

Used passenger vehicles can only be imported within one year of their manufacture date, while electric vehicles may be imported up to three years after manufacture. There are small exceptions for returning Egyptian expatriates (UNEP, 2017). Vehicles must be from the same year as the importation, and the importer must be the original owner. All diesel vehicles are prohibited. Prior authorization is required before importing any vehicle¹⁵¹.

Control group

Defining a control group for the Egypt case study was challenging. On the one hand, its socioeconomic indicators are higher than those of most African and Northern African countries. On the other hand, it is a very large country with high population and high numbers of fatalities, and finding a control group of sufficient size was not straightforward. Table 6.6 shows a comparison of the geographical and socioeconomic indicators of Egypt vs. those of several neighbouring countries (i.e. Morocco, Algeria and Chad) that did not regulate used vehicles imports within the study period, as well as vs. those of India. It is noted that for several indicators, Egypt is closer to India than to its neighbouring countries, while the opposite is the case for some other indicators. It was decided to conduct two before-and-after analyses, one with India as the control group, and another one with the 3 neighbouring countries as control group.

¹⁵¹ <https://blog.japanesecartrade.com/45-egypt-import-regulation-for-japan-used-cars/>

Table 6.6

Selection of Control Group on the Basis of Socioeconomic and Geographical Indicators – EGYPT

Indicators	Egypt	India	Morocco	Algeria	Chad
Population density	117	488	85	20	16
Share of population living in urban areas (%)	41.3	36.4	65.1	75.3	24.4
GDP per capita	15,091	8,379	9,519	13,210	1,168
Income group	Lower middle income	Lower middle income	Lower middle income	Lower middle income	Low income
Road network density (km/ sq.km)	32	204	8	6	3
Share of paved roads (%)	74	67	79	92	63
Vehicle ownership (1000 inhabitants)	70	57	111	184	74
PTW ownership (1000 inhabitants)	31	157	5	3	0
Mean age of the population (years)	24.4	29.8	30.6	29.1	16.7
Mean annual precipitation (mm/year)	18	1,083	346	89	322
Mean annual temperature (oC)	23.1	25	18.4	23.6	27.6
HDI (Human Development Index)	0.728	0.644	0.698	0.745	0.394
Hospital beds per population (1000 inhabitants)	1.1	1.6	0.7	1.6	0.5

Source: WHO, 2023

Before-and-after analysis

The results of before-and-after analysis for the introduction of used vehicle imports regulations in Egypt is presented in Table 6.7. Treatment year is 2017, with a 5-year before and a 5-year after period. It is noted that, for the after period, data for the period 2020-2022 were collected for Egypt, Morocco and India from national web resources^{152, 153}. A cross-checking was carried out to ensure that the data are comparable with the WHO data of the rest of the study period; in case of discrepancy, the additional national data were corrected through a calculated coefficient to align with the estimates of WHO data. It was not possible to collect any recent data (after 2020) for Algeria and Chad, therefore their fatalities trend was considered to remain unchanged after 2020.

Table 6.7 shows that, with India as a control group, the policy resulted in a Crash Modification Factor (CMF) of 0.674, corresponding to a 32.6% reduction in fatalities in Egypt. The upper and lower bounds of the estimate are both positive, which is statistically significant¹⁵⁴ at 95% confidence level. With the 3 neighbouring countries as a control group, the policy resulted in a very similar Crash Modification Factor (CMF) of 0.646, corresponding to a 35.3% reduction in fatalities in Egypt, an estimate which is statistically significant too.

These findings suggest a very positive effect of the policy on road safety. However, because the policy was introduced rather recently, and the available data may not reflect the full impact of the change, it is recommended that this case study is repeated when data after the treatment are fully available and validated by WHO.

¹⁵² <https://english.ahram.org.eg/NewsContent/1/1235/472098/Egypt/Urban--Transport/Traffic-injuries-down-by--in-Egypt-in--CAPMAS.aspx>

¹⁵³ https://morth.nic.in/sites/default/files/RA_2022_30_Oct.pdf

¹⁵⁴ For a statistically significant estimate, both the upper and lower bound should be within the positive values range.

Table 6.7
Before-and-After Analysis Results - Egypt

	Control group (i): India			Control group (ii): Morocco, Algeria & Chad	
	Before	After		Before	After
Number of fatalities	2012-2016	2018-2022	Number of fatalities	2012-2016	2018-2022
Treatment group	60,347	33,584	Treatment group	60,347	33,584
Control group	1,251,299	1,033,461	Control group	92,504	79,637
CMF	Lower bound	Upper bound	CMF	Lower bound	Upper bound
0.674	0.666	0.682	0.646	0.638	0.655
Safety effect (% reduction)					
32.62	31.84	33.38	35.36	34.46	36.24
Significant effect - positive			Significant effect - positive		

Source: UNECE, 2025

6.4. Case study 4: Romania

General background

Romania, located in Eastern Europe, is a unitary semi-presidential republic divided into 41 administrative counties and the municipality of Bucharest, which functions as a separate administrative region. Romania is classified within the Upper Middle-Income country group by the World Bank and belongs to the WHO European Region. According to recent data, Romania recorded 1,778 road fatalities in 2021, with a distribution of approximately 76% male and 24% female among the fatalities. The country has an estimated 86,336 kilometres of paved roads. Romania has 9,422,452 registered vehicles¹⁵⁵, or around 49,424 vehicles per 100,000 people (INS, 2022; WHO, 2023).

Road safety context and development

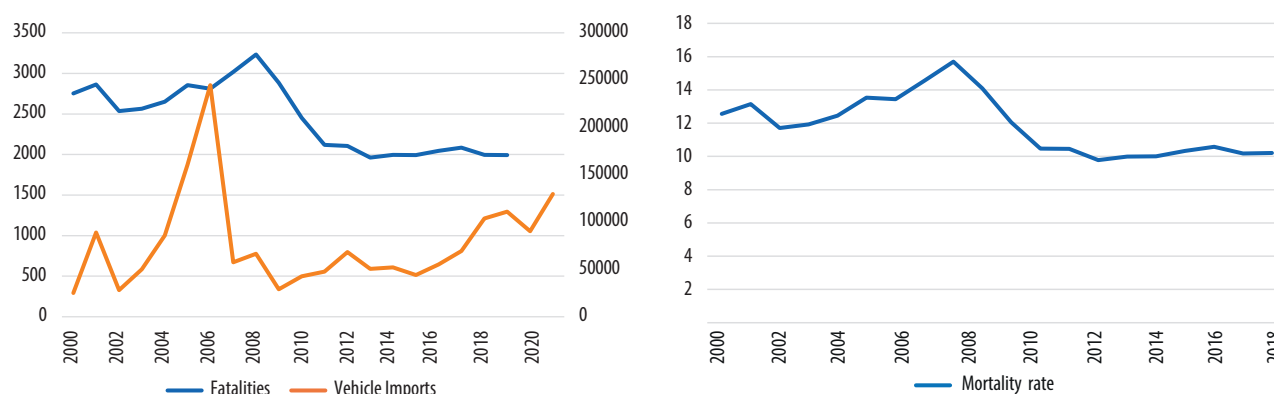
Romania has one of the oldest passenger car fleets in the EU. Moreover, Romania is one of the few countries in the European Union with a zero-alcohol limit for all drivers. The Romanian road infrastructure is characterised by low road density. The per capita GDP in Romania is below the EU average, but the percentage of GDP dedicated for road infrastructure is higher than the percentage observed in the EU (European Commission, 2023).

Road fatalities presented an increasing trend in Romania before 2007, followed by a sharp reduction thereafter. However, since 2012 fatalities have plateaued in the country and no significant reduction has been reported. Used vehicle imports boosted between 2002-2006, however after the country's accession to the EU, and the adoption of relevant UN core vehicle regulations, the number of these imports had a significant drop. An increasing trend on imported used vehicles in observed after 2015.

¹⁵⁵ <https://www.ceicdata.com/en/indicator/romania/number-of-registered-vehicles>

Figure 6.4

Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Romania, 2000-2022



Source: WHO, 2023; ITF, 2020

Used vehicles policies

Romania has been reported to impose an 8-year age limit to used vehicle imports. A VAT of 22% has to be paid and, for cars originating outside the European Community, a 30% duty has to be paid as well¹⁵⁶.

Romania allows the import of used cars that meet at least Euro 4 emission standards. While Euro 4 is the minimum, vehicles meeting higher standards (Euro 5, Euro 6) are increasingly preferred and may face lower taxes. All imported vehicles must be right-hand drive to be road-legal in Romania. Left-hand drive vehicles are not permitted for regular use on Romanian roads, which is an important consideration when sourcing cars for import.

Cars that have been previously declared as total losses or have undergone major structural repairs may face difficulties during the import process. Romanian authorities require a clean vehicle history to ensure road safety. All imported vehicles must undergo a technical inspection in Romania before they can be registered. This inspection ensures that the vehicle meets Romanian road safety standards. While a recent inspection from another EU country can be helpful, it doesn't exempt the vehicle from undergoing the Romanian inspection process¹⁵⁷.

Control group

Initially, Russia was considered as a large and representative enough control group for the Romania case study. Although Russia is an immense country, with a low population density and road density, the cultural and political links between the two countries in their history can counterbalance their differences on certain transport infrastructure, geography and climate indicators (see Table 6.8).

¹⁵⁶ <https://blog.japanesecartrade.com/552-romania-import-regulation-for-japan-used-cars/>

¹⁵⁷ <https://ecarstrade.com/blog/how-to-import-a-car-to-romania>

Table 6.8

Selection of Control Group on the Basis of Socioeconomic and Geographical Indicators – Romania

Indicators	Romania	Russia
Population density	83	9
Share of population living in urban areas (%)	54.7	75.3
GDP per capita	41,888	36,485
Income group	Upper middle income	Upper middle income
Road network density (km/sq.km)	36	9
Share of paved roads (%)	73	44
Vehicle ownership (1000 inhabitants)	477	361
PTW ownership (1000 inhabitants)	6	16
Mean age of the population (years)	45.5	41.9
Mean annual precipitation (mm/year)	637	460
Mean annual temperature (°C)	10.18	-3.8
HDI (Human Development Index)	0.827	0.821
Hospital beds per population (1000 inhabitants)	7.1	7

Source: WHO, 2023

Before-and-after analysis

The results of before-and-after analysis for the introduction of UN core vehicle regulations as a result of the EU accession in 2007, are presented in

Table 6.9. Treatment year is 2007, with a 5-year before and a 5-year after period. All the relevant data were available from WHO.

Table 6.9

Before-and-After Analysis Results – Romania

	Before	After
Number of fatalities	2001-2006	2008-2012
Treatment group	16,280	12,791
Control group	253,734	254,603
CMF	Lower bound	Upper bound
0.783	0.765	1.019
Safety effect (% reduction)		
21.70	22.12	23.25
Significant effect - positive		

Source: UNECE, 2025

Table 6.9 shows that the policy resulted in a Crash Modification Factor (CMF) of 0.783, corresponding to a ~22% reduction in fatalities in the 'after' period. The upper and lower bounds of the estimate are [22.12, 23.25], which is highly marginally significant¹⁵⁸ at 95% confidence level.

A separate analysis was conducted by considering a 5-year time lag in the impact of the policy. The results are presented in Table 6.10. In this case, the 'after' period is taken to be 2011-2015. Table 6.10 shows that the policy resulted in a Crash Modification Factor (CMF) of 1.11, corresponding to an increase of 11% in fatalities. The upper and lower bounds of the estimate are [-8.64, -13.6], which is significant. This is a counter-intuitive finding, which suggests that 5 years after the EU accession, road safety started to deteriorate.

Table 6.10
Before-and-After Analysis Results with 5-year Time Lag in the Effect of Policy – Romania

	Before	After
Number of fatalities	2001-2006	2011-2015
Treatment group	13,469	10,174
Control group	214,308	145,713
CMF	Lower bound	Upper bound
1.11	1.082	1.030
Safety effect (% reduction)		
-11.10	-8.64	-13.6
Significant effect - negative		

Source: UNECE, 2025

The two analyses combined, suggest that there was a significant effect of EU accession on road safety, which can be certainly attributed to the adoption of UN core vehicle regulations and the regulation of used vehicle imports, to a large extent. This can be confirmed by the sharp decrease in the number of imported used vehicles as of 2007. Therefore, it can be concluded that there was an immediate effect of the policy on road safety. On the other hand, it is likely that the economic growth that followed the EU accession resulted in increased traffic, and eventually increased traffic fatalities. Unfortunately there is no traffic data available for Romania to enable validation of this interpretation.

6.5. Case study 5: Georgia

General background

Georgia is a transcontinental country in Eastern Europe and West Asia. It is a Unitary parliamentary republic comprising 9 administrative regions and 2 autonomous republics, as well as the capital with special administrative status. Georgia belongs to the Upper Middle income country group, and in the European WHO region. Based on 2021 data, Georgia suffered 449 fatalities (77% male, 23% female). The country has a total of 6970 paved kilometres, while total vehicle registration data is not available (WHO, 2023).

Road safety context and development

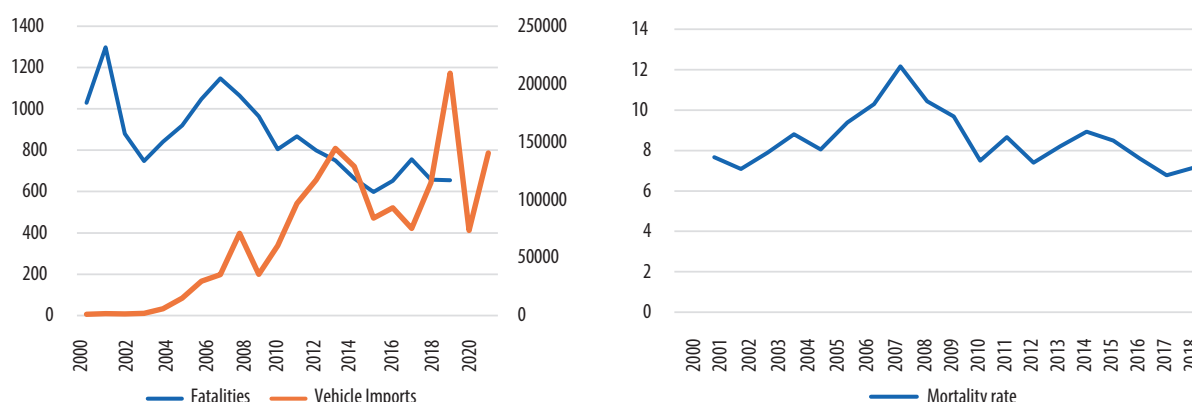
Georgia has a national law on universal access to emergency care, but does not have a national law guaranteeing free-of-charge access to rehabilitative care for all injured individuals. The country has adopted national strategies to promote alternatives to individuals use of powered vehicles (WHO, 2023).

¹⁵⁸ For a statistically significant estimate, both the upper and lower bound should be within the positive values range.

Georgia has some road safety and transport systems in place. The country has a Civil Registration and Vital Statistics system as of 2021. Georgia has systems in place for monitoring driving under the influence of alcohol or drugs, along with related road traffic fatalities and injuries. There are also systems in place to track speeding violations and speeding-related injuries and fatalities. The country also monitors mobile phone use while driving. However, Georgia does not monitor safety belt and child-restraint systems use, powered 2- and 3- wheeler helmet use or the frequency and distribution of journeys by modal type (WHO, 2023).

Figure 6.5 shows that traffic fatalities have been decreasing in Georgia over the last 20 years. At the same time, used vehicles imports have been increasing, in particular with a sharp increase on 2017, i.e. the year on which Georgia regulated used vehicle imports.

Figure 6.5
Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Georgia, 2000-2022



Source: WHO, 2023;ITF, 2020

Used vehicles policies

There are no age restrictions on the importation of used vehicles, and no pre-shipment inspection is required¹⁵⁹. In Georgia, imported cars are subject to two types of taxation based on engine volume and age. An excise tax is applied according to the vehicle's age, and an import tax of 0.05 GEL per cubic centimetre of engine displacement is charged. Additionally, an extra 5% of this amount is applied for each year of the vehicle's use (UNEP, 2017).

Control group

Initially, Russia was considered as a representative enough control group for the Georgia case study. Although Russia is an immense country, with a low population density and road density, the cultural and political links between the two countries in their history can counterbalance their differences in size, as well as on certain economic and climate indicators (see Table 6.11). A second analysis will be carried out with Ukraine as the control group; Ukraine has a smaller size, but is also culturally similar on several aspects, and appears a closer match on certain key socioeconomic indicators.

¹⁵⁹ <https://blog.japanesecartrade.com/538-georgia-import-regulation-for-japan-used-cars/>

Table 6.11

Selection of Control Group on the Basis of Socioeconomic and Geographical Indicators – Georgia

Indicators	Georgia	Russia	Ukraine
Population density	53	9	65
Share of population living in urban areas (%)	61.0	75.3	70.1
GDP per capita	8,825	36,485	12,671
Income group	Upper middle income	Upper middle income	Upper middle income
Road network density (km/ sq.km)	29	9	28
Share of paved roads (%)	unknown	44	98
Vehicle ownership (1000 inhabitants)	418	361	280
PTW ownership (1000 inhabitants)	unknown	16	22
Mean age of the population (years)	37	41.9	44.9
Mean annual precipitation (mm/year)	1,026	460	565
Mean annual temperature (oC)	9	-3.8	9.3
HDI (Human Development Index)	0.814	0.821	0.734
Hospital beds per population (1000 inhabitants)	4.9	7	6.3

Source: WHO, 2023

Before-and-after analysis

The results of before-and-after analysis for the introduction of used vehicle imports regulations in Georgia is presented in Table 6.12. Treatment year is 2017, with a 5-year before and a 5-year after period. It is noted that, for the after period, data for the period 2020-2022 were collected for Georgia, Russia and Ukraine from national and international web resources¹⁶⁰. A cross-checking was carried out to ensure that the data are comparable with the WHO data of the rest of the study period; in case of discrepancy, the additional national data were corrected through a calculated coefficient to align with the estimates of WHO data.

Table 6.12 shows that, with Russia as a control group, the policy resulted in a Crash Modification Factor (CMF) of 1.284, corresponding to a 28.44% increase in fatalities in Georgia after the policy. The upper and lower bounds of the estimate are [-22.63, -34.52], which is statistically significant at 95% confidence level. With Ukraine as a control group, the policy resulted in a very similar Crash Modification Factor (CMF) of 1.368, corresponding to a 36.8% increase in fatalities. The upper and lower bounds of the estimate are [-30.36, -43.55], which is statistically significant at 95% confidence level.

¹⁶⁰ https://unece.org/sites/default/files/2024-05/2325852_E_pdf_web.pdf

Table 6.12
Before-and-After Analysis Results – Georgia

Control group (i): Russia			Control group (ii): Ukraine		
	Before	After		Before	After
Number of fatalities	2012-2016	2018-2022	Number of fatalities	2012-2016	2018-2022
Treatment group	2,972	2,298	Treatment group	2,972	2,298
Control group	136,399	82,114	Control group	31,743	17,942
CMF	Lower bound	Upper bound	CMF	Lower bound	Upper bound
1.284	1.226	1.345	1.368	1.304	1.436
Safety effect (% reduction)					
-28.44	-22.63	-34.52	-36.80	-30.36	-43.55
Significant Effect - negative			Significant Effect - negative		

Source: UNECE, 2025

These findings reveal that the introduction of a policy on used vehicle imports had a negative effect on road safety in Georgia. It is noted, however, that the policy in Georgia was based on taxation, and not on the introduction of technical and environmental restrictions. It is therefore possible that this type of policy is not effective. On the other hand, it is also possible that the policy has not yet exhibited its impact. The case study analysis should be repeated in the future, especially once credible data on traffic fatalities are available for a period of at least 5 years after its introduction.

6.6. Case study 6: Nigeria

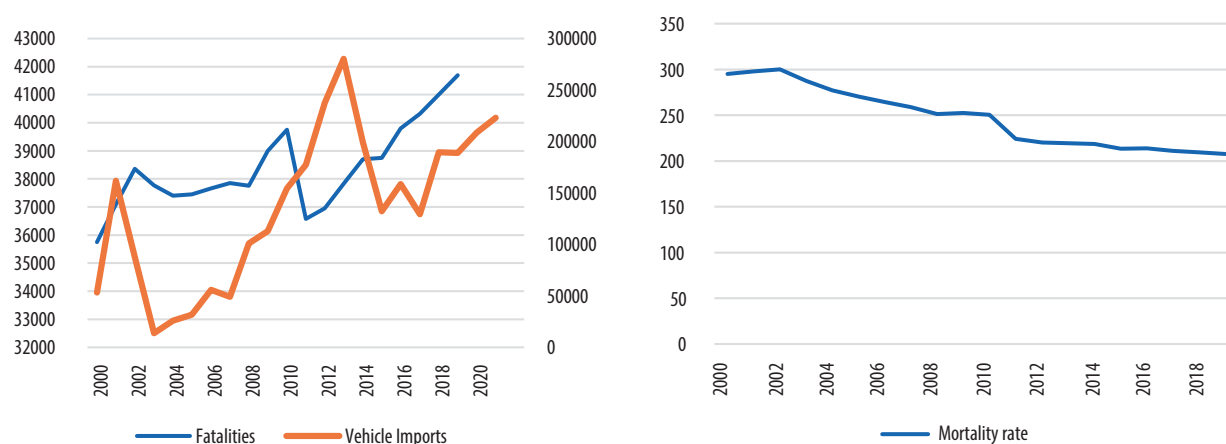
General background

Nigeria is a country in West Africa, a federal republic modelled after the United States, with 36 states and capital Abuja as an independent unit. It is a multinational state inhabited by more than 250 ethnic groups speaking 500 distinct languages. Nigeria's economy is the fourth largest in Africa, with a GDP per capita of \$6,340. Nigeria is classified within the Lower Middle-Income country group by the World Bank and belongs to the WHO African region. According to recent data, Nigeria recorded 41,693 road fatalities in 2021, with a distribution of approximately 19% male and 81% female among the fatalities. The country has an estimated 84,000 kilometres of paved roads and 15,189,136 (WHO, 2023).

Road safety context and development

Figure 6.6 shows the trends of traffic fatalities and imported used vehicles between 2000-2022. It is noted that used vehicles imports showed an increasing trend over the years, with a lot of fluctuation, and were boosted between 2004-2013. Between 2010-2013, used and new vehicle imports were regulated for the first time. At the same time, traffic fatalities showed a fluctuating trend over the whole examined period, with a notable increase after 2010 and steep increase thereafter. However, the traffic mortality rate in Nigeria shows a constantly decreasing trend. It is important to note that the country exhibited a huge growth in population over these twenty years; the population literally doubled over this period. This demographic growth, together with the increased number of vehicle imports, likely contributed to increased motorisation and more congestion, which counterbalanced the epidemiological traffic risk.

Figure 6.6

Development of Traffic Fatalities and Used Vehicle Imports (left panel) and Mortality Rates (right panel), Nigeria, 2000-2022

Source: WHO, 2023; ITF, 2020

Used vehicles policies

The vehicle policy of Nigeria has been diverse, and at times inconsistent. A regulation used vehicles was introduced as early as 2010, setting a limit of 15 years of age for all used imported vehicles. This was followed by new regulations in 2013 promoting the import of vehicle parts, rather than new vehicles, in order to reinforce the national industry¹⁶¹. According to web resources, the results from this policy started to be visible after a few years. Recently, Nigeria has established a ban on imports of vehicles that are more than 8 years old (Ayetor et al., 2021; Boateng & Klopp, 2022), however this recent effort cannot be evaluated yet because there is not a sufficient 'after' period. The present case study will look into the early introduction of a (lenient) age limit on vehicle imports.

Control group

Defining a control group for the Nigeria case study was challenging. On the one hand, its socioeconomic indicators are higher than those of most African countries. At the same time, it is a very large country with high population and high numbers of fatalities, and finding a control group of sufficient size was not straightforward. It was decided to use a large group of African countries as control group, namely neighbouring countries (e.g. central and northern Africa) that did not introduce any regulation in the imports of used vehicles over the examined period, and are also of sufficient size (e.g. their annual fatalities are a relatively high number). The selected control group countries are: Mali, Algeria, Chad, Burkina Faso, Cameroon, Central Africa, Cote d'Ivoire, Ethiopia and Ghana.

Before-and-after analysis

The results of before-and-after analysis for the introduction of used vehicle imports regulations in Nigeria is presented in Table 6.13 Table 6.2. Treatment year is 2010, with a 5-year before and a 5-year after period. All the relevant data were available from WHO.

¹⁶¹ <https://shipsandports.com.ng/nigeria-auto-policy-a-timeline-of-inconsistency/>

Table 6.13
Before-and-After Analysis Results – Nigeria

	Before	After
Number of fatalities	2005-2010	2011-2015
Treatment group	229,472	188,803
Control group	248,361	291,586
CMF	Lower bound	Upper bound
0.701	0.696	0.706
Safety effect (% reduction)		
29.98	29.44	30.99
Significant effect - positive		

Source: UNECE, 2025

Table 6.4 shows that the policy resulted in a Crash Modification Factor (CMF) of 0.701, corresponding to a ~30% reduction in fatalities in Nigeria. The upper and lower bounds of the estimate are both positive, which is significant at 95% confidence level.

A separate analysis was conducted by considering a 5-year time lag in the impact of the policy. The results are presented in Table 6.14. In this case, the 'after' period is taken to be 2015-2019. The results show that, with the time lag in the effect, the policy resulted in a Crash Modification Factor (CMF) of 0.654, corresponding to a ~35% reduction in fatalities. The upper and lower bounds of the estimate are [34.17, 35.04], which is also statistically significant.

Table 6.14
Before-and-After Analysis Results with 5-Year Time Lag in the Effect of Policy – Pakistan

	Before	After
Number of fatalities	2000-2005	2015-2019
Treatment group	229,472	201,577
Control group	248,361	333,643
CMF	Lower bound	Upper bound
0.654	0.650	1.004
Safety effect (% reduction)		
34.6	34.17	35.04
Significant effect - positive		

Source: UNECE, 2025

These findings suggest a very positive effect of the policy on road safety. However, because there is not sufficient information about other possible confounding factors that may have affected road safety trends in Africa over the examined period, the consultation with national experts is required in order to validate these findings.

6.7. Global effect of used vehicles' policies

6.7.1. Calculation of Weighted Mean Effect

Table 6.15 summarises the results of the six case studies that were implemented within this project. On the basis of the methodology presented in section 3.2.2, a Weighted Mean Effect was calculated, which brings together the individual case study estimated safety effects of their policies, by applying a statistical weighting which takes into account the different sample sizes and uncertainty levels of their estimates. In this calculation, we considered only the 'immediate effect' results of the policies, i.e. the cases with no time lag in the effect of the policy. Moreover, in case two control groups were considered in the analysis, we include the smallest (most conservative) estimate of the two.

The Weighted Mean Effect of all the estimated Crash Modification Factors (CMFs) was found to be 0.812, corresponding to **a global safety effect of 18.8% reduction in fatalities within 5 years from the introduction of a regulation on the imports of used vehicles**. The lower and upper bounds of this effect are [18.5, 19.2], therefore the estimate has minimal uncertainty – which is expected, given the large numbers of fatalities that were used in the case studies.

Table 6.15

Summary of Case Studies – Global Effect of Used Vehicles Policies on Road Safety with no Time Lag in the Effect

Country	Year of regulation	Type of regulation	Control groups	CMF	Safety effect (% reduction)	Statistical significance
Malaysia	2017	Age limit (<5 years)	India	0.843	15.7	Significant
Pakistan	2006	Age limit (<3 years)	India	0.994	0.6	Non-significant
Egypt	2017	Ban (<1 year)	i. India, ii. Several African countries	0.674	32.6	Significant
Romania	2007	UN core vehicle regulations	Russia	0.783	21.7	Significant
Georgia	2017	Age and vehicle size taxation	i. Russia, ii. Ukraine	1.368	-36.8	Significant
Nigeria	2010	Age limit (<15 years)	Several African Countries	0.701	29.9	Significant
			Weighted Mean Effect	0.812	Lower bound 0.805	Upper bound 0.812
			Global Effect (% reduction)	18.8	18.5	19.2

Highly significant

Source: UNECE, 2025

6.7.2. Discussion and limitations

The estimated global effect should be considered with some caution, as it brings together six quite diverse case studies. On the other hand, all 6 countries have also progressively adopted UN vehicle standards, therefore it is possible that part of the safety effect is also due to these regulations too. Romania regulated their vehicle imports as part of their accession to the EU, therefore the short-term improvement may be due to a number of additional transport and safety regulations that took place at the same time.

Overall, it can be said that **countries that introduced an age limit on vehicle imports had a positive effect on road fatalities**, even if this age limit was quite lenient (i.e. >15 years in Nigeria). On the other hand, taxation-based regulation (e.g. Georgia) does not seem to be effective in the short term.

It is also worth noting that **the estimated safety effect may also depend on the trend and magnitude of vehicle imports before the regulation**. For instance, Pakistan introduced the regulation at a time when used vehicles imports started to increase sharply, and the influx of vehicles possibly limited the effectiveness of the policy. On the other hand, Nigeria introduced additional policies to reinforce national vehicle industry at approximately the same time as the used vehicles regulation, and this led to a sharp decrease in the imports, which probably facilitated the impact of the policy.

For three out of the six case studies, the policy was introduced in 2017, and **may be too early to fully evaluate**. In these cases, data after 2019 had to be collected from informal web resources (national or international), and in some cases converted to meet the WHO time series trend. Therefore, their results (either positive, as in Malaysia and Egypt, or negative, as in Georgia) should be considered with caution, and validated in future studies.

7. CONCLUSIONS & RECOMMENDATIONS

7.1. Summary of findings

The aim of the present study was to compile and review existing regulations, standards, and processes for used imported vehicle in terms of road safety and environmental standards that have been adopted and are presently in force by importing LMICs globally, as well as to analyse road safety data and other relevant data linked to the safety impacts of the imports of used vehicles. Moreover, the work set up and conducted the relevant case studies in different regions, in order to assess the impacts of imports of used vehicles and their regulation.

A systematic review was presented on the existing regulations and standards in place for the imported used vehicles in LMICs globally. First, the international and regional regulations and frameworks were described, following by analyses on the regional level and subsequently on the country level. At the same time, the adherence of LMICs to the UN core requirements on vehicle safety was examined. The results were summarised in a comparative analysis, in which **the main features of existing regulations and standards for 136 countries** are included (see Table 2.1), namely:

- year of regulation
- age limit adopted
- technical standards included
- environmental standards included
- adherence to the UN core vehicle safety regulations

Subsequently, an **analysis framework** was developed for the next stages of the project. The analysis methods that were implemented were:

- **A macroscopic analysis**, aiming to associate the share of imported used vehicles in the fleet with the traffic mortality rates of countries around the world.
- **A cross-sectional before-and-after analysis** with control group, aiming to estimate the safety effect (percentage reduction of fatalities) that can be credibly attributed to the used vehicle imports' policy intervention, without any of the bias that may be involved in a simple before/after comparison.

The second type of analysis was implemented on **six case studies** for countries that fulfilled the following criteria (namely: Malaysia, Pakistan, Egypt, Romania, Georgia & Nigeria).

- i. Country has implemented a policy on used vehicles imports at least 5 years earlier than the year of its last available road safety data
- ii. Country adheres to (most of) the core UN requirements for vehicle safety
- iii. UN regions are adequately represented
- iv. Availability of (time series of) road fatality data, vehicle fleet data and used vehicle imports data for at least 10 years before and 5 years after the policy introduction

The main findings of the study can be summarised as follows:

- In all countries, each 1 unit increase of total registered vehicles per 108 of population leads to an increase of about 0.091 of the crash mortality rate.
- In LMICs, each 1 unit increase of total registered vehicles per 108 of population leads to an increase of about 0.086 of the crash mortality rate.
- In all countries, safety belt compliance above 70% decreases mortality rates by more than 2.9 on average.

7. CONCLUSIONS & RECOMMENDATIONS

- In all countries, each 1 unit increase of average used vehicle imports of the most recent 5 years ago per 1000 population (years 2017-2021) leads to an increase of about 0.004 of the crash mortality rate.
- **In LMICs, each 1 unit increase of average used vehicle imports of the most recent 5 years ago per 1000 population (years 2017-2021) leads to an increase of about 0.476 of the crash mortality rate.**
- In the **Malaysia** case study, it was found that an age limit of <5 years in the imports of used vehicles, introduced in 2017, resulted in a significant reduction in traffic fatalities by ~15%, compared to what would have happened without the intervention over the next 5 years.
- In the **Pakistan** case study, an age limit of <3 years in the imports of used vehicles, introduced in 2006, had in a non-significant effect on traffic fatalities. This is attributed to the fact that the imports of used vehicles were boosted after the regulation, together with an overall growth of the population and motorization in the country.
- In the **Egypt** case study, an age limit of <1 years in the imports of used vehicles (practically serving as a ban in the imports of used vehicles), introduced in 2017, resulted in a significant reduction in traffic fatalities by ~32%, compared to what would have happened without the intervention over the next 5 years.
- In the **Romania** case study, the accession to the EU on 2007, with the adoption of all relevant at the time UN core vehicle regulations, as well as other regulations that may have benefited road safety, resulted in a significant reduction of ~22% of traffic fatalities, compared to what would have happened without the intervention over the next 5 years. This is also associated with a large drop in the imports of used vehicles after 2007. It should be noted, however, that when considering a 5-year time lag on the impact of the policy, showed a negative effect of the policy (fatalities increased). This is attributed to the overall economic growth that developed in the medium term, which counterbalanced the immediate impacts of vehicle regulations.
- In the **Georgia** case study, a taxation-based regulation of vehicle imports on 2017 was associated with a negative impact on road safety, as fatalities increased by ~37%, compared to what would have happened without the intervention over the next 5 years. This type of regulation, which practically allows the imports of all used vehicles is indicated to be ineffective, at least in the short-term.
- Finally, in the **Nigeria** case study, it was found that an age limit of <15 years in the imports of used vehicles, introduced in 2010, together with policies that aim to reinforce the national vehicle industry, resulted in a significant reduction in traffic fatalities by ~30%, compared to what would have happened without the intervention over the next 5 years. A 5-year time lag in the effect of the policy indicated an even larger benefit of ~35% fatalities reduction, compared to African countries with no regulation at all (control group).

From the above results, a Weighted Mean Effect was calculated. It was found that **the introduction of a policy on the imports of used vehicles may result in a 18.8% reduction of traffic fatalities within 5 years of implementation**, on average. This indicates a substantial benefit to be expected from introducing such regulations, while also adopting the UN 'core' vehicle regulations. Of course, individual case studies suggest that the effect can be higher or lower, depending on the timing of regulation, the magnitude of vehicle imports and the time of regulation, and the other accompanying policies that may affect road safety at the same time.

Finally, it should be underlined that, the stakeholders' survey revealed a gap between the perceived safety of imported used vehicles and their actual safety; the survey participants highlighted that the lack of concrete evidence on the road safety impacts of imported used vehicles makes it difficult to isolate their contribution to the overall road safety level of the country. The present report is **a contribution to this gap of knowledge, providing for the first time a quantitative estimate of the impact of imported used vehicle policies on road safety**, and its statistical significance. It is also noted that the survey participants made a number of useful suggestions for policy directions namely:

- Preserving the contribution of vehicle imports to the mobility and economic growth of LMICs
- Placing the policy focus on vehicle technical inspection and UN core vehicle standards in LMICs
- Sharing responsibility between exporters and importers as regards the quality of used vehicles
- Assisting LMICs in capacity building for the enforcement of regulations on imported used vehicles and their use.

7.2. Recommendations

On the basis of international literature review on the effects of vehicle characteristics on road safety, it was concluded that vehicle age is the key variable to assess the state of imported used vehicles in LMICs and its safety implications. Older vehicles are associated with increased crash occurrence and fatality risk. Nevertheless, vehicle age alone is not sufficient (and it is seldom available in road safety databases); its impact on crash risk and crash severity is strongly associated with other factors, such as the presence and use of restraint systems by the vehicle occupants, the crashworthiness of the vehicle and active safety / driver assistance systems in place.

Recommendation 1: Countries should monitor the state of the vehicle fleet in the country, by promoting newer, environmentally friendly and technically robust vehicles:

- through the adoption of core UN vehicle regulations,
- through the adoption of high environmental standards or requirements (i.e. Level 4 or above),
- through the implementation and monitoring of Periodic Technical Inspections (PTI) for vehicles

At the same time, it was found that increased imports of used vehicles per population are associated with increased mortality rates, not only in LMICs (stronger effect) but in all countries around the world. The magnitude of imports itself is a significant determinant of the road safety performance in any country. The Pakistan and Georgia case studies demonstrate that a regulation of imports may be ineffective if the influx of used vehicles is not controlled by it. On the contrary, the Romania and Nigeria case studies demonstrate that, if the policy results in a reduction of the influx of used vehicles imports, positive safety effects are to be expected.

Recommendation 2: Countries should control the influx of used vehicles imports:

- through regulation of the number of imported vehicles per population, in order to meet the transport demand with safe and sustainable transport modes.
- through the adoption of a national strategy for addressing the transport demand, including the provision of safe and sustainable public and private transport options in the medium-term, to reduce the low-quality private transport.
- through the adoption of minimum technical and environmental standards for imported used vehicles.

The regulation of imported used vehicles in the countries has been very diverse. Existing policies may or may not include age limits in the imported vehicles, and these age limits may vary between countries from 1 to 15 years. They may or may not include environmental standards, and these may vary from Euro 3 to Euro 6. They may or may not require technical inspections pre-shipment or upon arrival. All these elements alone are important for an effective regulation of used vehicles imports. The case studies demonstrated that the regulation of used vehicles imports with age limits and UN core vehicle regulations can result in substantial safety benefits on the short- to medium-term.

Recommendation 3: Countries, especially LMICs, should introduce effective regulations of imported used vehicles, with at least all of the following elements:

- through the adoption of the UN core regulations for all imported used vehicles,
- through the adoption of age limit of 8 years or lower for all imported used vehicles,
- through the adoption of Level 4 or higher environmental standards for all imported used vehicles.
- through mandatory technical inspections of all used vehicles prior to their deployment on the country's road network.

At the same time, it is important that standards are established at the international level for the imports of used vehicles, to prevent the large diversity of national standards, and facilitate the countries in making evidence-based decisions in this respect.

Recommendation 4: International standards should be established for the imports of used vehicles in LMICs:

- international organisations should carry out relevant studies and stakeholder consultations to consolidate the national experience, identify best practice and facilitate the transfer of knowledge
- international organisations should take leadership in developing appropriate standards, and promoting these standards among their member states¹⁶²
- countries should work closely with the UNECE, UNEP and other international stakeholders to benefit from international evidence and receive strategic and technical assistance in the development of their used vehicle policies

The results of the present analysis also confirmed the well-known effect of restraint systems on road safety. The modelling results suggest that there are substantial safety benefits in countries with higher safety belt wearing rates among the population.

Recommendation 5: Countries should establish and monitor the presence and use of restraint systems and protective devices in all vehicles in the country:

- through the adoption of the UN core regulation for safety belts and safety belt anchorages,
- through the ban of used vehicles imports that do not comply with UN safety belt regulations,
- through the establishment of a national law on safety belt, helmet and standardised child restraints use for all passengers (front and rear seats)
- through the systematic enforcement of this law by means of systematic checks, random in time and space, by means of traffic police roadside checks, and effective collection of fines for offenders

One of the main limitations of the present study was the unavailability or poor comparability of the necessary data. Data on vehicle imports and their regulations were gathered from various international sources, in order to have a complete picture of the country status; a part of this information could not be fully validated. Moreover, the WHO and UN road safety data, that were used as primary sources in this study, were incomplete for a number of countries in general, and for several LMICs in particular.

Recommendation 6: Countries should work on improving their data systems regarding road safety, vehicle fleet and used vehicle imports:

- through the creation of databases on the basis of international definitions (e.g. the UN Glossary for Transport statistics), and the regular update of these databases.
- through the reporting of their data to international organisations' databases or relevant questionnaires, through national contact points with good knowledge of data quality and limitations

¹⁶² The World Forum for Harmonisation of Vehicle Regulations (WP.29) has established an Informal Working Group on 'Safer and Cleaner Used and New Vehicles (IWG on SCUNV) for Developing Countries' (<https://wiki.uncece.org/pages/viewpage.action?pagelD=164397099&pre-view=/164397099/223019213/ToR%20SCUNV.pdf>)

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APPENDICES

A. Generalised Linear models (Negative Binomial) for road safety

The **Negative Binomial Model** is formulated as follows:

$$y = \log(p) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where p the expected integer of Mortality Rate values, β_0 is the intercept, $\beta_1, \dots, \beta_n$ are the coefficients, and $X_1, \dots, X_n$ are the predictor variables.

The logarithmic link is used to ensure non-negative predicted values for count data. The independent predictor variables X_1 used in each regression must not be mutually correlated, which would bias the results through multicollinearity. Multicollinearity occurs when two or more predictor variables in a model are highly correlated, meaning they provide redundant information. This can make it difficult to assess the independent effect of each predictor on the dependent variable and can inflate the variances of the estimated regression coefficients, leading to less reliable statistical inferences.

To detect multicollinearity in regression models, Variance Inflation Factor (VIF) is a useful statistical measure. For each predictor X_i , VIF is calculated as a fraction:

$$VIF(X_i) = \frac{1}{1 - R_i^2}$$

Where R_i^2 is the R-squared value from the regression of X_i on all the other predictor variables in the model. Empirically, VIF values below 2 suggest very low multicollinearity effects, while values above 5 warrant attention.

Moreover, **Marginal Effects to the Mean (MEM)** are used for increased interpretability of results. MEMs measure **the change in the expected count of the dependent variable in response to a change in an explanatory variable, while holding other variables constant**. Marginal effects help interpret how predictors influence the outcome variable's mean in a non-linear model like NB, where relationships are not as straightforward as in linear models.

For a predictor X_i , the marginal effect on the expected count is calculated as the partial derivative of the mean expected value p_j with respect to X_i . Specifically for NB models:

$$\frac{\partial E(Y|X_j)}{\partial X_i} = p_j * X_i$$

In practical terms, for continuous variables, the value of each MEM describes how much the expected count changes for a one-unit change in the predictor (e.g. for each unit of a million population). For discrete variables, the MEM value reflects the difference in the predicted expected count when the variable changes between two groups (e.g., male vs. female).

B. Observational before-and-after studies

The evaluation of the treatment effect is performed by means of an odds-ratio. In this case, the safety effect of the treatment at site i is estimated as:

$$\text{Estimated effect } (\theta_i) = [\mu_A / \mu'_B] / [C_A / C'_B]$$

Where:

μ_A - the number of road crashes / fatalities observed at the treatment area (site i) in the "after" period

μ'_B - the number of road crashes / fatalities at the treatment area (site i) in the "before" period

C_A - the number of road crashes / fatalities observed at the control group area in the "after" period

C'_B - the number of road crashes / fatalities at the control group area in the "before" period

The applicable value of the safety effect, i.e. the best estimate of crash / fatality reduction associated with the treatment (in percent), is calculated as $(1-\theta_i)*100$.

In case there are multiple treatment sites, each with their own estimated safety effect θ_i , it is possible to calculate the Weighted Mean effect for all sites, as well as its confidence interval.

The statistical weight of the estimate for treatment site (i) is then:

$$w_i = \frac{1}{\frac{1}{\mu_{Ai}} + \frac{1}{\mu'_{Bi}} + \frac{1}{C_{Ai}} + \frac{1}{C'_{Bi}}}$$

The global effect (for all treated sites) is estimated by the weighted mean:

$$\text{Weighted Mean Effect (WME)}_i = \exp \left[\frac{\sum_i w_i \ln(\theta_i)}{\sum_i w_i} \right]$$

with 95% confidence interval for the effect estimated as follows:

$$\left[WME * \exp \frac{Z_{\frac{\alpha}{2}}}{\sqrt{\sum_i w_i}}, WME * \exp \frac{Z_{1-\frac{\alpha}{2}}}{\sqrt{\sum_i w_i}} \right]$$

The applicable value of the safety effect, i.e. the best estimate of accident reduction associated with the treatment (in percent), is calculated as $(1-WME)*100$.

C. Road safety data collected

Table A.1

Number of fatalities (Country-Reported and WHO-Estimated) and Mortality Rate (Number of Fatalities per 100,000 Population) in LMICs on 2010, 2016, 2021

Country name	Reported 2021	WHO estimated 2021	WHO-estimated 2016	WHO-estimated 2010	Mortality rate 2021	Mortality rate 2016	Mortality rate 2010
Afghanistan	738	9684	7106	5546	24.1	20.5	19.7
Albania	197	309	370	388	10.8	12.8	13.3
Algeria	3322	8106	8536	8039	18.3	21.2	22.4
Antigua and Barbuda	7	7	8	5	7.5	8.8	5.8
Argentina	3863	3983	5755	5252	8.8	13.2	12.8
Armenia	368	379	501	473	13.6	17.5	16.1
Azerbaijan	706	1769	2096	2242	17.2	21	24.3
Bangladesh	5084	31578	24944	25697	18.6	15.6	17.3
Belarus	524	706	1007	1627	7.4	10.4	16.7
Belize	70	70	95	45	17.5	25.9	14
Benin	1124	3225	2167	1517	24.8	19.2	16.1
Bhutan	71	95	141	125	12.2	18.8	17.7
Bolivia (Plurinational State of)	1325	2127	1911	1901	17.6	17	18.6
Bosnia and Herzegovina	263	447	507	647	13.7	14.6	17
Botswana	413	426	464	397	16.5	19.7	19
Brazil	31468	33586	38665	43876	15.7	18.7	22.3
Bulgaria	561	578	730	803	8.4	10.1	10.6
Burkina Faso	1272	6137	4376	3672	27.8	22.7	22.8
Burundi	592	1546	2014	2222	12.3	18.5	24.3
Cabo Verde	39	97	129	107	16.5	23.1	20.5
Cambodia	1794	3113	4220	4369	18.8	27	30.4
Cameroon	930	2870	4898	4576	10.6	20.7	23
Central African Republic	1370	1412	1199	1129	25.9	24.4	24.2
Chad	254	4533	3958	3270	26.4	27.1	27.5
China	62218	248099	256180	279005	17.4	18.3	20.7
Colombia	7238	8341	7771	6380	16.2	16.3	14.2
Comoros	32	238	156	138	29	20.9	21
Congo	223	488	1024	850	8.4	19.7	19.2
Cook Islands	3	3	1	2	17.6	5.7	11.6
Costa Rica	722	798	1028	705	15.5	20.8	15.3
Côte d'Ivoire	1614	5670	4846	4695	20.6	20	22.2
Cuba	589	613	1004	911	5.4	8.9	8.1
Democratic Republic of the Congo	3364	15615	18037	17624	16.3	22.2	26.5
Dominica	8	8	5	7	11	7.1	10.2
Dominican Republic	2967	3048	3203	2190	27.4	30.4	22.4
Ecuador	2131	4173	2850	3399	23.4	17.3	22.7
Egypt	7101	10263	11870	12617	9.4	11.9	14.5

Country name	Reported 2021	WHO estimated 2021	WHO-estimated 2016	WHO-estimated 2010	Mortality rate 2021	Mortality rate 2016	Mortality rate 2010
El Salvador	1358	1358	1215	1159	21.5	19.4	19
Eritrea	100	640	645	709	17.7	19.2	22.5
Eswatini	229	295	230	224	24.7	20.1	20.4
Ethiopia	3971	21258	21115	18380	17.7	20.1	20.6
Gabon	89	293	353	326	12.5	16.9	19.1
Gambia	200	582	418	307	22	18	15.8
Georgia	449	476	621	706	12.7	16.5	18.4
Ghana	2890	8494	5973	5487	25.9	20.2	21.5
Guatemala	2154	2221	2123	1958	12.6	13	13.5
Guinea	682	5061	3092	2556	37.4	25.9	24.9
Guinea-Bissau	100	629	567	479	30.5	30.9	30.6
Guyana	99	122	134	115	15.2	17.7	15.4
Haiti	134	3579	2394	2202	31.3	22.3	22.4
Honduras	1743	1904	1723	1415	18.5	18.2	16.7
India	153972	216618	202999	212091	15.4	15.2	17.1
Indonesia	25266	31063	32231	32216	11.3	12.3	13.2
Iran (Islamic Republic of)	16778	18156	16456	25096	20.6	19.8	33.3
Iraq	5563	9344	8184	5746	21.5	21.1	18.4
Jamaica	487	502	391	329	17.8	14	12
Jordan	589	1514	1522	862	13.6	15.3	12.4
Kazakhstan	2270	2340	2910	4074	12.2	16.1	24.5
Kenya	4579	14926	11172	9928	28.2	23.3	23.9
Kiribati	8	8	2	6	6.2	1.7	5.6
Kyrgyzstan	869	869	1069	924	13.3	17.8	16.8
Lao People's Democratic Republic	831	1217	1639	1710	16.4	23.8	27
Lebanon	378	544	1106	1057	9.7	17.7	21.2
Lesotho	282	492	615	646	21.6	28.7	31.9
Liberia	232	794	1033	932	15.3	22	23.2
Libya	2218	2287	2489	2576	34	39.6	39.7
Madagascar	300	6512	7579	5698	22.5	29.7	26.2
Malawi	1444	4023	3048	2448	20.2	17.5	16.6
Malaysia	4539	4680	7374	7085	13.9	23.4	24.7
Maldives	5	7	7	13	1.3	1.5	3.6
Mali	736	4429	3721	3460	20.2	19.9	22.3
Mauritania	99	438	744	756	9.5	18.4	22.1
Mauritius	108	126	164	163	9.7	12.7	12.7
Mexico	14721	15267	16842	16559	12	13.9	14.7
Mongolia	383	414	494	506	12.4	16.3	18.7
Montenegro	55	57	67	98	9.1	10.6	15.5
Morocco	3685	6900	6917	7972	18.6	19.7	24.6
Mozambique	944	6451	6610	6002	20.1	23.9	26
Myanmar	5325	10405	14753	16994	19.3	28.4	34.4
Namibia	540	557	760	556	22	32.7	26.5

Country name	Reported 2021	WHO estimated 2021	WHO-estimated 2016	WHO-estimated 2010	Mortality rate 2021	Mortality rate 2016	Mortality rate 2010
Nepal	2883	8479	6530	6797	28.2	23.4	25
Nicaragua	904	932	816	765	13.6	12.8	13.1
Niger	1152	6278	4693	3903	24.9	22.4	23.4
Nigeria	6205	36722	30528	28202	17.2	16.2	17.5
Niue	0	0	0	0	0	0	0
North Macedonia	116	120	170	167	5.7	8.1	8
occupied Palestinian territory, including east Jerusalem	186	239	320	170	4.7	7	4.3
Pakistan	5816	27568	27543	26078	11.9	12.9	13.4
Paraguay	1396	1405	1214	1265	21	19.4	21.9
Peru	3032	4284	4056	3827	12.7	13	13.1
Philippines	8746	11062	14051	10630	9.7	13.4	11.2
Republic of Moldova	257	278	332	486	9.1	10.3	13.2
Russian Federation	14874	15335	22043	32364	10.6	15.2	22.6
Rwanda	655	1563	1812	1944	11.6	15.2	18.9
Saint Lucia	14	16	16	16	8.9	9.1	9.4
Samoa	19	21	21	22	9.6	10.2	11.3
Sao Tome and Principe	25	26	25	34	11.7	12.2	18.7
Senegal	877	3502	2816	2630	20.8	19.1	21
Serbia	521	537	626	680	7.4	8.4	8.9
Sierra Leone	336	1165	1567	1564	13.8	20.9	24.3
Somalia	70	3440	4021	3575	20.2	28.1	29.7
South Africa	12541	14528	16300	16180	24.5	28.9	31.2
South Sudan	350	2500	2859	3004	23.3	25.8	30.9
Sri Lanka	2513	2513	3017	2805	11.5	14.1	13.6
Sudan	3024	8967	8847	8902	19.6	22.5	26.4
Suriname	96	99	76	90	16.2	13.1	16.5
Syrian Arab Republic	549	6383	4276	3891	29.9	22.5	17.4
Tajikistan	395	1352	1622	1663	13.9	18.6	21.8
Thailand	16957	18218	24728	31245	25.4	35	45.8
Timor-Leste	59	159	195	202	12	15.9	18.6
Togo	680	1961	1598	1540	22.7	20.9	23.4
Tunisia	2273	2004	2852	2388	16.3	24.4	21.9
Türkiye	5362	5528	7526	5238	6.5	9.3	7.2
Uganda	4159	7315	5850	6401	16	15.1	19.8
United Republic of Tanzania	1368	10052	10746	9137	15.8	19.8	20.3
Uzbekistan	2197	3155	3040	2969	9.3	9.7	10.4
Venezuela (Bolivarian Republic of)		3718	6784	8744	13.2	22.1	30.5
Viet Nam	5699	17229	24970	29160	17.7	26.8	33.4
Yemen	3409	9821	8320	6279	29.8	28.4	25.4
Zambia	2163	3338	2440	2439	17.1	14.6	17.7
Zimbabwe	1902	4782	3326	3465	29.9	23	27

Source: WHO, 2023

D. Vehicle fleet data collected

Table A.2

Total Registered Vehicles and Percentage of Powered Light Vehicles in the Fleet, 2021

Country name	Total registered vehicles	Registered cars and 4-wheeled light vehicles	% Powered light vehicles in the fleet	Number of vehicles per 1,000 inhabitants
Afghanistan				
Albania	740,669	593,280	57.4	259
Algeria	8,720,015	5,432,965		197
Argentina	26,413,085	16,754,756	31.6	583
Armenia			68.2	
Azerbaijan	1,591,786	1,361,560	51.8	154
Bangladesh	5,013,908	767,797	41.5	30
Belarus			48.5	
Belize			18.8	
Benin				
Bhutan	119,636		94.9	154
Bolivia (Plurinational State of)	2,226,662	1,163,386	60.8	184
Bosnia and Herzegovina	1,152,743	985,070	52.6	352
Botswana	914,730	876,595	63.1	353
Brazil	111,446,870	73,134,262	22.3	520
Bulgaria	3,571,571	2,830,594	48.48	519
Burkina Faso				
Burundi				
Cabo Verde	81,000	57,510		138
Cambodia	6,356,653		6.2	383
Cameroon	31,594			1
Central African Republic				
Chad	32,995	130,291		2
China	395,264,200	257,973,900	12.91	277
Colombia	16,780,651	5,957,616	7.8	326
Comoros			34.79	
Congo	25,227	1,466		4
Cook Islands	1,291	34	20	76
Costa Rica	2,631,200	1,686,979	24	511
Côte d'Ivoire			11.1	
Cuba	663,898	267,773	15	59
Democratic Republic of the Congo			36.1	
Dominica	39,388		17	544
Dominican Republic	5,152,448	1,683,453	10	463
Ecuador	2,535,853	1,658,243	2	142
Egypt	10,909,456	5,383,886	37.7	100
El Salvador	1,477,912	903,674	13.8	234

Country name	Total registered vehicles	Registered cars and 4-wheeled light vehicles	% Powered light vehicles in the fleet	Number of vehicles per 1,000 inhabitants
Eritrea	73,583	38,850	51	20
Eswatini	16,230	12,542	53.7	14
Ethiopia				
Gabon	7,451	5,821		3
Gambia	94,962			36
Georgia	1	1	44.9	0
Ghana	3,314,215		14.1	101
Guatemala	4,472,894	2,145,881	12	254
Guinea				
Guinea-Bissau				
Guyana			13.3	
Haiti				
Honduras	2,249,986		0	219
India	326,300,000	43,650,000	12.9	232
Indonesia	147,110,303	19,687,228	2.22	537
Iran (Islamic Republic of)	36,767,152		49.2	418
Iraq	7,260,000		64.8	167
Jamaica	783,321		0	277
Jordan	1,752,104	1,620,045	71.3	157
Kazakhstan	4,338,639		63.9	226
Kenya	4,224,356	71,264	26.49	80
Kiribati			40	
Kyrgyzstan	1,396,462	1,103,445	27.6	214
Lao People's Democratic Republic			18.7	
Lebanon	2,535,914	2,437,081	24.3	453
Lesotho	215,273		54	94
Liberia	59,239	41,397	33.7	11
Libya	298,180		75	44
Madagascar	599,010	143,468	31.1	21
Malawi				
Malaysia			19.5	
Maldives	94,210	12,070	0	181
Mali			27.9	
Mauritania	400,000			87
Mauritius	622,988	393	16.7	480
Mexico	53,115,396	35,460,804	13.9	419
Mongolia	1,234,701	1,093,500	65.5	369
Montenegro	254,409	221,768	65.37	405
Morocco	6,574,280	2,663,078	25.2	177
Mozambique	1,165,705		44.5	36
Myanmar	7,562,508	596,252	10.6	141
Namibia	401,476	346,978	65.3	159

Country name	Total registered vehicles	Registered cars and 4-wheeled light vehicles	% Powered light vehicles in the fleet	Number of vehicles per 1,000 inhabitants
Nepal	3,987,267	344,687	35.1	133
Nicaragua			23.2	
Niger			67	
Nigeria	15,189,136			71
Niue			100	
North Macedonia	549,653	477,820	50.02	261
occupied Palestinian territory, including east Jerusalem	435,584	368,747	40.32	85
Pakistan	34,301,159	4,517,158	16.1	148
Paraguay	2,684,358	964,120	13.5	400
Peru	7,688,373	2,277,257	24	228
Philippines	13	4,389,676	0.68	
Republic of Moldova	928,832	716,906	42.79	303
Russian Federation	60,141,889	50,304,104	59.4	414
Rwanda	270,673		14.6	20
Saint Lucia	75,194		46.7	419
Samoa	29,756		21.05	136
Sao Tome and Principe			25	
Senegal			55.4	
Serbia	2,373,492		44	325
Sierra Leone	60,361			7
Somalia	131,523	121,523		8
South Africa	7,652,077	2,671,314	57	129
South Sudan	32,910			3
Sri Lanka	8,331,702	1,355,597	14.55	383
Sudan	1,867,458		0	41
Suriname	244,016		39	398
Syrian Arab Republic	2,596,542		18.8	122
Tajikistan	527,332		57.4	54
Thailand	42,313,968	17,934,311	0	591
Timor-Leste	232,012	5,149		176
Togo	93,944	20,313	11.47	11
Tunisia	2,327,348	2,070,143	35.11	190
Türkiye	25,249,119	13,706,065	46.7	298
Uganda	1,391,510		14.04	30
United Republic of Tanzania	3,225,017	4,682	50.14	51
Uzbekistan				
Venezuela (Bolivarian Republic of)			2.6	
Viet Nam	75,353,485	428,841	35.25	773
Yemen				
Zambia	899,186	27,979	35	46
Zimbabwe	15,630,004		26.7	977

Source: WHO, 2023

E. Used Vehicles Imports in LMIC

Table A.3

Number of Imported Used Vehicles per LMIC Country 2012 – 2021

Country	Year										Total
	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	
Afghanistan	2,585	573	744	742	625	1,259	1,775	693	2,951	2,018	112,173
Albania	12,882	9,866	11,046	10,760	16,053	18,196	22,421	23,442	30,675	38,815	210,695
Algeria	655	590	518	401	1,417	12,444	16,243	3,180	2,852	4,449	44,615
Angola	2,166	1,659	920	653	202	393	490	976	1,384	555	43,165
Argentina	1,789	5,967	1,922	1,910	3,014	2,568	2,181	1,850	1,196	565	27,401
Armenia	32,007	39,797	48,531	21,142	15,151	47,615	65,327	198,395	4,964	3,540	517,766
Azerbaijan	4,445	5,173	4,079	2,005	420	1,589	7,624	10,798	8,396	8,202	57,563
Bangladesh	7,086	13,074	19,789	29,215	32,276	31,189	25,076	22,087	17,448	26,943	269,423
Belarus	64,203	72,444	38,521	5,182	1,245	4,144	4,330	14,028	16,314	23,992	639,749
Benin	339,551	349,736	314,107	212,675	35,890	36,456	56,901	76,442	60,500	68,073	2,021,785
Bolivia_ Plurinational State of	1,520	4,142	4,624	552	305	195	201	127	95	53	15,085
Bosnia and Herzegovina	11,567	12,408	14,393	11,354	6,855	6,754	8,520	7,061	3,819	5,029	104,881
Bosnia Herzegovina	28,483	38,580	48,925	47,181	53,679	54,321	60,752	49,467	31,488	45,793	505,106
Botswana	1,947	8,916	598	311	287	589	2,463	450	251	188	56,828
Brazil	27,411	22,717	24,916	20,634	15,050	16,223	25,961	33,907	16,188	22,323	294,659
Cambodia	20,510	19,562	26,033	44,572	42,027	45,715	50,372	58,112	31,895	18,235	390,619
Cameroon	32,830	33,608	33,621	36,958	48,944	29,989	27,986	26,549	38,935	40,696	402,236
Central African Republic	480	228	226	325	471	581	504	411	454	751	5,307
Chad	154	147	127	222	238	155	99	323	222	132	2,012
Chile	70,403	85,844	78,357	78,035	90,759	112,603	119,179	101,609	87,982	93,874	1,081,198
China	13,312	21,045	23,514	22,270	18,452	16,449	11,954	8,913	3,826	4,778	173,671
Colombia	280	316	296	393	474	645	822	920	586	911	6,572
Congo	16,204	17,036	15,602	12,015	12,149	8,616	5,564	5,879	10,342	10,506	133,008
Congo_ the Democratic Republic of the	1,584	1,697	1,626	1,531	776	4,584	2,773	4,880	14,008	3,438	38,930
Costa Rica	10,301	5,505	6,151	7,771	7,176	6,252	3,809	3,482	3,984	3,469	75,056
Cote d'Ivoire	13,864	18,801	22,322	25,009	41,839	46,468	26,694	3,177	4,944	7,122	228,174
Cuba	183	154	32	138	106	63	59	46	59	275	1,494
Demokratische Republik Kongo	1,579	290	176	175	118	175	168	225	222	200	5,642
Dominica	51	38	106	360	177	163	108	176	128	70	1,529

Year											
Country	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Dominican Republic	19,725	19,487	21,516	39,703	55,543	47,815	45,892	47,546	39,006	38,494	427,595
Ecuador	361	578	631	528	630	1,010	821	909	300	3,122	9,987
Egypt	7,886	7,089	7,794	17,758	14,627	17,271	21,415	17,887	20,722	15,637	158,218
El Salvador	9,583	8,529	6,089	6,573	7,909	8,384	6,248	6,709	9,394	18,668	98,629
Eritrea	303	639	361	25	8	4	7	5	5	3	1,749
Eswatini	50	57	20	30	10	12	14	35	38	13	402
Fiji	4	13	1	8	27	11	12	7	11	4	104
Fyr Macedonia	10,058	13,575	15,240	13,367	13,550	12,666	15,981	18,649	15,674	24,487	180,094
Gabon	9,269	9,093	1,801	476	760	824	971	1,278	886	429	35,310
Gambia	5,308	5,048	7,369	7,916	9,289	12,290	14,734	13,367	14,802	25,205	124,989
Gaza Strip	4										4
Georgia	117,114	144,432	128,827	84,171	93,135	75,324	114,680	209,399	73,586	140,499	1,338,445
Ghana	92,377	90,547	37,243	43,939	61,584	58,464	62,759	58,160	77,926	52,959	747,112
Guinea	22,578	29,822	26,954	28,017	36,304	35,716	40,611	48,813	50,769	49,170	402,202
Guinea-Bissau	1,233	1,007	1,872	2,401	3,216	3,663	2,572	2,419	3,088	4,159	28,528
Guyana	677	600	538	1,402	383	370	493	585	660	920	7,964
Haiti	1,965	1,689	2,483	2,728	2,846	2,918	4,215	5,584	9,114	10,941	47,117
Hong Kong	12,232	6,792	6,968	4,833	7,216	7,069	7,882	2,232	2,045	289	90,891
India	140	136	80	59	4,011	66	132	111	67	725	5,875
Indonesia	774	580	179	107	25	36	44	34	201	33	2,467
Iran_ Islamic Republic of	407	769	923	639	1,047	1,238	553	45	35	47	7,950
Iraq	3,898	3,906	9,057	8,930	16,766	24,631	20,885	5,091	8,358	9,028	117,073
Jordan	35,732	49,611	32,440	63,111	58,043	65,562	67,938	67,045	72,639	32,954	605,510
Kazakhstan	7,799	10,478	4,624	959	124	424	1,100	1,731	3,560	540	130,075
Kenya	54,507	67,654	71,572	80,889	59,997	80,442	82,864	81,808	70,502	76,092	989,151
Kosovo	6,488	5,542	5,079	6,446	7,250	7,253	7,655	10,411	10,188	21,766	100,959
Kyrgyzstan	28,740	35,115	31,091	3,760	3,199	3,918	10,905	20,948	5,253	347	177,208
Lao People's Democratic Republic	75	49	19	514	285	207	192	177	69	51	1,744
Lebanon	38,573	40,033	34,504	37,035	41,324	37,829	29,602	19,065	8,069	34,154	425,976
Lesotho	32	35	26	28	20	20	22	17	6	12	263
Liberia	7,936	4,835	2,713	3,444	4,010	2,606	3,384	2,774	4,820	5,551	55,902
Libyan Arab Jamahiriya	216,819	118,315	68,354	87,518	65,054	126,483	271,009	453,294	207,295	170,568	1,854,216
Macedonia_ the former Yugoslav Republic of	1,693	1,457	2,163	2,016	1,964	1,687	5,599	2,360	1,764	2,134	28,554
Madagascar	3,988	4,214	3,718	2,939	6,632	7,176	5,098	5,630	3,693	1,252	48,401

Year											
Country	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Malawi	539	445	410	299	12,446	497	668	693	468	420	22,373
Malaysia	30,949	34,889	31,909	24,475	23,489	21,032	23,609	22,282	22,301	23,941	326,106
Maldives	9	3	11	7	4		4	5	21	3	87
Mali	4,792	5,632	6,577	7,620	8,899	8,053	6,866	5,305	6,917	10,588	77,261
Mauritania	7,643	6,236	5,666	978	2,626	2,522	3,925	5,296	6,686	20,324	67,413
Mauritius	634	332	196	357	2,038	7,045	151	144	147	61	11,979
Mexico	64,466	108,919	110,564	80,505	87,151	62,060	52,612	54,911	42,339	68,519	841,079
Moldova_ Republic of	13,589	11,956	9,153	5,389	5,071	6,597	8,807	13,206	19,696	16,045	132,363
Mongolia	33,161	36,869	39,614	39,020	39,430	53,836	74,190	75,140	63,711	64,311	579,624
Montenegro	6,424	8,041	7,926	5,750	8,689	10,567	14,472	16,048	9,626	16,668	114,740
Morocco	13,980	10,547	9,212	9,440	9,454	9,651	8,712	10,481	5,212	7,457	108,829
Mozambique	12,037	17,921	24,063	21,512	8,537	10,404	14,186	17,864	17,264	16,747	186,231
Myanmar	121,881	136,254	161,614	144,025	127,473	104,370	69,501	65,428	32,801	14,458	1,005,155
Namibia	27,059	7,219	12,679	3,032	3,916	8,959	4,326	3,139	3,393	6,177	222,451
Nauru	3	7	3		1		1	4	3	2	25
Nepal	5	5	1	3	16	8	3	6	1	2	58
Nicaragua	653	479	544	2,145	2,673	3,764	1,796	1,537	1,331	621	16,923
Niger	27,526	31,450	27,852	20,938	4,035	5,604	9,312	11,556	13,344	15,109	186,678
Nigeria	237,554	280,312	199,243	132,072	158,448	129,295	189,394	188,851	208,697	223,157	2,278,653
Pakistan	67,207	29,531	38,986	50,287	66,711	87,581	55,260	11,131	24,439	38,986	548,345
Palestinian Territory_ Occupied	259	209	225	449	317	341	385	385	423	3,241	6,520
Panama	2,134	2,377	2,063	2,796	2,335	1,396	1,224	1,172	407	514	21,777
Peru	1,266	209	182	485	705	1,938	2,299	1,665	406	591	14,550
Philippines	24,070	25,637	28,231	38,122	41,552	46,483	43,818	43,970	32,954	39,801	407,862
Republic Democratic of Congo	8,285	4,660	2,474	2,602	1,577	1,793	2,532	2,828	3,770	6,078	45,610
Rwanda	3,027	1,373	1,423	1,742	1,569	1,044	874	1,283	1,118	892	15,894
Senegal	9,408	10,893	15,057	18,664	31,671	33,969	35,639	37,486	40,375	41,096	285,027
Serbia	6,739	6,927	5,782	6,019	4,600	4,107	10,317	10,301	9,841	11,371	86,460
Seychelles	11	31	5	12	20	18	16	22	12	20	1,208
Sierra Leone	5,515	6,719	5,158	5,794	7,605	7,309	7,595	6,310	9,129	11,852	95,177
Somalia	329	4,044	255	302	292	19,431	299	333	298	404	26,455
South Africa	62,455	64,409	59,307	67,067	48,588	72,243	91,182	53,095	58,430	44,209	782,662
South Sudan	10	31	29	26	9	14	16	27	29	425	619
Sudan	306	259	88	1,221	1,915	4,654	3,401	6,536	4,701	108	23,675
Suriname	660	458	336	265	193	291	392	401	372	421	4,973
Syrian Arab Republic	441	236	50	468	2,908	4,768	3,493	6,344	3,212	4	22,604
Tajikistan	39,453	51,432	34,622	10,647	4,232	11,509	8,049	12,445	19,658	11,759	262,330

Year											
Country	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Tanzania_ United Republic of	28,476	34,877	44,158	46,356	41,247	50,132	115,571	63,778	53,350	65,163	596,487
Thailand	13,002	9,643	10,064	11,816	5,975	8,230	10,528	21,669	23,958	27,967	237,106
Timor-Leste	4	2		6	1	4	9		1	1	28
Togo	34,340	32,688	28,168	27,749	31,227	32,416	31,323	32,935	36,126	43,482	384,759
Tonga	8	5	2	4	4	4	4	10	3	10	68
Tunisia	37,865	14,764	12,997	14,287	12,746	11,139	6,266	6,405	15,450	7,943	168,636
Turkey	11,129	20,607	6,419	17,858	4,412	7,431	8,022	39,969	16,886	7,610	153,200
Turkmenistan	5,700	991	801	377	91	42	61	117	12	13	23,922
Uganda	25,736	28,494	27,816	24,013	26,038	27,602	29,240	31,483	28,352	27,876	345,006
Ukraine	19,992	18,959	12,986	11,928	30,068	81,161	112,281	197,593	288,749	306,979	1,103,273
Uruguay	798	959	497	371	306	315	444	210	150	163	5,339
Uzbekistan	40	24	26	516	501	329	2,017	9,645	2,520	109	15,796
Uzbekistan	38	57	29	49	21	36	54	383	341	109	1,188
Venezuela_ Bolivarian Republic of	524	276	351	648	848	620	769	933	808	251	6,751
Viet Nam	1,753	879	953	6,995	8,738	3,094	820	1,738	664	190	35,974
West Bank								1			1
Western Sahara					8		2			6	16
Western Samoa	1						2	1	2	1	7
Yemen	15,012	24,124	19,235	18,385	15,729	21,721	11,432	18,303	41,704	1,760	206,466
Zaire	608	406	330	363	243	177	223	378	387	565	4,241
Zambia	21,363	25,505	22,382	14,795	6,814	18,470	16,975	9,586	7,425	14,602	314,050
Zimbabwe	38,753	6,035	8,240	1,614	1,182	1,582	24,135	64,038	25,868	1,981	410,562
Grand Total											28,559,658

Source: ITF Dashboard, 2023

Table A.4
Number of Vehicles Exported per Exporting Country and Importing LMIC Country, 2021

Importing country	Exporting country											Total
	EU	Argentina	Brazil	Mexico	Canada	United States	United Kingdom	China	Japan	Korea	Thailand	
Afghanistan	2,257					1						2,018
Albania	33,034				4,201	1,697	299					38,815
Algeria	4,440				65	14						4,449
Angola	711				35	87	2					555
Argentina	83		370		3	109	3					565
Armenia	107				4	33						3,540
Azerbaijan	8,505				6	5						8,202
Bangladesh	4					2	30		26,909			26,943
Belarus	23,869				2	379	2					23,992
Benin	43,896				3,340	22,919	50					68,073
Bolivia_ Plurinational State of	49					6	5					53
Bosnia and Herzegovina	6,482				8	15	656					5,029
Bosnia Herzegovina	49,877											45,793
Botswana	4				2	1	181					188
Brazil	849	12,392		5,908	2	502	12	32	1,002	220	1,412	22,323
Cambodia	609				190	17,439						18,235
Cameroon	35,970				2,040	1,360	1,742					40,696
Central African Republic	856				32	4	2					751
Chad	80				49	4	4					132
Chile	495		1,097		219	3,402	5		88,697			93,874
China	857				463	3,450	32					4,778
Colombia	81		307		3	137	409					911
Congo	8,916				402	497	769					10,506

Importing country	Exporting country											Total
	EU	Argentina	Brazil	Mexico	Canada	United States	United Kingdom	China	Japan	Korea	Thailand	
Congo_ the Democratic Republic of the	67				1,018		2,386					3,438
Costa Rica	36		37		112	3,290						3,469
Cote d'Ivoire	2,491				3,435	1,251	74					7,122
Cuba	458											275
Demokratische Republik Kongo	200											200
Dominica	6				30	32	3					70
Dominican Republic	60		1		314	38,121						38,494
Ecuador	2,737		260		7	121						3,122
Egypt	15,507				81	456	39					15,637
El Salvador	5				66	18,597						18,668
Eritrea	3											3
Eswatini	0						13					13
Fiji	0					1	3					4
Fyr Macedonia	25,209											24,487
Gabon	173				165	101	9					429
Gambia	14,804				385	1,377	9,199					25,205
Georgia	62,554				1,860	69,602	2,401		8,612			140,499
Ghana	27,340				6,253	20,543	1,081					52,959
Guinea	46,549				3,584	2,941	582					49,170
Guinea-Bissau	4,165				1	4	5					4,159
Guyana	27				11	123	760					920
Haiti	1,442		20		1,511	7,970						10,941
Hong Kong	80				29	15	165					289
India	678				6	31	11					725
Indonesia	18					4	15					33
Iran_ Islamic Republic of	75											47

Importing country	Exporting country											Total
	EU	Argentina	Brazil	Mexico	Canada	United States	United Kingdom	China	Japan	Korea	Thailand	
Iraq	1,179				702	6,052	1,454					9,028
Jordan	3,356				1,875	27,962	111					32,954
Kazakhstan	636				5	140						540
Kenya	209				11	28	2,420		73,425			76,092
Kosovo	22,522					3	10					21,766
Kyrgyzstan	405				8	11						347
Lao People's Democratic Republic	9				9	33	3					51
Lebanon	6,395				8,297	19,870	4					34,154
Lesotho	4						8					12
Liberia	2,886				136	2,790	14					5,551
Libyan Arab Jamahiriya	177,352				8,525	23,514	35					170,568
Macedonia_ the former Yugoslav Republic of	2,843				67		60					2,134
Madagascar	1,236				11	11	16					1,252
Malawi	5					3	412					420
Malaysia	23				1	25	3,191		20,704			23,941
Maldives	6											3
Mali	8,923				786	976	34					10,588
Mauritania	15,255				2,556	2,561	4					20,324
Mauritius	7						54					61
Mexico	146		2		15	66,848	1,519					68,519
Moldova_ Republic of	16,724				17	232	1,656					16,045
Mongolia	100				3	711	9		63,500			64,311
Montenegro	18,377				78	57						16,668
Morocco	8,097				277	121	47					7,457
Mozambique	34					5	216		16,494			16,747

Importing country	Exporting country											Total
	EU	Argentina	Brazil	Mexico	Canada	United States	United Kingdom	China	Japan	Korea	Thailand	
Myanmar	27				550	686	2		13,193			14,458
Namibia	10					21	6,146					6,177
Nauru	2					1						2
Nepal	4											2
Nicaragua	12		4		17	591						621
Niger	15,687				768	58						15,109
Nigeria	81,491				22,132	108,014	15,675					223,157
Pakistan	30				18	93	400		38,446			38,986
Palestinian Territory – Occupied	3,313											3,241
Panama	16		27		13	456	2					514
Peru	26		514		3	50						591
Philippines	154				32	128			39,533			39,801
Republic Democratic of Congo	6,212											6,078
Rwanda	844				196	8	25					892
Senegal	25,642				8,859	6,110	678					41,096
Serbia	18,506				44	44	715					11,371
Seychelles	0				1	1	18					20
Sierra Leone	6,765				494	5,352	306					11,852
Somalia	379				20	7	175					404
South Africa	180				3	24	373		43,665			44,209
South Sudan	22						412					425
Sudan	57					2	64					108
Suriname	325					68	180					421
Syrian Arab Republic	3						2					4

Importing country	Exporting country											Total
	EU	Argentina	Brazil	Mexico	Canada	United States	United Kingdom	China	Japan	Korea	Thailand	
Tajikistan	12,803					1						11,759
Tanzania_ United Republic of	1,173				187	109	1,392		62,306			65,163
Thailand	82				18	101	198		27,568			27,967
Timor- Leste	0				1							1
Togo	33,853				3,028	6,799	101					43,482
Tonga	1				4		5					10
Tunisia	7,490				407	44	108					7,943
Turkey	2,350				677	1,176	3,507					7,610
Turkmenistan	8					6						13
Uganda	30				11	5	372		27,460			27,876
Ukraine	216,223				3,102	106,489	2,789					306,979
Uruguay	81		16		2	69						163
Uzbekistan	90				2	33						109
Uzbekistan	109											109
Venezuela_ Bolivarian Republic of	27				39	186						251
Viet Nam	35				4	151						190
Western Sahara	7											6
Western Samoa	0					1						1
Yemen	1				7	1,752						1,760
Zaire	0					565						565
Zambia	20					3	759		13,820			14,602
Zimbabwe	8					3	1,970					1,981
Grand Total	1,147,542	12,392	2,655	5,908	93,952	607,803	68,600	32	565,334	220	1,412	2,398,411

Source: ITF Dashboard, 2023

F. Stakeholders' survey questionnaire

**UNECE**

Stakeholders' survey on road safety and imported used vehicles in LMICs

This survey aims to collect information and experiences from stakeholders around the world on the **impact of imported used vehicles on road safety**, in order to understand the road safety impacts of the trade of used vehicles and draft recommendations for policies aiming to improve road safety in LMICs.

The study is implemented within the **UNSFRR Project "Safer and Cleaner Used Vehicles for Africa"**. More specifically, after the successful 1st phase of the project focusing on Africa (ECOWAS and EAC), the current 2nd phase of the project expands its focus to the Central and Southern African regions, as well as selected Asian countries. The objective of the present study is the analysis of road safety data linked to used vehicles globally, and the development of recommendations based on the analysis of the policy impacts on trade of used vehicles.

Disclaimer: The information collected in this survey will be handled according to the highest ethical standards of participant privacy. The collected data will be handled and processed only by the dedicated UNECE staff and the researcher undertaking the study. Only aggregate results will be published by the UNECE, in the form of graphs and/or summary Tables. Individual responses provided in free text form may be published or quoted anonymously, and they will not be associated to participants names, affiliations or other identifying information. By participating in this survey, you agree to these terms of data handling, processing and publication.

For any further information, please **contact** Dr. Eleonora Papadimitriou <nopapadi@mail.ntua.gr>

* Required

1. Last Name *

2. Name *

3. Affiliation *

4. What is your organisation's main role in the used vehicles trade market? *

- ☐ Importing country / agency
- ☐ Exporting country / agency
- ☐ Regulatory body - national
- ☐ Regulatory body - international
- ☐ Consultancy / advisory
- ☐ Research / academic
- ☐ Industry / manufacturer
- ☐ Other

5. Please indicate how much you agree with each one of the following statements, regarding the **impacts of importing used vehicles in LMICs** *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The import of used vehicles is necessary to support mobility demand in LMICs	☐	☐	☐	☐	☐
The import of used vehicles is detrimental for road safety	☐	☐	☐	☐	☐
The import of used vehicles leads to more congestion	☐	☐	☐	☐	☐
The import of used vehicles leads to increased pollution	☐	☐	☐	☐	☐
The imports of used vehicles contribute to economic growth in LMICs	☐	☐	☐	☐	☐

6. In your opinion, how do the imports of used vehicles affect the number of traffic fatalities in a country? *

Enter your answer

7. In your opinion, what are the most important vehicle-related factors that may result in negative impacts of imported used vehicles on traffic safety? *

- ☐ Age of vehicle
- ☐ Non adoption of UN vehicle regulations
- ☐ Lack of passive safety systems
- ☐ Lack of active safety systems
- ☐ Lack of technical inspection
- ☐ Other

8. Do you have an estimate of the share of imported used vehicles on the roads in your country? *

Enter your answer

9. Do you have an estimate of the share of traffic crashes with involvement of an imported used vehicle? *

Enter your answer

9. Do you have an estimate of the share of traffic crashes with involvement of an imported used vehicle? *

Enter your answer

10. Please indicate how much you agree with the following statements, regarding the **effective policies on used vehicles imports** *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Used vehicle imports should be banned	☐	☐	☐	☐	☐
There should be an age limit in the imported used vehicles	☐	☐	☐	☐	☐
There should be a mandatory technical inspection of imported used vehicles	☐	☐	☐	☐	☐
Imported used vehicles should meet the UN core vehicle regulations	☐	☐	☐	☐	☐
Used vehicles imports should be regulated at national level	☐	☐	☐	☐	☐

11. What are your recommendations for policy actions from the importing countries, in order to improve the safety of imported used vehicles? *

Enter your answer

12. What are your recommendations for actions from the side of the exporting countries / agencies, in order to support the safety of imported used vehicles in LMICs? *

Enter your answer

13. Any other comment or remark that you would like to add? *

Enter your answer

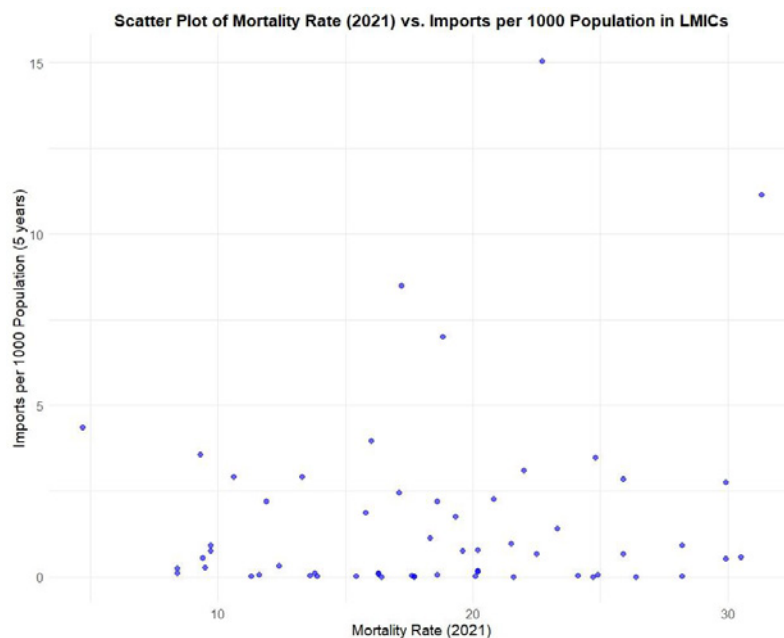
Submit

G. Descriptive macroscopic analysis

LMICs

Figure A. 1

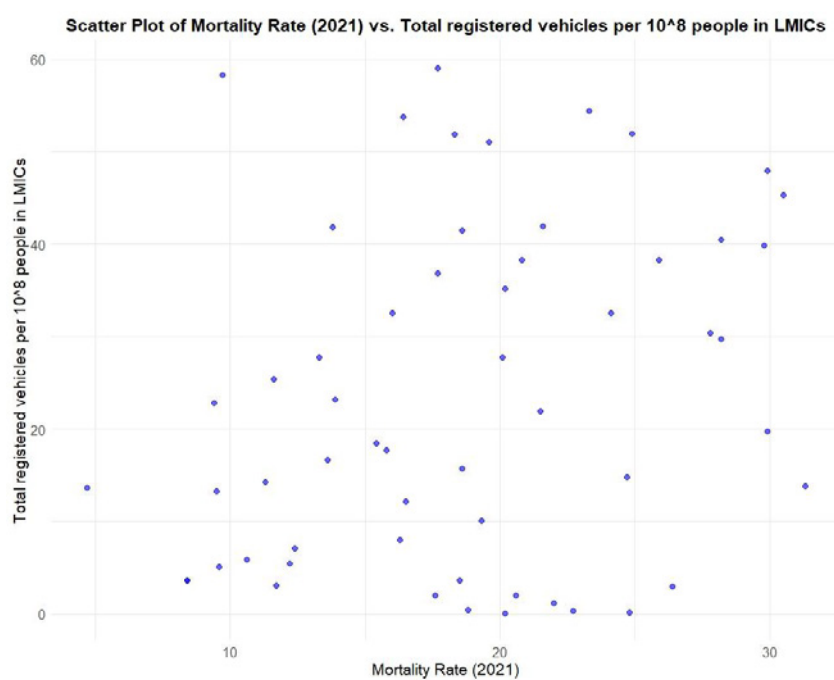
Mortality rates vs. average imports per population



Source: WHO, 2023 and ITF 2023, UNECE 2025

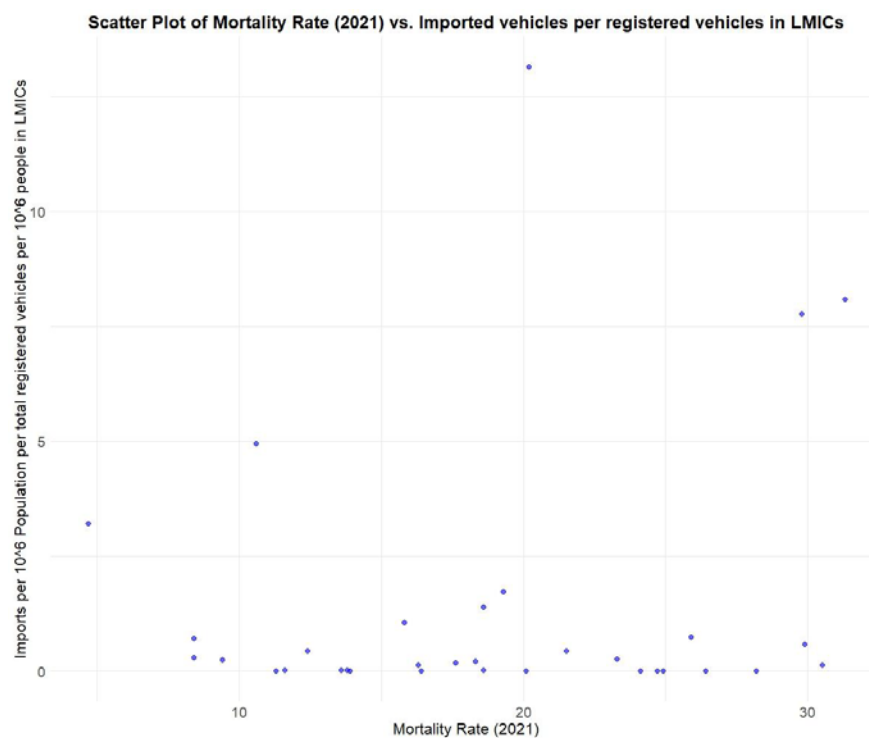
Figure A. 2

Mortality rates vs. passenger cars



Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 3
Mortality rates vs. imports per passenger cars



Source: WHO 2023, ITF 2023, UNECE, 2025

For this graph, there are issues with order of magnitude.

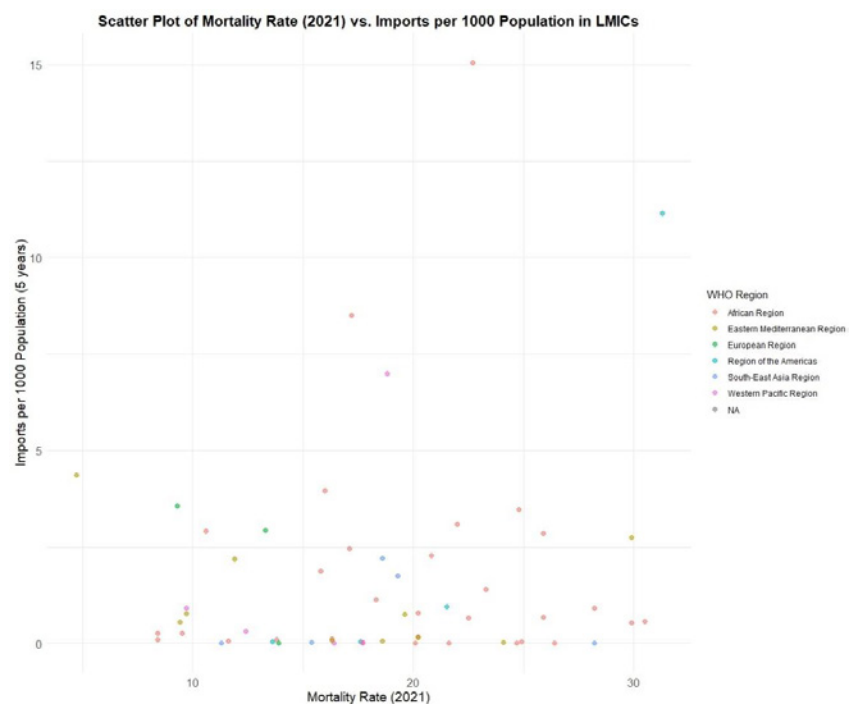
To solve this, a new variable is created:

$$\text{IMPpervch} = (\text{IMPORT_17_21_per10}^6\text{_pop} / \text{Tot_reg_veh_per10}^6\text{_pop})$$

I.e. Used car imports between 2017-2021 per million people divided by

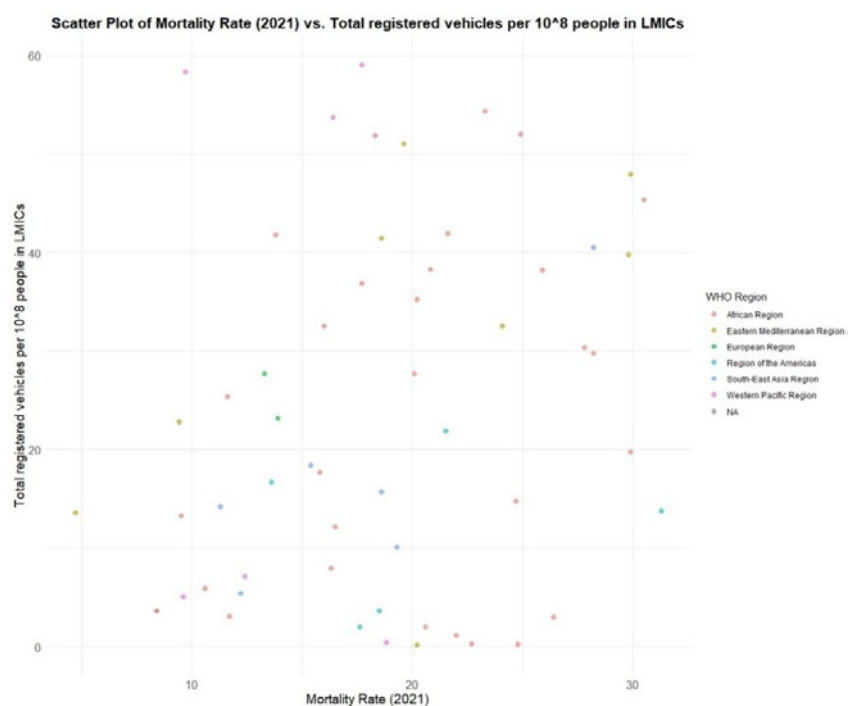
the total registered vehicles per million people (used in all scatterplots **but not models**)

Figure A. 4

Mortality rates vs. average imports per population per UN region

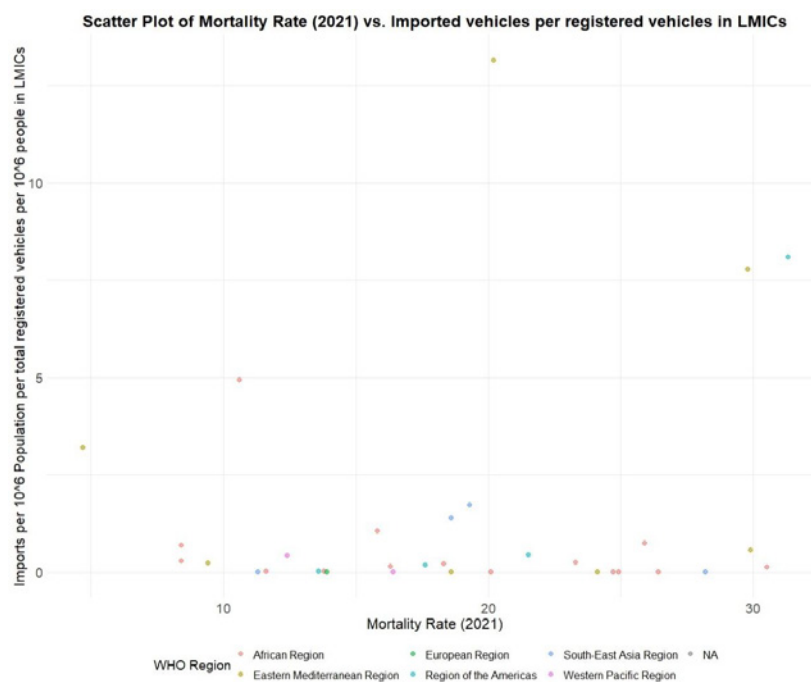
Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 5

Mortality rates vs. passenger cars per UN region

Source: WHO 2023, ITF 2023, UNECE, 2025

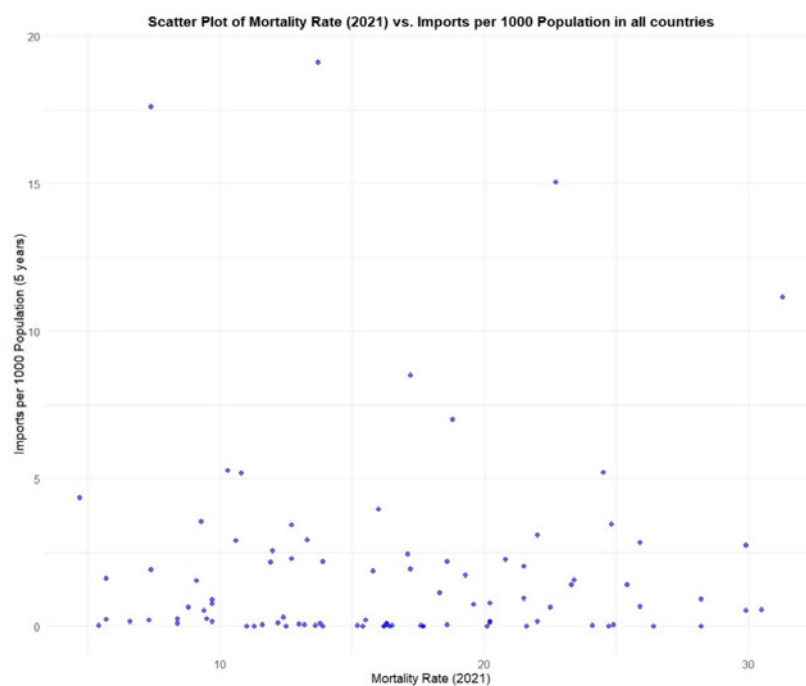
Figure A. 6
Mortality rates vs. imports per passenger cars per UN region



Source: WHO 2023, ITF 2023, UNECE, 2025

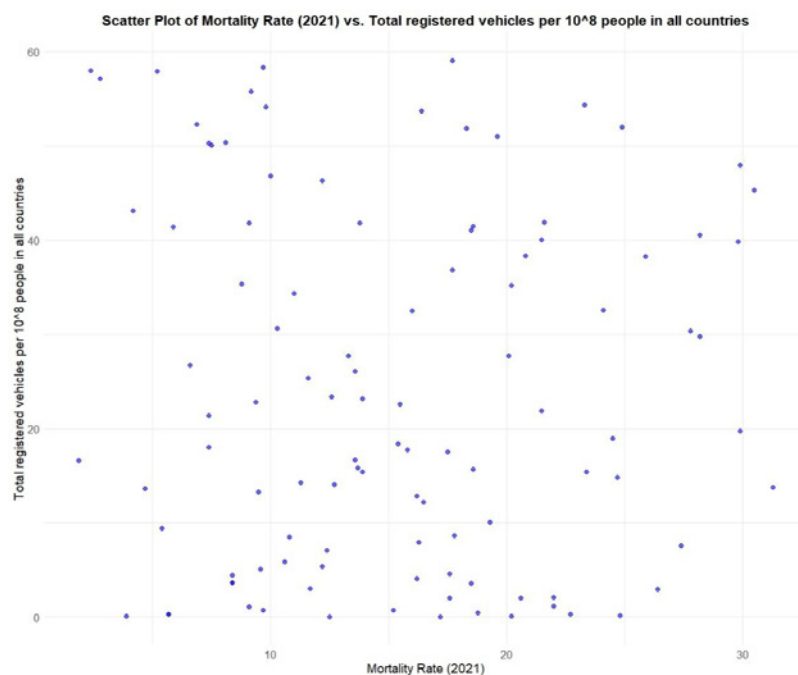
All countries

Figure A. 7
Mortality rates vs. average imports per population



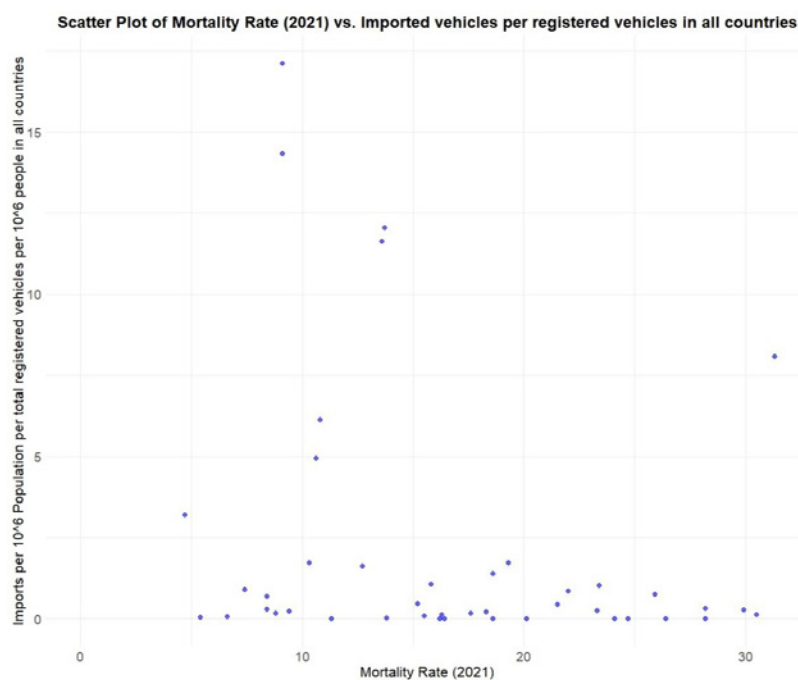
Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 8
Mortality rates vs. passenger cars



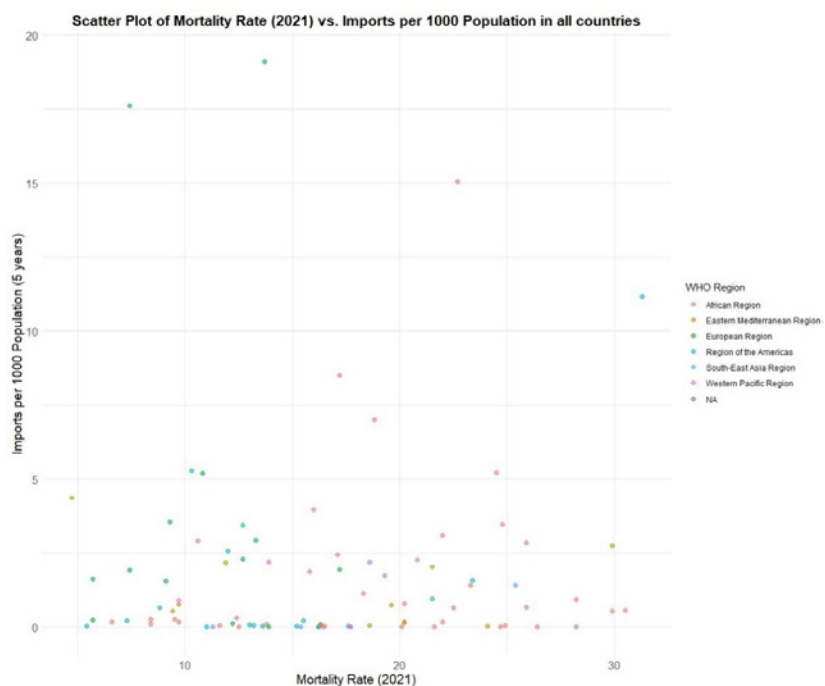
Source WHO 2023, ITF 2023, UNECE, 2025

Figure A. 9
Mortality rates vs. imports per passenger cars



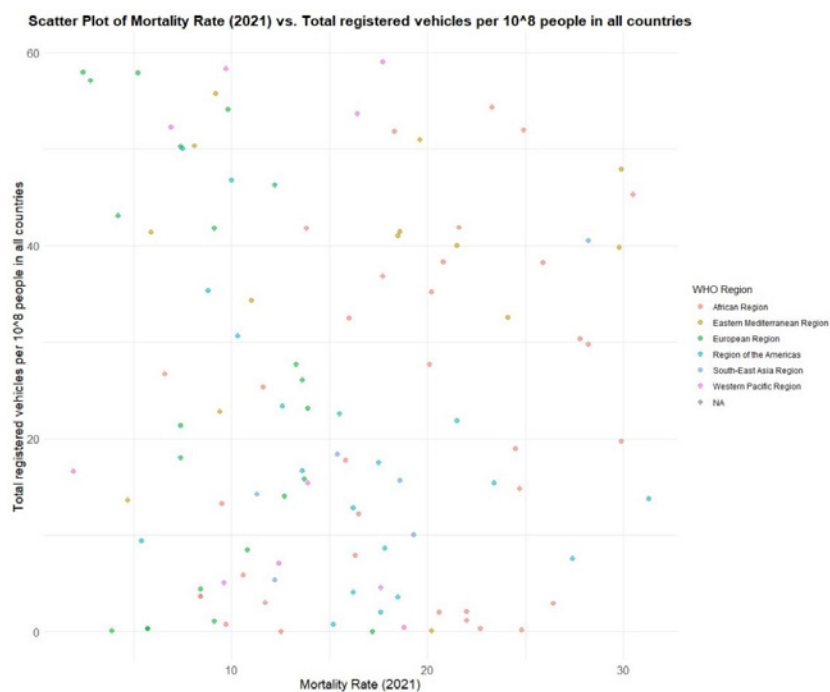
Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 10
Mortality rates vs. average imports per population per UN region



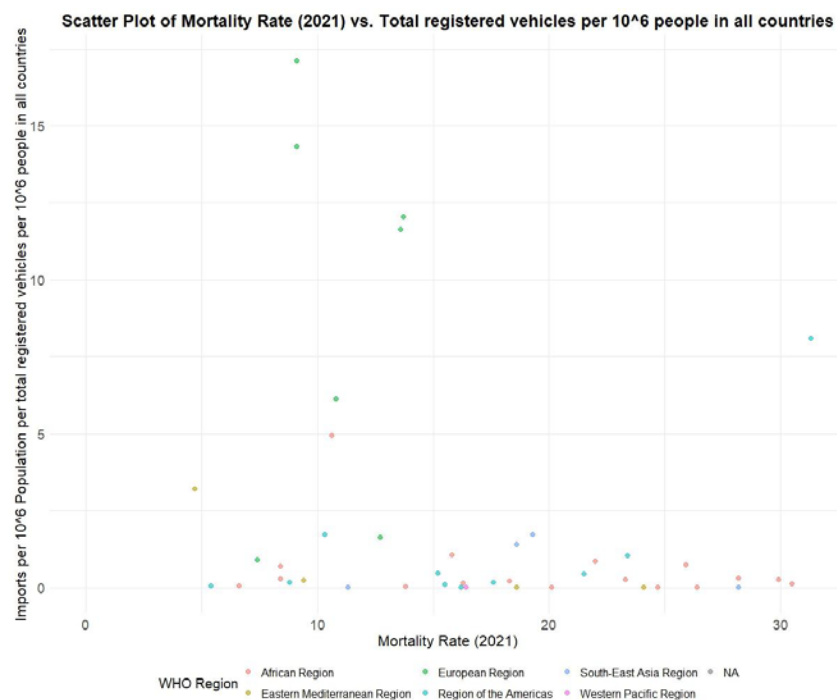
Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 11
Mortality rates vs. passenger cars per UN region



Source: WHO 2023, ITF 2023, UNECE, 2025

Figure A. 12

Mortality rates vs. imports per passenger cars per UN region

Source: WHO 2023, ITF 2023, UNECE, 2025

Analysis of the impact of imported used vehicles on road safety and the effectiveness of policies

Part of the WP.29 "How it works – How to join it" series

Part of the WP.29 "How it works-How to join it" series, this publication aims to analyse the policy impacts on trade of used vehicles and provide an analysis of road safety data linked to used vehicles globally.

This publication aims to develop recommendations based on the analysis of the policy impacts on trade of used vehicles, and the analysis of road safety data linked to used vehicles globally. It includes review of the impact of safer and cleaner vehicle policies that have been implemented in LMICs; and a review of road safety accidents and mortality data globally linked to used vehicles, with a dedicated analysis aiming to estimate the road safety impacts of relevant policies.

Whether or not a regulatory approach is pursued, or whether additional policy levers are employed, this publication highlights the case for using the UN Regulations to ensure that importing countries will only receive quality used vehicles.

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